Chapter (23):- Electric Fields:-

23.1 Properties of charges:-

1- Kinds of charges :- +ve and -ve charges

Charges of the same sign repel each other and opposite sign attract each other.

2- Conservation of electric charges:-

when one object rubbed against another, one object gains +ve charge while the other gains an equal amount of -ve charge

like rod of glass (+ve) rubbed by silk (-ve).

the total charge on isolated system is conserved.

3- Quantization of electric charge :-

charge appears as some integral multiple of a fundamental amount of charge (e)

$q = ne$ [n :- integer
e :- charge of electron $e = 1.6 \times 10^{-19}$ C]

[23.2] charging objects by induction :-

materials classified according to ability of electrons to move through it to :-

a- Conductors :- materials in which some of the electrons are free electrons that are not bound to atom and can move freely through the material.

b- insulators :- materials in which all electrons are bound to atoms and cannot move freely through the material.

c- semi conductors :- their electrical properties are between insulators and conductors.

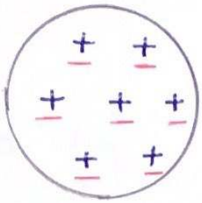
methods of charging :-

1- charging by contact.

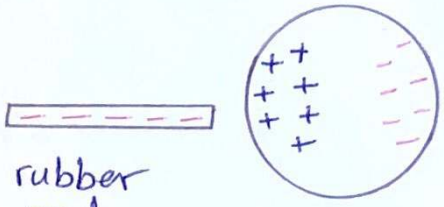
2- charging by rubbing.

3- charging by induction.

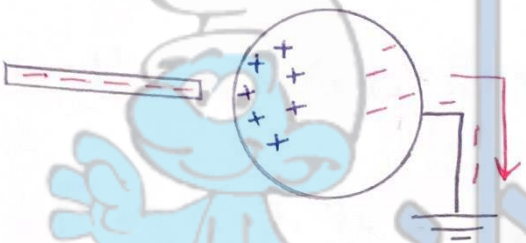
Charging by induction:-



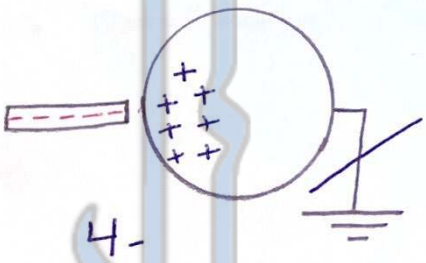
1- neutral conducting sphere.



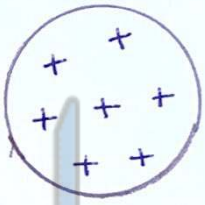
2-



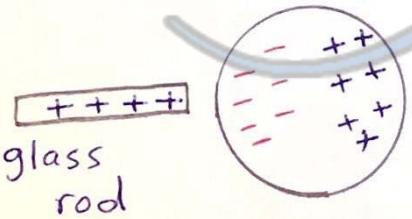
3-



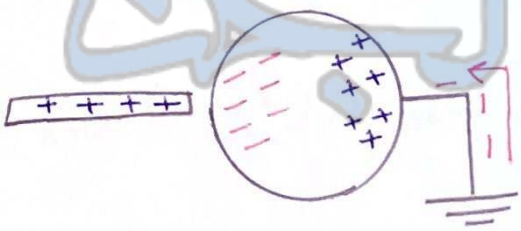
4-



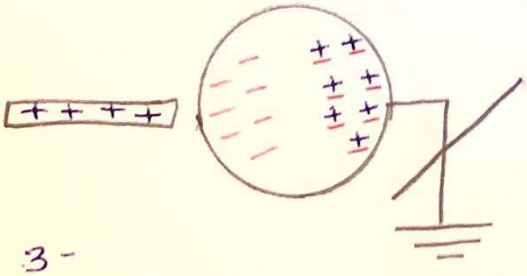
5- Charged conductor



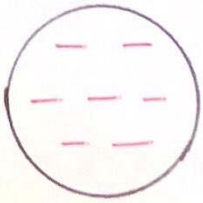
1-



2-



3-



4-

23.3 Coulomb's Law:-

Properties of electric force

$F \propto \frac{1}{r^2}$ $\quad$ $F \propto q_1 \cdot q_2$

- F is attractive force for opposite sign charges and repulsive force for the same sign charges
- the electric force is conservative force.

$F = \frac{K q_1 q_2}{r^2}$, where

K is Coulomb constant $K = 8.987 \times 10^9 \ N.m^2/C^2$

$K = \frac{1}{4\pi\epsilon_0}$, ϵ_0 is permittivity of vacuum

$$\epsilon_0 = 8.85 \times 10^{-12} \ C^2/N.m^2$$

Ex(23.1):- Hydrogen atom:-

$r = 5.3 \times 10^{-11}$ m $\quad$ $q_p = 1.6 \times 10^{-19}$ C

$q_e = -1.6 \times 10^{-19}$ C

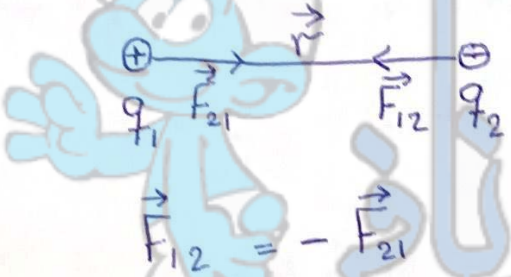
$$F_e = \frac{K q_1 q_2}{r^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2} = 8.2 \times 10^{-8} \ N$$

$$F_g = \frac{G m_e m_p}{r^2}$$

$$= \frac{6.67 \times 10^{-11} \times 9.11 \times 10^{-31} \times 1.67 \times 10^{-27}}{(5.3 \times 10^{-11})^2}$$

$$= 3.6 \times 10^{-47} \text{ N}$$

$F_g \ll F_e$

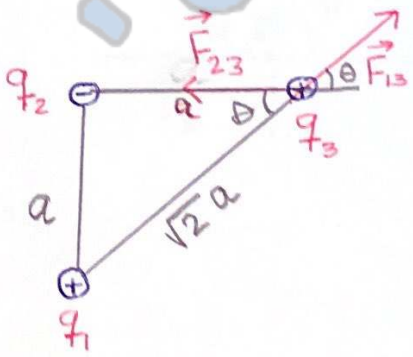


$$\vec{F}_{12} = -\vec{F}_{21}$$

Ex(23-2) Find the resultant Force

$q_1 = q_3 = 5\ \mu C$ $\quad q_2 = -2\ \mu C$ $\quad a = 0.1\ m$

Find the resultant force on q_3?



$$F_{13} = \frac{9 \times 10^9 \times (5 \times 10^{-6})^2}{(0.1 \times \sqrt{2})^2} = 11\ N$$

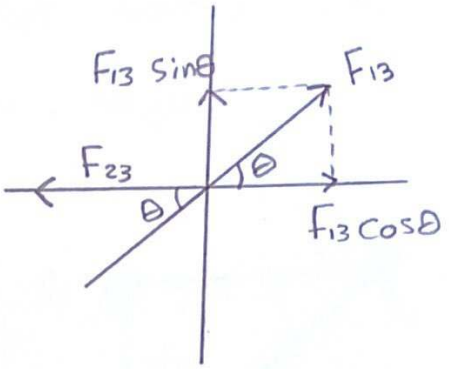
$$\vec{F}_{23} = \frac{9 \times 10^9 \times 5 \times 10^{-6} \times 2 \times 10^{-6}}{(0.1)^2}$$

$$= -9\hat{i}\ N$$

$$(\vec{F}_{13})_x = F_{13}\cos\theta$$

$$= 11 \times \frac{\cancel{a}}{\cancel{a}\sqrt{2}}$$

$$= \frac{11}{\sqrt{2}}\hat{i}\,N$$



$$(\vec{F}_{13})_y = F_{13}\sin\theta$$

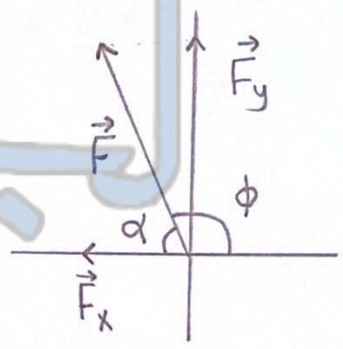
$$= 11 \times \frac{\cancel{a}}{\cancel{a}\sqrt{2}} = \frac{11}{\sqrt{2}}\hat{j}\,N$$

$$\Sigma \vec{F}_x = (\vec{F}_{13})_x + (\vec{F}_{23})$$

$$= \frac{11}{\sqrt{2}}\hat{i} - 9\hat{i} = -1.1\,N$$

$$\Sigma F_y = 7.8\hat{j}\,N$$

$$\vec{F} = F_x\hat{i} + F_y\hat{j} = -1.1\hat{i} + 7.8\hat{j}$$



$$|\vec{F}| = \sqrt{(F_x)^2 + (F_y)^2}$$

$$= \sqrt{(-1.1)^2 + (7.8)^2} = 7.9\,N$$

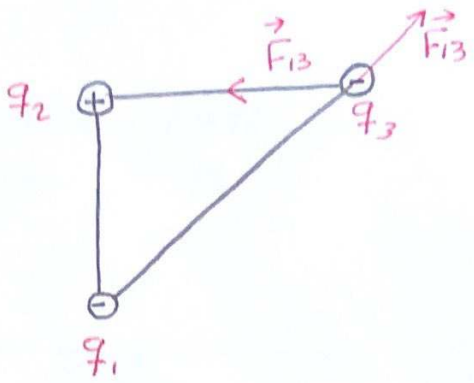
$$\tan\alpha = \frac{F_y}{F_x} = \frac{7.8}{-1.1} = -7$$

$$\alpha = \tan^{-1}(-7) = -82^\circ$$

$$\phi = 180 - 82 = 98^\circ$$

What if ?? the signs of all three charges were charged to opposite charge ?

the final result for $\Sigma \vec{F}$ would be the same.



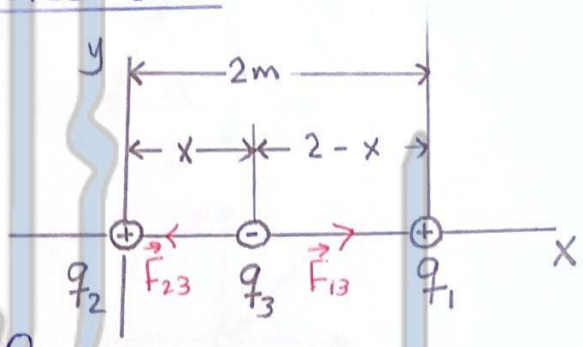
Ex(23.3):- where is the net force zero?

$q_1 = 15\ \mu C$ at $x = 2\ m$

$q_2 = 6\ \mu C$ at the origion

the resultant force at $q_3 = 0$

what is the x- coordinate of q_3



$$|\vec{F}_{13}| = |\vec{F}_{23}|$$

$$\frac{K q_1 q_3}{r_{13}^2} = \frac{K q_2 q_3}{r_{23}^2}$$

$$\sqrt{\frac{15 \times 10^{-6}}{(2-x)^2}} = \sqrt{\frac{6 \times 10^{-6}}{x^2}}$$

$$\frac{3.87}{2-x} = \frac{2.45}{x}$$

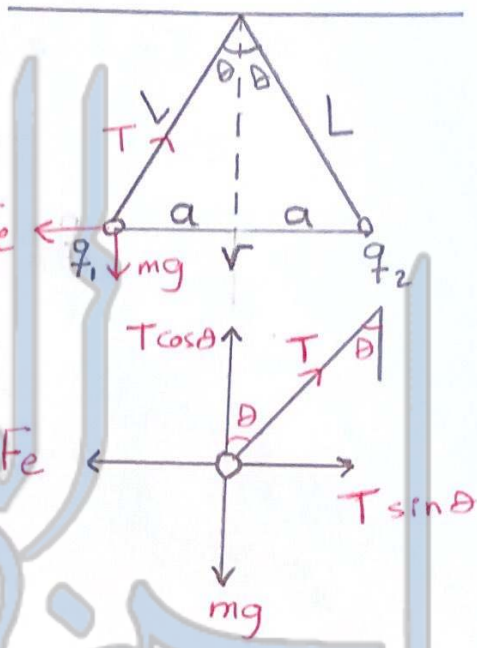
$3.87x = 4.9 - 2.45x$

$x = 0.77\ m$

Ex(23.4):- Find the charges on the spheres?

$m_1 = m_2 = 3\times10^{-2}\ kg \qquad L = 0.15\ m \qquad \theta = 5^\circ$

$q_1 = q_2 = ??$



at equalibrium $\Sigma F_x = 0$

$\Sigma F_y = 0$

$F_e = T\sin\theta$ ---①

$mg = T\cos\theta$ ---②

$\frac{F_e}{mg} = \tan\theta = \tan 5^\circ$

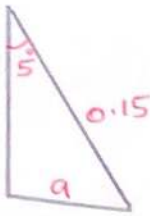
$F_e = 3\times10^{-2}\times10\times0.087$

$= 0.026\ N$

$F_e = \frac{K\,q_1 q_2}{r^2}$

$0.026 = \frac{9\times10^{9}\times q^2}{r^2}$

$q^2 = \frac{0.026\ r^2}{9\times10^{9}}$



$$\sin 5^\circ = \frac{a}{0.15}$$

$$a = 0.013 \text{ m}$$

$$r = 2a = 2 \times 0.013$$

$$= 0.026 \text{ m}$$

$$q^2 = \frac{0.026 \times (0.026)^2}{9 \times 10^9} = 1.97 \times 10^{-15}$$

$$q = 4.42 \times 10^{-8} \text{ C}$$

$$= 44.2 \times 10^{-9} \text{ C}$$

$$= 44.2 \text{ nC}$$

Quick Quiz (23.1), Page 692:-

Quick Quiz 23.1 Three objects are brought close to each other, two at a time. When objects A and B are brought together, they repel. When objects B and C are brought together, they also repel. Which of the following are true? **(a)** Objects A and C possess charges of the same sign. **(b)** Objects A and C possess charges of opposite sign. **(c)** All three objects possess charges of the same sign. **(d)** One object is neutral. **(e)** Additional experiments must be performed to determine the signs of the charges.

A repel B , B repel C

answer :- a , c , e

Quick Quiz (23.2), Page 693 :-

Quick Quiz 23.2 Three objects are brought close to one another, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. Which of the following are necessarily true? **(a)** Objects A and C possess charges of the same sign. **(b)** Objects A and C possess charges of opposite sign. **(c)** All three objects possess charges of the same sign. **(d)** One object is neutral. **(e)** Additional experiments must be performed to determine information about the charges on the objects.

A attract B, B repel C

suppose that A is +ve then B is -ve

and C is -ve too.

answer :- b , e

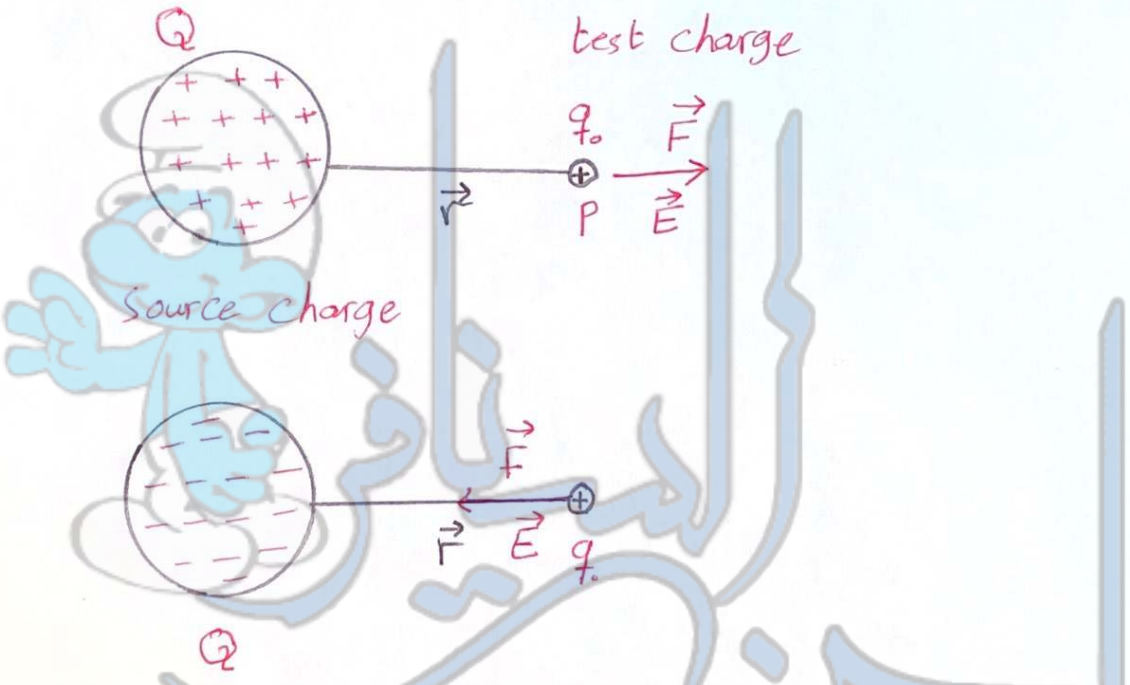
Quick Quiz (23.3), Page 696 :-

Quick Quiz 23.3 Object A has a charge of $+2\ \mu C$, and object B has a charge of $+6\ \mu C$. Which statement is true about the electric forces on the objects? **(a)** $\vec{F}_{AB} = -3\vec{F}_{BA}$ **(b)** $\vec{F}_{AB} = -\vec{F}_{BA}$ **(c)** $3\vec{F}_{AB} = -\vec{F}_{BA}$ **(d)** $\vec{F}_{AB} = 3\vec{F}_{BA}$ **(e)** $\vec{F}_{AB} = \vec{F}_{BA}$ **(f)** $3\vec{F}_{AB} = \vec{F}_{BA}$

answer :- (b)

23.4 Analysis Model : Particle in a electric field :-

Electric field :- electric force acting on a +ve test charge placed at the point divided by magnitude of the test charge q_0.



$$\vec{E} = \frac{\vec{F}}{q_0} \equiv [N/C]$$

$$\vec{F} = \frac{K Q q_0}{r^2} \hat{r}$$

$$\vec{E} = \frac{K Q}{r^2} \hat{r}$$

$\vec{E} = K \sum_i \frac{q_i}{r_i^2} \hat{r}_i \equiv$ total electric field due to a group of charges.

Ex(23.5) A suspended water droplet:-

$m = 3 \times 10^{-12}$ Kg $\quad \vec{E} = -6 \times 10^3 \hat{j}$ N/C

$\Sigma F_y = 0$

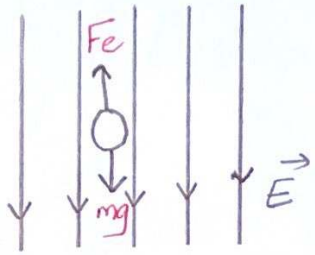
$F_e - F_g = 0$

$q(E) - mg = 0$

$q = \frac{mg}{E}$

$= \frac{-3 \times 10^{-12} \times 9.8}{6 \times 10^3}$

$= -4.9 \times 10^{-15}$ C



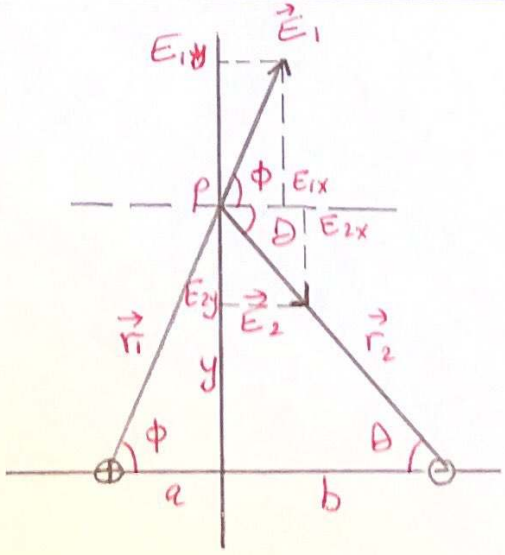
Ex(23.6):- electric field due to two charges:-

A) $E_1 = \frac{K q_1}{r_1^2}$

$r_1 = \sqrt{a^2 + y^2}$

$E_1 = \frac{K q_1}{a^2 + y^2}$

$\vec{E}_1 = \frac{K q_1}{a^2+y^2} \cos\phi \hat{i} + \frac{K q_1}{a^2+y^2} \sin\phi \hat{j}$



$$E_2 = \frac{K q_2}{r_2^2}$$

$$r_2 = \sqrt{b^2 + y^2}$$

$$E_2 = \frac{K q_2}{b^2 + y^2}$$

$$\vec{E}_2 = \frac{K q_2}{b^2+y^2} \cos\theta \, \hat{i} - \frac{K q_2}{b^2+y^2} \sin\theta \, \hat{j}$$

$$E_x = E_{1x} + E_{2x}$$

$$= \frac{K q_1}{a^2+y^2} \cos\phi + \frac{K q_2}{b^2+y^2} \cos\theta$$

$$E_y = E_{1y} + E_{2y}$$

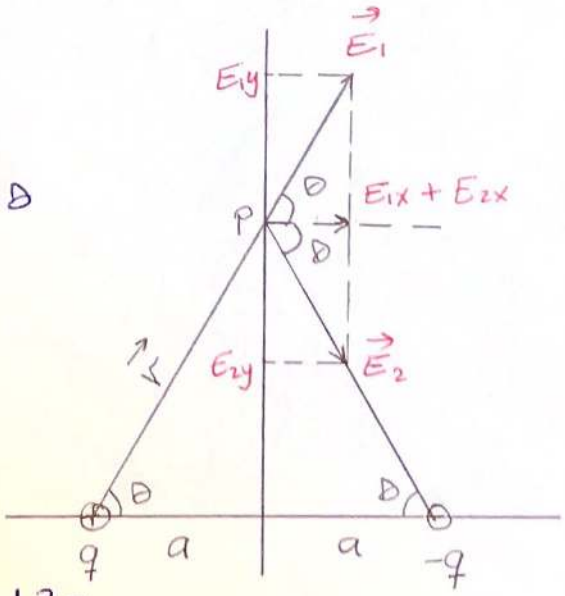
$$= \frac{K q_1}{a^2+y^2} \sin\phi - \frac{K q_2}{b^2+y^2} \sin\theta$$

B) $q_1 = q_2$, $a = b$

$$E_x = \frac{K q}{a^2+y^2} \cos\theta + \frac{K q}{a^2+y^2} \cos\theta$$

$$= \frac{2 K q}{a^2+y^2} \cos\theta$$

$$\cos\theta = \frac{a}{r} = \frac{a}{\sqrt{a^2+y^2}}$$



$$E_x = \frac{2Kq}{a^2+y^2} \cdot \frac{a}{\sqrt{a^2+y^2}}$$

$$E_x = \frac{2Kaq}{(a^2+y^2)^{3/2}}$$

$$E_y = \frac{Kq}{a^2+y^2}\sin\theta - \frac{Kq}{a^2+y^2}\sin\theta$$

$$= 0$$

C) $y >>> a$

$$E \approx \frac{2Kaq}{y^3}$$

Example:-

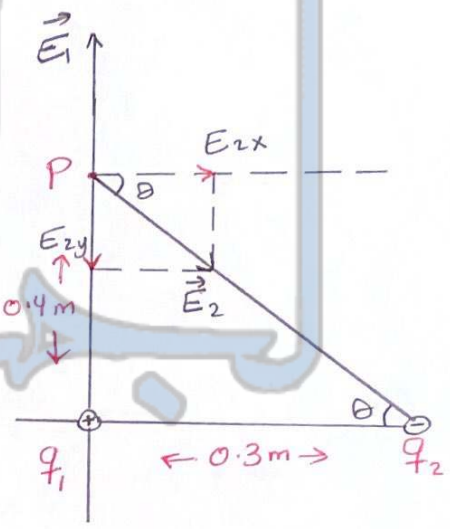
$q_1 = 7\mu C \qquad q_2 = -5\mu C$

$$\vec{E}_1 = \frac{Kq_1}{r^2} = \frac{9\times10^{9}\times7\times10^{-6}}{(0.4)^2}$$

$$= 3.94\times10^{5}\,\hat{j}\ N/C$$

$$E_2 = \frac{Kq_2}{r^2} = \frac{9\times10^{9}\times5\times10^{-6}}{(0.5)^2}$$

$$= 1.8\times10^{5}\ N/C$$

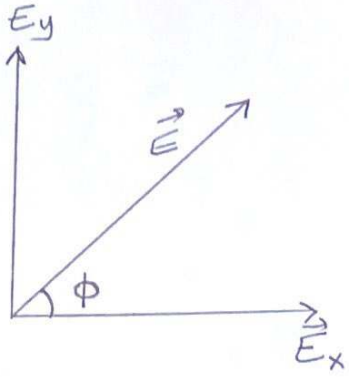


$$E_{2x} = E_2 \cos\theta = 1.8 \times 10^5 \times \frac{3}{5} = 1.1 \times 10^5 \text{ N/C}$$

$$E_{2y} = E_2 \sin\theta = 1.8 \times 10^5 \times \frac{4}{5} = 1.44 \times 10^5 \text{ N/C}$$

$$\Sigma E_y = (3.94 - 1.44) \times 10^5 = 2.5 \times 10^5 \text{ N/C}$$

$$\vec{E} = [1.1 \times 10^5 \hat{i} + 2.5 \times 10^5 \hat{j}] \text{ N/C}$$



$$|\vec{E}| = \sqrt{(1.1 \times 10^5)^2 + (2.5 \times 10^5)^2}$$

$$= 2.7 \times 10^5 \text{ N/C}$$

$$\phi = \tan^{-1}\left(\frac{E_y}{E_x}\right)$$

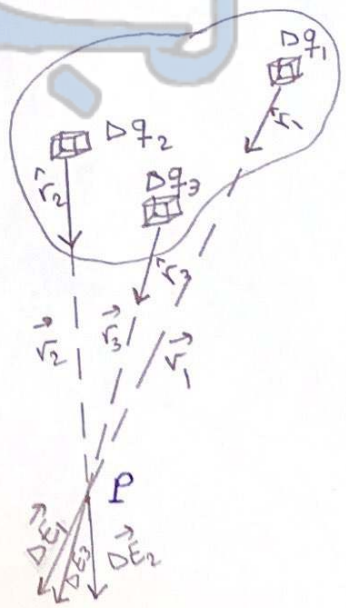
$$= \tan^{-1}\left(\frac{2.5}{1.1}\right) = 66^\circ$$

23.5 Electric field of a continuous charge distribution:-

$$\Delta\vec{E} = \frac{K\,\Delta q}{r^2}\,\hat{r}$$

r : distance from the element charge to the point P.

$\hat{r}$:- unit vector directed from element charge to ward P.



total electric field

$$E = K \lim_{\Delta q_i \to 0} \sum_i \frac{\Delta q_i}{r_i^2}\,\hat{r}_i \Rightarrow$$

$$\vec{E} = K_e \int \frac{dq}{r^2} \hat{r}$$

if a charge is uniformly distributed through a volume V, ρ is the volume charge density

$$\rho = \frac{Q}{V} \equiv [C/m^3]$$

Or a surface of area A, σ is the surface charge density

$$\sigma = \frac{Q}{A} \equiv [C/m^2]$$

or along a line of length L, λ is the length charge density

$$\lambda = \frac{Q}{L} \equiv [C/m]$$

but when a charge is nonuniformly distributed over a volume $dQ = \rho\, dV$

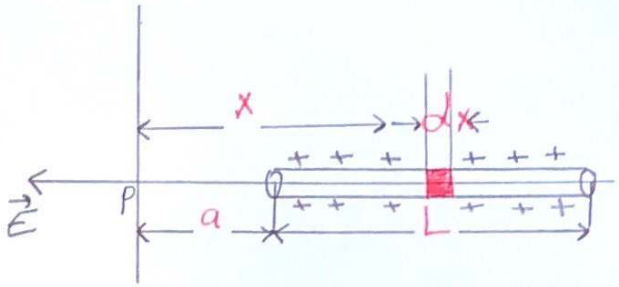
a surface $dQ = \sigma\, dA$

a length $dQ = \lambda\, dL$

Ex(23.7):- The electric field due to a charged rod:

Q :- total charge

λ :- Linear charge density



$$dq = \lambda\, dx$$

$$dE = \frac{K\, dq}{x^2} = \frac{K \lambda\, dx}{x^2}$$

$$E = K \int_a^{L+a} \frac{\lambda\, dx}{x^2}$$

$$= K\lambda \int_a^{L+a} x^{-2}\, dx = K\lambda \left[\frac{-1}{x}\right]_a^{L+a}$$

$$= K\lambda \left[\frac{1}{a} - \frac{1}{a+L}\right]$$

$$= K\lambda \left[\frac{a+L-a}{a(a+L)}\right]$$

$$= K\lambda \left[\frac{L}{a(a+L)}\right]$$

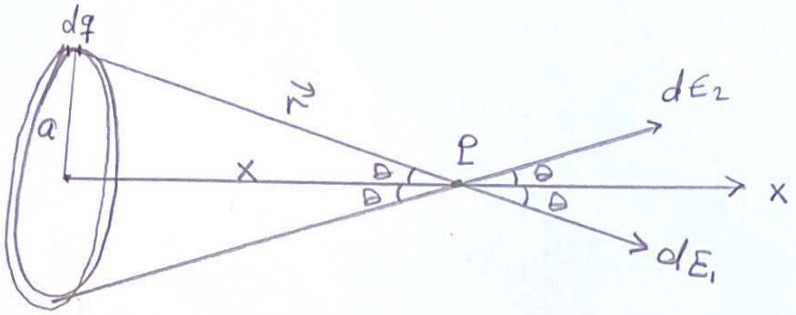
$$= K \frac{Q}{L} \left[\frac{L}{a(a+L)}\right]$$

$$E = \frac{K Q}{a(a+L)}$$

when $a >>> L \Rightarrow E = \frac{KQ}{a^2}$

the rod becomes point charge

Ex (23.8):- the electric field of a uniform ring of charge:-



$$dE = \frac{K\,dq}{r^2} = \frac{K\,dq}{x^2+a^2}$$

$dE_y = 0$, by symmetry

$$dE_x = dE\cos\theta$$

$$= \frac{K\,dq}{x^2+a^2}\cdot\frac{x}{(x^2+a^2)^{1/2}}$$

$$= \frac{K\,x\,dq}{(x^2+a^2)^{3/2}}$$

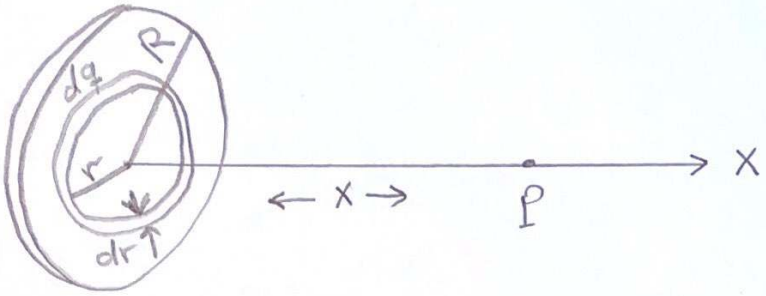
$$E = \int dE_x = K\int \frac{x\,dq}{(x^2+a^2)^{3/2}}$$

$$= \frac{K\times x\times Q}{(x^2+a^2)^{3/2}}$$

for $x >>> a \Rightarrow E = \frac{KQ}{x^2}$ (ring becomes point charge)

for $x = 0 \Rightarrow E = 0$

Ex(23.9):- The electric field of a uniformly charged disk:-



$$dq = \sigma\, dA = \sigma (2\pi r\, dr)$$

$$dE_x = \frac{K\, x\, dq}{(x^2+r^2)^{3/2}}$$, from previous example r instead of a

$$E_x = \int_0^R \frac{K\, x\, (2\pi r)\sigma\, dr}{(x^2+r^2)^{3/2}} = 2\pi\sigma K x \int_0^R \frac{r\, dr}{(x^2+r^2)^{3/2}}$$

let $u = x^2 + r^2$

$$du = 2r\, dr \Rightarrow dr = \frac{du}{2r}$$

$$\int \frac{r\, dr}{(x^2+r^2)^{3/2}} = \int \frac{r}{u^{3/2}} \cdot \frac{du}{2r} = \frac{1}{2}\int u^{-3/2}\, du$$

$$= \frac{1}{2}(-2u^{-1/2}) = \frac{-1}{u^{1/2}} = \frac{-1}{(x^2+r^2)^{1/2}}\Bigg]_0^R$$

$$= -\left[\frac{1}{(x^2+R^2)^{1/2}} - \frac{1}{x}\right]$$

$$= \frac{1}{x} - \frac{1}{(x^2+R^2)^{1/2}}$$

$$E_x = 2\pi\sigma K x\left[\frac{1}{x} - \frac{1}{(x^2+R^2)^{1/2}}\right]$$

$$= 2\pi\sigma K\left[1 - \frac{x}{(x^2+R^2)^{1/2}}\right]$$

when P too near the surface ; $R \gg x$

$$E = 2\pi\sigma K$$

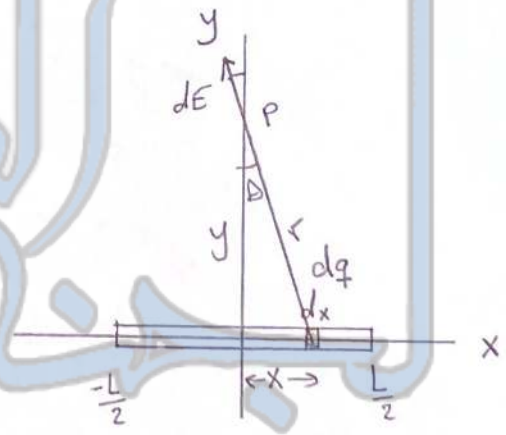
$$= \frac{2\pi\sigma}{4\pi\epsilon_0} = \frac{\sigma}{2\epsilon_0}$$

for infinite sheet $\Rightarrow$ uniform electric field

Example:-

$$dE = \frac{K\,dq}{r^2}$$

$$= \frac{K\lambda\,dx}{x^2+y^2}$$



$$dE_y = dE\cos\theta$$

$$= \frac{K\lambda\,dx}{x^2+y^2} \cdot \frac{y}{\sqrt{x^2+y^2}}$$

$$= \frac{K\lambda y\,dx}{(x^2+y^2)^{3/2}}$$

$$E_y = \int_{-\frac{L}{2}}^{\frac{L}{2}} K\lambda y\,\frac{dx}{(x^2+y^2)^{3/2}}$$

$$E_y = \frac{K\lambda y\, x}{y^2\sqrt{x^2+y^2}}\Bigg]_{-\frac{L}{2}}^{L/2}$$

$$= \frac{K\lambda L}{y\sqrt{(\frac{L}{2})^2+y^2}}$$

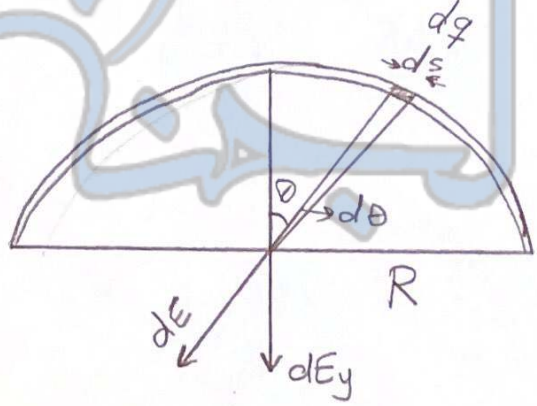
$$E_y = \frac{K Q}{y\sqrt{(\frac{L}{2})^2+y^2}}$$

for $y \gg\gg L \Rightarrow E_y = \frac{KQ}{y^2}$

for $y \ll L \Rightarrow E_y = \frac{K\lambda L}{y\,\frac{L}{2}}$

$E_y = \frac{2K\lambda}{y}$, for infinite rod.

Example:-



$$dE = \frac{K\,dq}{R^2}$$

$$dE_y = dE\cos\theta$$

$$= \frac{K\,dq}{R^2}\cos\theta$$

$$\sin(d\theta) = \frac{ds}{R} = d\theta \quad \text{(for small angle)}$$

$$\therefore ds = R\,d\theta$$

$$dq = \lambda\, ds$$
$$= \lambda R\, d\theta$$

$$dE_y = \frac{K\, dq}{R^2}\cos\theta$$
$$= \frac{K\lambda}{R}\cos\theta\, d\theta$$

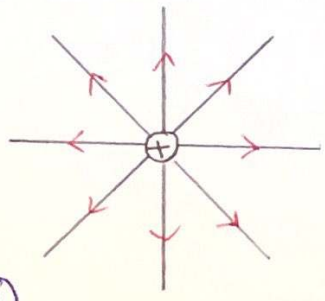
$$E_y = \frac{2K\lambda}{R}\int_0^{\pi/2}\cos\theta\, d\theta = \frac{2K\lambda}{R}\sin\theta\Big]_0^{\pi/2}$$

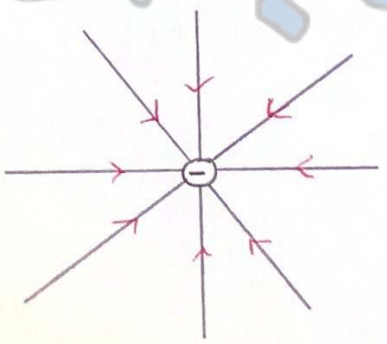
$$= \frac{2K\lambda}{R}(1-0) = \frac{2K\lambda}{R}$$

in general for arc $E = \frac{2K\lambda}{R}\sin\theta$

where $\theta = \frac{1}{2}$ angle of arc.

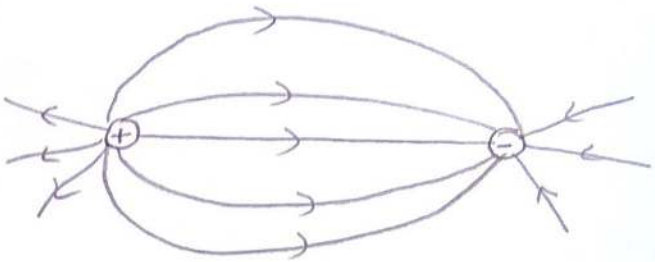
[23.6] Electric Field Lines:-





① For +ve charge, $\vec{E}$ lines are directed radially outward.

For -ve charge, $\vec{E}$ lines are directed radially inward.

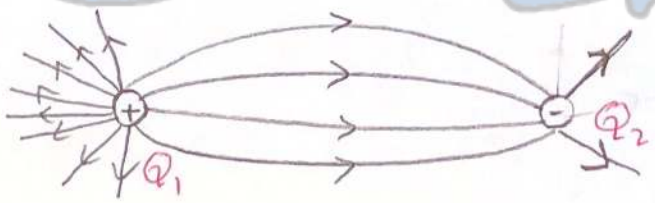


② the direction of $\vec{E}$ line is the direction of the tangent for curved line and the direction of the straight line.

③ No two field Lines can cross.

④ the number of electric field Lines are directly Proportional to the magnitude of the charge.

Ex: for $Q_1 = 2q$ and $Q_2 = q$ draw the $\vec{E}$ lines diagram.



$N_1 = 12$ $\quad N_2 = 6$

$$\frac{N_1}{N_2} = \frac{Q_1}{Q_2} = \frac{2q}{q}$$

$$N_1 = 2N_2$$

23.7 Motion of a charged particle in a uniform $\vec{E}$

for particle of charge q and mass m, $\vec{E}$ act on the charge by force qE

$$\Sigma \vec{F} = m\vec{a} = q\vec{E}$$

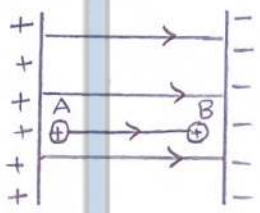
$$\vec{a} = \frac{q\vec{E}}{m}$$

$+q$ accelerates in the direction of $\vec{E}$.

$-q$ accelerates in the opposite direction of $\vec{E}$.

EX(23.10):- an accelerating positive charge:-

$V_A = 0$ $\quad X_A = 0$



a) $\vec{a} = \frac{q\vec{E}}{m}$ = constant

$\Delta X = \cancel{V_A t}^{\,0} + \frac{1}{2}at^2 = X_B - \cancel{X_A}^{\,0}$

$X_B = \frac{1}{2}at^2 = \frac{1}{2}\frac{qE}{m}t^2$

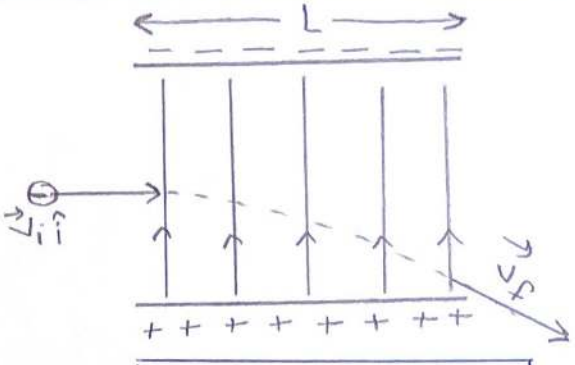
$V_B^2 = \cancel{V_A^2}^{\,0} + 2a(X_B - X_A) = 0 + 2a(d-0) = 2ad$

$$V_B = \sqrt{2ad} = \sqrt{\frac{2qEd}{m}}$$

b) $W = \Delta K \Rightarrow F\Delta X = K_B - K_A = \frac{1}{2}mV_B^2 - 0$

$$V_B = \sqrt{\frac{2F\Delta X}{m}} = \sqrt{\frac{2qEd}{m}}$$

Ex(23.11):- An accelerated electron:-



$v_i = 3 \times 10^6$ m/s

$E = 200$ N/C

$L = 0.1$ m

at $t = 0$

$\vec{v} = v_i \hat{i}$

$a_x = 0 \; ; \; F_x = 0$

$\vec{a}_y = -\frac{eE}{m_e} \hat{j}$

$v_{xi} = v_i \quad , \quad v_{yi} = 0$

$= \text{constant}$

$v_y = a_y t = -\frac{eE}{m_e} t$

$x_f = v_i t \Rightarrow t = \frac{x_f}{v_i}$

$y_f = \frac{1}{2} a_y t^2 = -\frac{1}{2} \frac{eE}{m_e} t^2 = -\frac{1}{2} \frac{eE}{m_e} \left(\frac{x_f}{v_i}\right)^2$

a) $a_y = -\frac{eE}{m_e} \hat{j} = -\frac{(1.6 \times 10^{-19}) \times 200}{9.11 \times 10^{-31}}$

$= -3.51 \times 10^{13} \hat{j} \; m/s^2$

b) $x_f = x_i + v_x t + \frac{1}{2} a_x t^2$

$x_f - x_i = v_x t \Rightarrow t = \frac{\Delta x}{v_x} = \frac{0.1}{3 \times 10^6} = 3.33 \times 10^{-8}$ sec

c) $y_f = \frac{1}{2} a_y t^2 = -\frac{1}{2} (3.51 \times 10^{13})(3.33 \times 10^{-8})^2$

$= -0.0195$ m

$= -1.95$ cm

Solved problems chapter (23)

Q9:- $q_1 = 7.5\,nC$ $q_2 = 4.2\,nC$ $r = 1.8\,m$

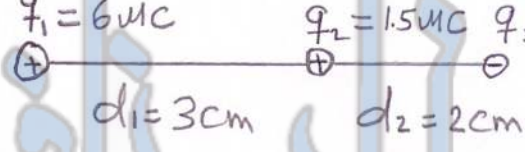
a- $F = \dfrac{K q_1 q_2}{r^2}$

$= \dfrac{9\times10^{9} \times 7.5\times10^{-9} \times 4.2\times10^{-9}}{(1.8)^2}$

$= 87.5\times10^{-9}\ N$

b- repulsive force.

Q12:-



a- on q_1, $F_{21} = \dfrac{9\times10^{9} \times 6\times10^{-6} \times 1.5\times10^{-6}}{9\times10^{-4}} = -9\times10^{-5}\,\hat{i}\,(N)$

$F_{31} = \dfrac{9\times10^{9} \times 6\times10^{-6} \times 2\times10^{-6}}{25\times10^{-4}} = 4.32\times10^{-5}\,\hat{i}\ (N)$

$\Sigma \vec{F}_1 = (-9 + 4.32)\times10^{-5} = -4.68\times10^{-5}\,\hat{i}\ (N)$

b- on q_2, $\vec{F}_{12} = -\vec{F}_{21} = 9\times10^{-5}\,\hat{i}\ (N)$

$F_{32} = \dfrac{9\times10^{9} \times 1.5\times10^{-6} \times 2\times10^{-6}}{4\times10^{-4}} = 6.75\times10^{-5}\,\hat{i}\ (N)$

$\Sigma \vec{F}_2 = (9 + 6.75)\times10^{-5}$

$= 15.75\times10^{-5}\,\hat{i}\ (N)$

C- on q_3

$$\vec{F}_{13} = -\vec{F}_{31} = -4.32 \times 10^{-5}\,\hat{i}\ (N)$$

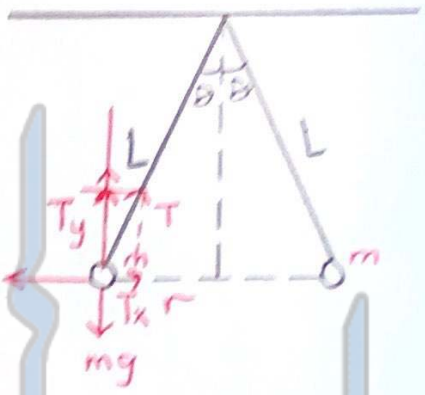
$$\vec{F}_{23} = -\vec{F}_{32} = -6.75 \times 10^{-5}\,\hat{i}\ (N)$$

$$\Sigma \vec{F}_3 = -11.07 \times 10^{-5}\,\hat{i}\ (N)$$

Q16:- $m = 0.2\ g$

$q_1 = q_2 = 7.2\ nC$

$\theta = 5°$ $\quad L = ??$



$$\Sigma F_x = \Sigma F_y = 0$$

$$T\cos\theta = mg \Rightarrow T = \frac{mg}{\cos\theta}$$

$$T = \frac{0.2 \times 10^{-3} \times 10}{\cos(5)} = 2 \times 10^{-3}\ N$$

$$\sin\theta = \frac{r}{L} \Rightarrow L = \frac{r}{\sin\theta}$$

$$F_e = T\sin\theta = 2 \times 10^{-3} \sin(5) = 1.75 \times 10^{-4}\ N$$

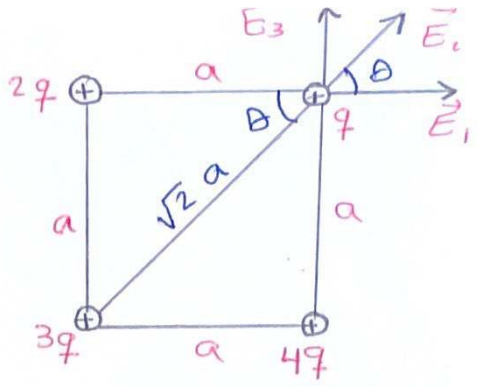
$$1.75 \times 10^{-4} = \frac{9 \times 10^{9} \times (7.2 \times 10^{-9})^2}{r^2}$$

$$r = 0.052\ m$$

$$L = \frac{r}{\sin\theta} = \frac{0.052}{\sin(5)}$$

$$= 0.6\ m$$

Q 25 :-



$$E_1 = \frac{9\times10^9 \times 2q}{a^2} = \frac{18\times10^9}{a^2}\,q$$

$$E_2 = \frac{9\times10^9 \times 3q}{2a^2} = \frac{13.5\times10^9}{a^2}\,q$$

$$E_3 = \frac{9\times10^9 \times 4q}{a^2} = \frac{36\times10^9}{a^2}\,q$$

$$E_x = E_1 + E_{2x}$$

$$= \frac{18\times10^9}{a^2}\,q + \frac{13.5\times10^9}{a^2}\,q\cos\theta \quad , \cos\theta = \frac{1}{\sqrt{2}}$$

$$= \frac{27.55\times10^9}{a^2}\,q$$

$$E_y = E_3 + E_{2y}$$

$$= \frac{36\times10^9}{a^2}\,q + \frac{13.5\times10^9}{a^2}\,q\sin\theta \quad , \sin\theta = \frac{1}{\sqrt{2}}$$

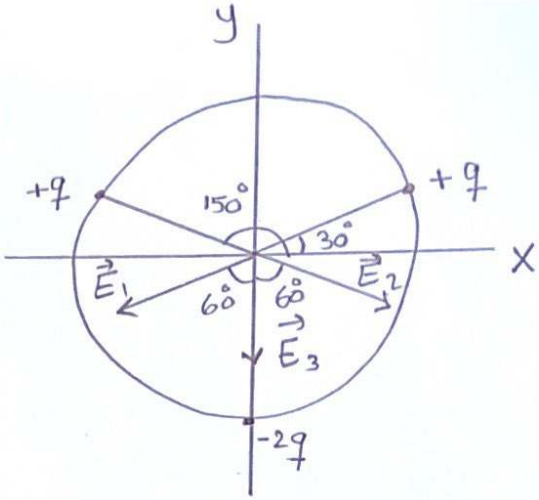
$$= \frac{45.55\times10^9}{a^2}\,q$$

$$E = \sqrt{(27.55)^2 + (45.55)^2}\times10^9\,\frac{q}{a^2}$$

$$= 53.23\times10^9\,\frac{q}{a^2}\ N/C$$

$$\phi = \tan^{-1}\left(\frac{45.55}{27.55}\right) = 58.8^\circ$$

Q26:-



$$E_1 = 9\times10^9 \frac{q}{r^2}$$

$$E_2 = 9\times10^9 \frac{q}{r^2}$$

$$E_{1x} = -E_{2x} \Rightarrow \Sigma E_x = 0$$

$$E_3 = 9\times10^9 \times \frac{2q}{r^2} = 18\times10^9 \frac{q}{r^2}$$

$$E_{1y} = E_{2y} = 9\times10^9 \frac{q}{r^2}\cos 60^\circ = 4.5\times10^9 \frac{q}{r^2}$$

$$E_y = (2\times 4.5\times10^9 + 18\times10^9)\frac{q}{r^2}$$

$$E_y = -27\times10^9 \frac{q}{r^2}\hat{j} \ (N/C)$$

Q37:- $L = 14\,cm$ $\quad Q = -22\,\mu C$ $\quad a = 36\,cm$

For a finite rod $E = \dfrac{KQ}{a(a+L)}$

$$E = \frac{9\times10^9 \times 22\times10^{-6}}{0.36(0.36+0.14)}$$

$$= 11\times10^5 \ N/C$$

Q38:- $R = 35\ cm \quad \sigma = 7.9 \times 10^{-3}\ C/m^2$

a- $X = 5\ cm$

for a disk $E = 2\pi\sigma K\left[1 - \frac{x}{(x^2+R^2)^{1/2}}\right]$

$E = 2\pi \times 7.9\times10^{-3} \times 9\times10^{9}\left[1 - \frac{5\times10^{-2}}{(25\times10^{-4}+0.1225)^{1/2}}\right]$

$= 38\ 3556680\ N/C$

c- $X = 50\ cm$

$E = 2\pi \times 7.9\times10^{-3} \times 9\times10^{9}\left[1 - \frac{0.5}{(0.25+0.1225)^{1/2}}\right]$

$= 807\ 55333.15\ N/C$

Q39:- $a = 10\ cm \quad Q = 75\ \mu C$

b- $X = 5\ cm$

for charged ring $E = \frac{KQX}{(x^2+a^2)^{3/2}}$

$E = \frac{9\times10^{9} \times 75\times10^{-6} \times 5\times10^{-2}}{(25\times10^{-4} + 0.01)^{3/2}}$

$= 24149534.16\ N/C$

Q 52:- $E = -6 \times 10^5 \hat{i}$ (N/C)

$t = 0$ $\quad \Delta x = 7\,cm$ $\quad v_f = 0$

a- $a = \frac{qE}{m} = \frac{1.6 \times 10^{-19} \times -6 \times 10^5}{1.67 \times 10^{-27}}$

$= -5.75 \times 10^{13}$ m/s^2

b- $v_f^2 = v_i^2 + 2a\Delta x$

$0 = v_i^2 - 2 \times 5.75 \times 10^{13} \times 0.07$

$v_i = 2837252.2$ m/s

c- $v_f = v_i + at$

$t = \frac{v_f - v_i}{a}$

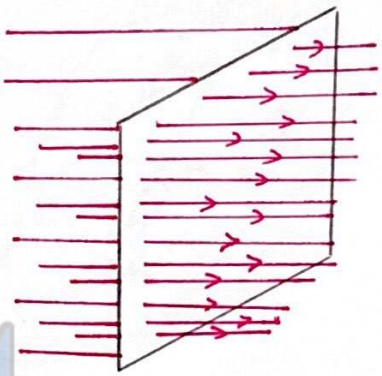
$= \frac{0 - 2837252.2}{-5.75 \times 10^{13}}$

$= 4.9 \times 10^{-8}$ sec.

Chapter (24) Gauss's Law:-

24.1 Electric Flux:-

Electric Flux :- is the number of electric field lines that Pass a plane of area (A) Perpendicular to the field.



$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

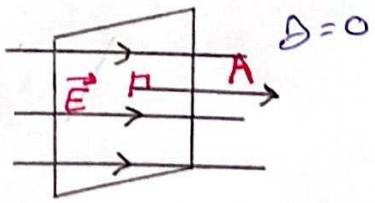
$$\Phi_E = [N/C] \cdot [m^2] = N \cdot m^2/C$$

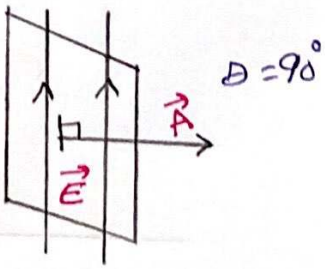
$\vec{A}$:- area vector that is has magnitude of area and direction perpendicular to the surface.

θ :- is the angle between $\vec{E}$ and $\vec{A}$.

$$\Phi_{max} = EA\cos(0) = EA$$

$$\Phi_{min} = EA\cos(90) = Zero$$

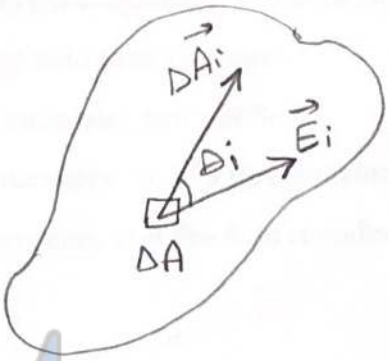




For nonuniform surface, we divide the surface into small equal areas each one = ΔA

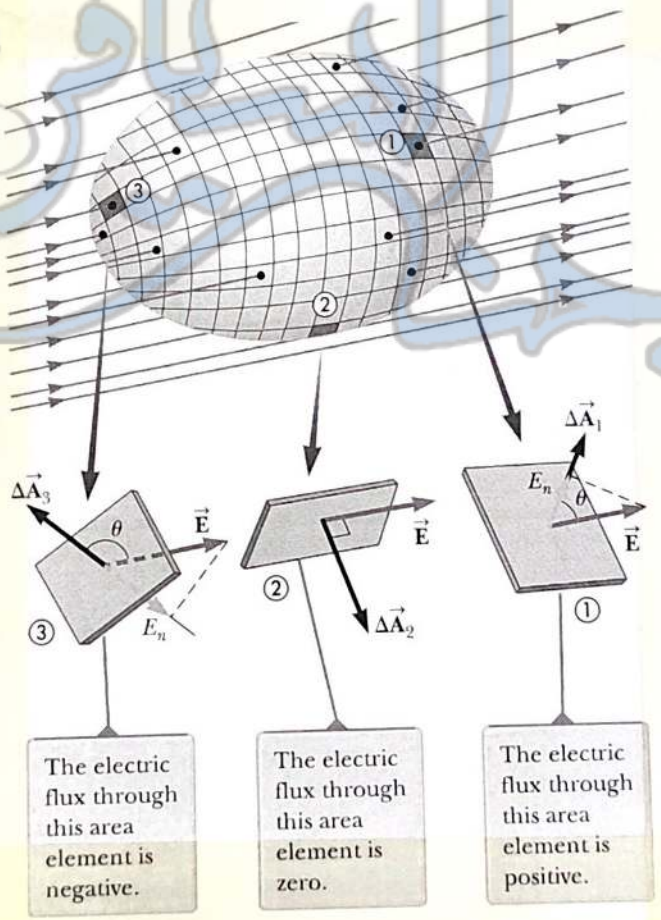
$$\Delta \Phi_E = E_i \, \Delta A_i \cos\theta_i$$

$$\Phi_E = \lim_{\Delta A_i \to 0} \sum \vec{E}_i \cdot \Delta \vec{A}_i$$



$$= \int_{\text{surface}} \vec{E} \cdot d\vec{A}$$

$$= \oint \vec{E} \cdot d\vec{A}$$, for closed surface.



Quick Quiz 24.1 Suppose a point charge is located at the center of a spherical surface. The electric field at the surface of the sphere and the total flux through the sphere are determined. Now the radius of the sphere is halved. What happens to the flux through the sphere and the magnitude of the electric field at the surface of the sphere? (a) The flux and field both increase. (b) The flux and field both decrease. (c) The flux increases, and the field decreases. (d) The flux decreases, and the field increases. (e) The flux remains the same, and the field increases. (f) The flux decreases, and the field remains the same.

the answer is (e), the flux will remain the same as the same number of electric field lines will pass through the surface. but $\vec{E}$ will decrease; $E = \frac{Kq}{r^2}$, when the radius of the sphere is halved E will be double twice.

$$E' = \frac{Kq}{(\frac{1}{2}r)^2} = 4\,\frac{Kq}{r^2} = 4E$$

$$\Phi = EA = \frac{Kq}{r^2} \cdot 4\pi r^2$$

$$\Phi' = E'A' = \cancel{4}\,\frac{Kq}{r^2} \cdot 4\pi\,\frac{r^2}{\cancel{4}} = \frac{Kq}{r^2} \cdot 4\pi r^2$$

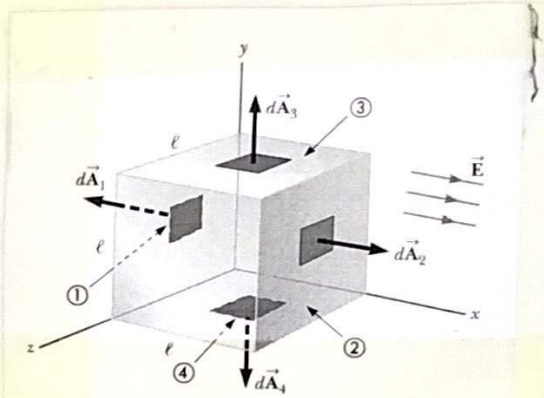
$$\therefore \quad \Phi = \Phi'$$

Example (24.1) :- Flux through a cube:-

$$\Phi_E = \int_1 \vec{E}\cdot d\vec{A} + \int_2 \vec{E}\cdot d\vec{A}$$

$$+ \int_3 \vec{E}\cdot d\vec{A} + \int_4 \vec{E}\cdot d\vec{A}$$

$$+ \int_5 \vec{E}\cdot d\vec{A} + \int_6 \vec{E}\cdot d\vec{A}$$



$$\Phi_E = \int_1 E\cos(180)\,dA + \int_2 E\cos(0)\,dA + \int_3 E\cos(90)\,dA$$

$$+ \int_4 E\cos(90)\,dA + \int_5 E\cos(90)\,dA + \int_6 E\cos(90)\,dA$$

$$= -EA + EA + 0 + 0 + 0 + 0$$

$$= 0$$

24.2 Gauss's Law:-

the net charge flux through any closed surface (gaussian surface) is equal to the net charge inside the surface divided by ϵ_0.

$$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

Consider +ve point charge in the center of sphere of radius r.

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{r}$$

$$\Phi = \oint E \, dA \cos(0) = EA$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \cdot 4\pi r^2$$

$$\Phi = \frac{Q}{\epsilon_0}$$

Notes:-

1- the net flux through any closed surface surrounding a point charge Q is $\frac{Q}{\epsilon_0}$.

2- Φ is independent of the shape of the surface.

3- Φ is independent of the radius (r).

4- Φ through closed surface without charge inside it equal zero.

Conceptual Example 24.2: Flux Due to a Point Charge

A spherical gaussian surface surrounds a point charge *q*. Describe what happens to the total flux through the surface if (A) the charge is tripled, (B) the radius of the sphere is doubled, (C) the surface is changed to a cube, and (D) the charge is moved to another location inside the surface.

A) $\Phi = \frac{Q_{in}}{\epsilon_0}$ $\qquad Q'_{in} = 3Q_{in}$

$\Phi' = \frac{Q'_{in}}{\epsilon_0} = 3\frac{Q_{in}}{\epsilon_0} = 3\Phi$

B) Φ is constant; beause it is not depend on(r).

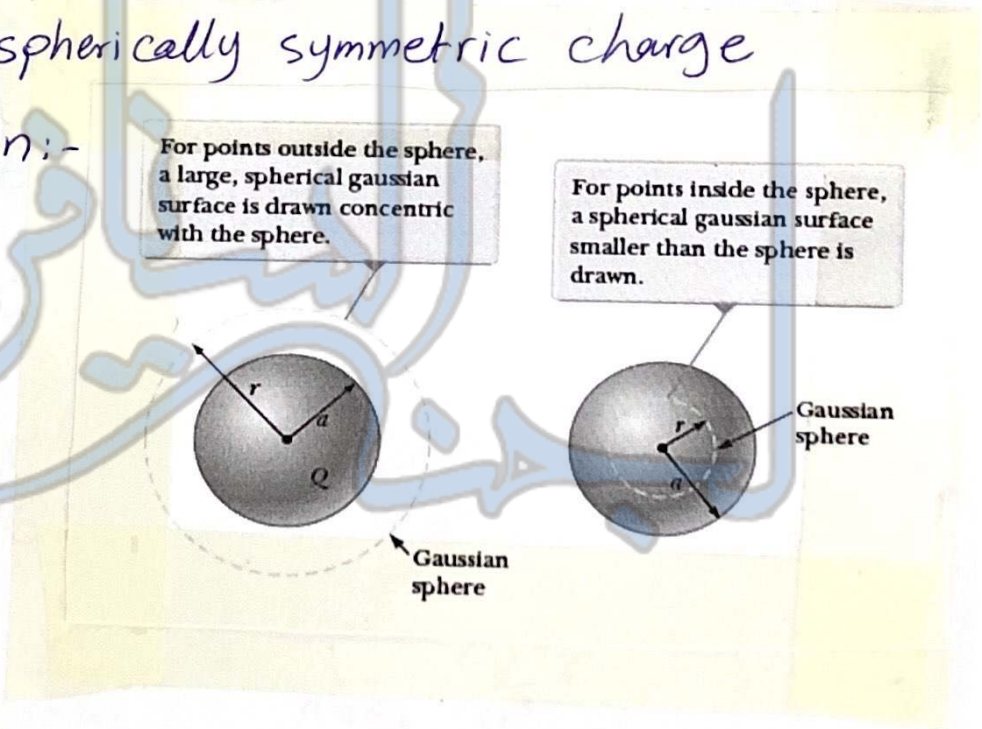
C) Φ is constant because it's independ on the shape.

D) Φ is constant because gauss'sLaw refer to the total charge Q inside the surface.

24.3 Application of Gauss's Law to various charge Distributions:-

Gauss's Law is useful for determining E, when the charge distribution is symmetric such as sphere, cylinder and plane sheet.

Ex(24.3):- a spherically symmetric charge distribution:-



A) E outside the sphere

$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

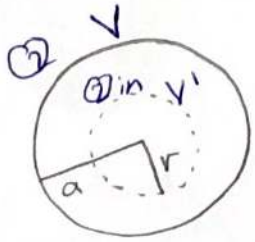
$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$E = K \frac{Q}{r^2}$$ the same for point charge.

(for $r > a$).

B) $\vec{E}$ inside the sphere.

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$



$$\rho = \frac{Q_{in}}{V'} = \frac{Q}{V}$$

$Q_{in} = \rho V'$, where V' is the volume of gauss's Law.

$$Q_{in} = \rho V' = \rho \frac{4}{3}\pi r^3$$

$$\oint \vec{E} \cdot d\vec{A} = E\cos(0) \oint dA = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$E = \frac{\rho r}{3\epsilon_0} \text{ , but } \rho = \frac{Q}{\frac{4}{3}\pi a^3}$$

$$E = \frac{Q}{\frac{4}{3}\pi a^3} \cdot \frac{r}{3\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{Q}{a^3} r$$

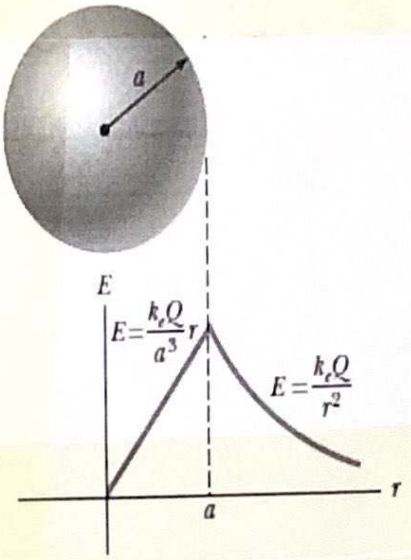
$$E = \frac{K Q r}{a^3} \quad (\text{for } r < a)$$

at $r = 0 \Rightarrow E = 0$

What if $r = a$??

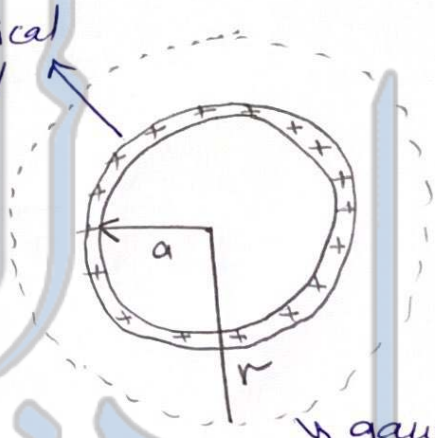
$$E_{outside} = \lim_{r \to a} \left(\frac{kQ}{r^2}\right) = \frac{KQ}{a^2}$$

$$E_{inside} = \lim_{r \to a} \frac{KQr}{a^3} = \frac{KQ}{a^2}$$



Example :- for spherical shell find $\vec{E}$ inside and outside the shell?

a) outside the shell



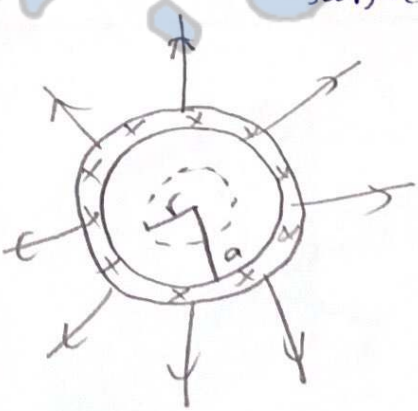
$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$E = \frac{KQ}{r^2}, \text{ for, } r > a$$

b- inside the shell

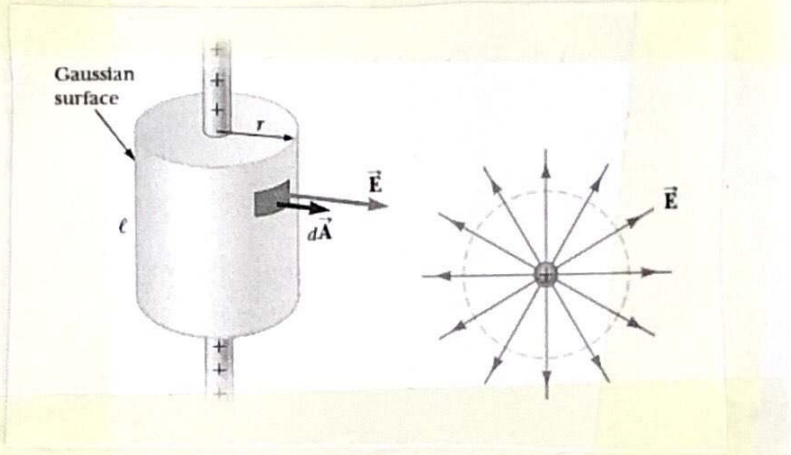


$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = 0$$

$E = 0$, inside the spherical shell where Q = zero.

Ex(24.4):- A cylindrically symmetric charge distribution:-



flux through upper and lower surfaces = zero;

$\theta = 90°$.

$$\Phi = \oint \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$E\cos(0) \oint dA = \frac{Q_{in}}{\epsilon_0}$$

$$E(2\pi r L) = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \cdot \frac{2}{2} = \frac{2\lambda}{4\pi\epsilon_0 r}$$

$$E = \frac{2K\lambda}{r} \quad , \quad E \propto \frac{1}{r}$$

Ex(24.5): A Plane of charge:-

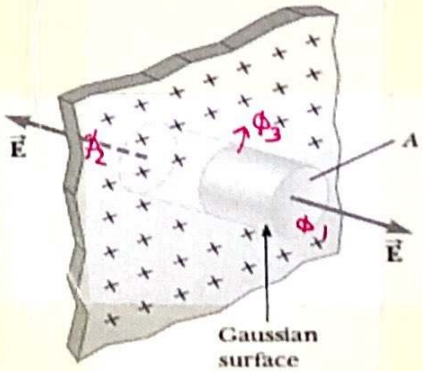
$$\Phi = \Phi_1 + \Phi_2 + \Phi_3$$

$$= EA + EA + 0$$

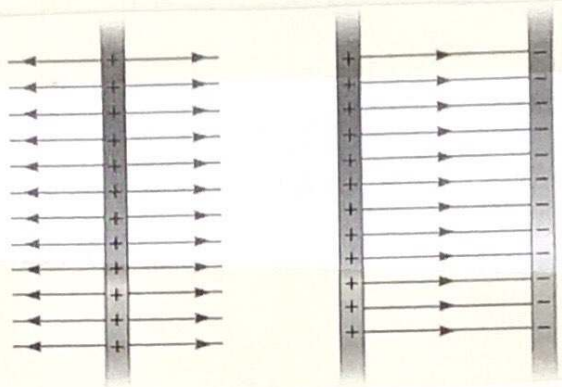
$$= 2EA$$

$$\Phi = \frac{Q_{in}}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} = 2EA$$

$$E = \frac{\sigma}{2\epsilon_0} \text{ , uniform } \vec{E}$$



What if we have two parallel plate with +ve and -ve charge



$\vec{E}$ between the Plates is uniform and zero elsewhare.

$$EA = \frac{Q_{in}}{\epsilon_0}$$

$$EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{\epsilon_0} \equiv \text{uniform Electric field.}$$

24.4 Conductors in electrostatic equalibrium :-

when there is no motion of charge with in the conductor, the conductor is in electrostatic equalibrium.

Properties of a conductor of electrostatic equalibrium :-

1- if an isolated conductor carries a charge the charge resides on the surface.

2- E is zero every where inside the conductor

wherer the conductor is solid or hollow.

3- the electric field just outside a charged conductor is perpendicular to the surface of conductor and has a magnitude of $\frac{\sigma}{\epsilon_0}$.

4- On an irregularity shaped conductor, the surface charge density is greatest at locations where the radius of curvature of the surface is smallest.

$\sigma = \frac{Q}{A}$, $A \propto r^2$

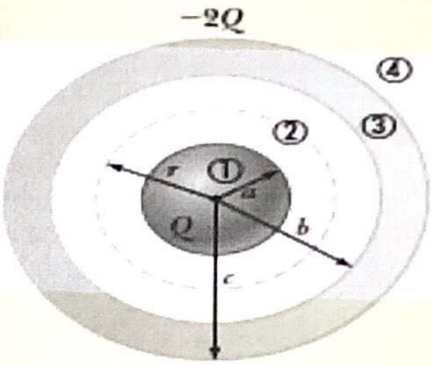
Smaller (r) $\Rightarrow$ Smaller (A) $\Rightarrow$ larger (σ)

Quick Quiz 24.3 Your younger brother likes to rub his feet on the carpet and then touch you to give you a shock. While you are trying to escape the shock treatment, you discover a hollow metal cylinder in your basement, large enough to climb inside. In which of the following cases will you not be shocked? (a) You climb inside the cylinder, making contact with the inner surface, and your charged brother touches the outer metal surface. (b) Your charged brother is inside touching the inner metal surface and you are outside, touching the outer metal surface. (c) Both of you are outside the cylinder, touching its outer metal surface but not touching each other directly

answer is (a).

Ex (24-7):- A sphere inside a spherical shell:-

in inner sphere is insulating sphere with radius ⓐ. with charge Q the spherical shell of 2-radii ⓐ and ⓑ and −2Q charge.



$$E_1 = \frac{KQ}{a^3} r \quad (\text{for } r < a)$$

$$E_2 = \frac{KQ}{r^2} \quad (\text{for } a < r < b)$$

$$E_3 = 0 \quad ; \quad Q_{in} = 0 \quad (\text{for } b < r < c)$$

$$E_4 = -\frac{KQ}{r^2} \quad (\text{for } r > c)$$

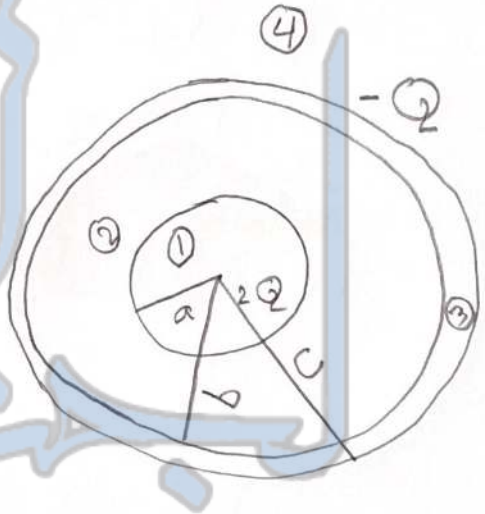
$$Q_{in} = -2Q + Q$$

$$= -Q$$

what if the inner sphere is conducting sphere ??

$E_1 = 0$; $Q_{in} = 0$.

Example:- conducting sphere of radius (a) and charge (2Q) surrounding by conducting spherical shell of inner and outer radii (b) and (c) and charge $-Q$ ↑ find E in the regions 1, 2, 3 and 4 ??



1- $E_1 = 0$; $Q_{in} = 0$

for $r < a$

2- $E_2 A = \frac{Q_{in}}{\epsilon_0}$

$E_2 (4\pi r^2) = \frac{2Q}{\epsilon_0}$

$E_2 = \frac{2KQ}{r^2}$ for $(a < r < b)$

3- $E_3 = 0$, $Q_{in} = 0$

4- $E_4 = \frac{KQ_{in}}{r^2} = \frac{K(2Q - Q)}{r^2} = \frac{KQ}{r^2}$

Solved problems chapter (24)

③ diameter = 40 cm $\Phi = 5.2 \times 10^5$ N·m²/C

$\Phi = E A \cos\Phi$

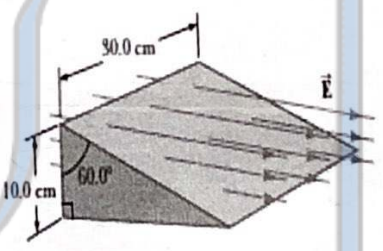
$5.2 \times 10^5 = E (\pi r^2) \cos(0)$

$r = \frac{1}{2} d = \frac{1}{2} \times 40 = 20$ cm

$E = \frac{5.2 \times 10^5}{\pi (0.2)^2} = 4.14 \times 10^6$ N/C

---

④ $E = 7.8 \times 10^4$ N/C



a) $\Phi_{vertical} = EA \cos 180$

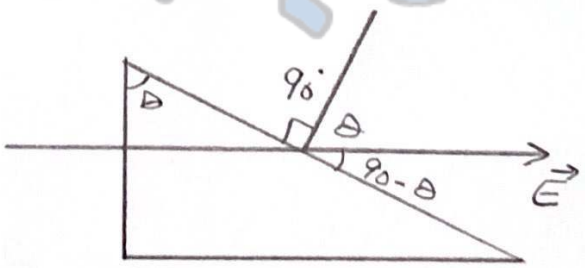
$\Phi_{ver} = (7.8 \times 10^4)(30 \times 10 \times 10^{-4})(-1)$

$\Phi_{ver} = -2.34 \times 10^3$ N·m²/C

b) $\Phi = EA \cos\theta$

$\Phi = 7.8 \times 10^4 \times 30 \times 10 \times 10^{-4} \times \frac{\cos 60}{\cos 60}$

$= 2.34 \times 10^3$ N·m²/C



c) Φ_{total} = Zero

Q_{in} = Zero

Q9) $Q_1 = 5\,\mu C$ $Q_2 = -9\,\mu C$ $Q_3 = 27\,\mu C$

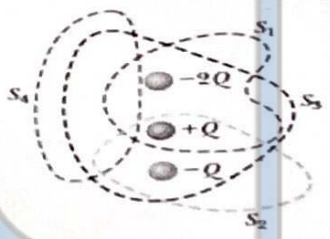
$Q_4 = -84\,\mu C$

a- $\Phi = \dfrac{Q_{in}}{\epsilon_0} = \dfrac{(5 - 9 + 27 - 84) \times 10^{-6}}{8.85 \times 10^{-12}}$

$= -6.9 \times 10^{6}$ $N \cdot m^2/C$

b- number of electric field lines entering the surface is larger than the number of electric field lines that leaving it.

Q11 :-



$\Phi_{S_1} = \dfrac{-2Q + Q}{\epsilon_0}$

$= \dfrac{-Q}{\epsilon_0}$

$\Phi_{S_2} = \dfrac{-Q + Q}{\epsilon_0} = zero$

$\Phi_{S_3} = \dfrac{-2Q + Q - Q}{\epsilon_0} = \dfrac{-2Q}{\epsilon_0}$

$\Phi_{S_4} = zero$

Q14:- $q = 12\ \mu C$ $\quad r = 22\ cm$

a) $\Phi = \frac{q}{\epsilon_0} = \frac{12 \times 10^{-6}}{8.85 \times 10^{-12}} = 1.36 \times 10^{6}\ N \cdot m^2/C$

b) $\Phi_{\text{half shell}} = \frac{1}{2}\Phi$

$= 0.68 \times 10^{6}\ N \cdot m^2/C$

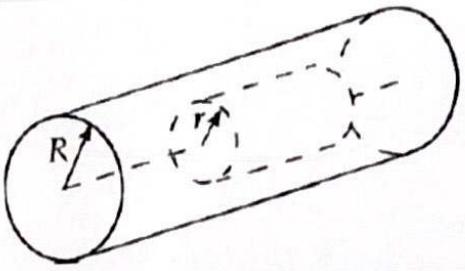
c) NO, Same number of electric field lines will pass through each surface regardless of the radius.

Q26:- 82 Proton $\quad r = 1.2 \times 10^{-15}\ m$

$E = \frac{KQ}{r^2} = \frac{9 \times 10^{9} \times 82 \times 1.6 \times 10^{-19}}{(1.2 \times 10^{-15})^2}$

$= 2.33 \times 10^{21}\ N/C$ outward

Q33:- $r < R$



$$\int \vec{E} \cdot d\vec{A} = \frac{Q_{in}}{\epsilon_0}$$

$$E(2\pi r L) = \frac{\rho \pi r^2 L}{\epsilon_0}$$

$$E = \frac{\rho r}{2\epsilon_0}$$

Q27:- $\sigma = 9 \mu C/m^2$, large flat sheet

for infinite sheet $E = \frac{\sigma}{2\epsilon_0}$

$$E = \frac{9 \times 10^{-6}}{2 \times 8.85 \times 10^{-12}} = 5.08 \times 10^5 \ N/C \ \hat{j}$$

Q24:- $\lambda = -90 \mu C/m$

a- $r = 10$ cm

$$E = \frac{2K\lambda}{r} = \frac{2 \times 9 \times 10^9 \times 90 \times 10^{-6}}{0.1} = 16.2 \times 10^6 \ N/C$$

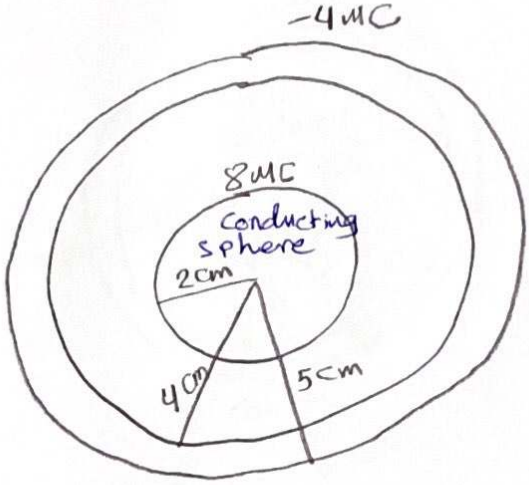
b- $r = 20$ cm

$$E = \frac{2 \times 9 \times 10^9 \times 90 \times 10^{-6}}{0.2} = 8.09 \times 10^6 \ N/C$$

c- $r = 100$ cm

$$E = \frac{2 \times 9 \times 10^9 \times 90 \times 10^{-6}}{1} = 1.62 \times 10^6 \ N/C$$

Q47:-



a- $r = 1\,cm$

$E = 0 \;;\; Q_{in} = 0$

b- $r = 3\,cm$

$$EA = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{Q_{in}}{\epsilon_0}$$

$$E = \frac{K Q_{in}}{r^2}$$

$$E = \frac{9\times10^{9} \times 8\times10^{-6}}{(3\times10^{-2})^2} = 8\times10^{7}\ N/C$$

c- $r = 4.5\,cm$

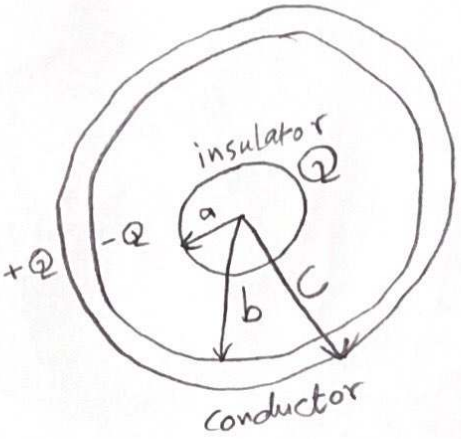
$E = 0 \;;\; Q_{in} = 0$

d- $r = 7\,cm$

$$E = \frac{K Q_{in}}{r^2} = \frac{9\times10^{9} \times 4\times10^{-6}}{49\times10^{-4}} = 0.74\times10^{7}\ N/C$$

$Q_{in} = 8 - 4 = 4\,\mu C$

Q54:-



a- [1] $(r < a)$

$$\int E \cdot dA = \frac{Q_{in}}{\epsilon_0}$$

$$E(4\pi r^2) = \frac{\rho \frac{4}{3}\pi r^3}{\epsilon_0}$$

$$E = \frac{\rho r}{3\epsilon_0}$$

[2] $a < r < b$

$$E = \frac{KQ}{r^2}$$

[3] $b < r < c$

$$E = 0$$

[4] $r > c$ $\quad E = \frac{KQ}{r^2}$

b- $Q_{inner} + Q = 0$

$Q_{inner} = -Q$ by induction

$$\sigma_{inner} = \frac{-Q}{4\pi b^2}$$

$Q_{inner} + Q_{outer} = 0 \Rightarrow Q_{outer} = +Q$

$$\sigma_{outer} = \frac{Q}{4\pi c^2}$$

Chapter (25):- Electric potential

25.1 Electric potential and potential difference:-

for q_o (test charge) placed in $\vec{E} \Rightarrow \vec{F} = q_o\vec{E}$

dw is the work done by this force

$$dw = \vec{F} \cdot d\vec{s} = q_o\vec{E} \cdot d\vec{s}$$

where $d\vec{s}$ is infinitismal displacement.

the work done by a conservative force = -ve of the change in potential energy (dU)

$$dU = -q_o\vec{E} \cdot d\vec{s}$$

the change in potential energy for finite displacement of the charge from point A to B

$$\Delta U = -q_o \int_A^B \vec{E} \cdot d\vec{s}$$, this integral called line or path integral, because $q\vec{E}$ is conservative force, this integral doesint

depend on the path taken from A to B.

Electric potential (V):- the potential energy Per unit charge.

$$V = \frac{U}{q_0}$$

Electric potential difference (ΔV)

$$\Delta V = V_B - V_A$$

$$= \frac{\Delta U}{q_0} = -\int_A^B \vec{E} \cdot d\vec{s}$$

Electric potential equal zero in infinity

$$V_\infty = 0$$

if an external agent moves a test charge from A to B without changing K.E for the test charge ⇒ the agent perform a work $W = \Delta U = q \Delta V$

$\Delta V \equiv [\text{volt}]$

1 volt = 1 J/C

Volt :- 1 J work must be done to move 1 C charge through Potential difference = 1 volt

$E \equiv [N/C] \equiv [V/m]$

We can introduce $\vec{E}$ as a measure of the rate of change with position.

Potential

a unit of energy used in atomic and nuclear physics is electron volt (e.v).

electron volt :- is the energy gain or lose when a charge of magnitude e^- is moved through potential difference (1 volt).

$1\ e.v = 1.6 \times 10^{-19}\ C \times 1\ \text{volt}$

$= 1.6 \times 10^{-19}\ C.\text{volt} = 1.6 \times 10^{-19}\ J$

Example:- Calculate the work needed to move a +ve charge from A to B, where

$V_A = 40$ volt , $V_B = 60$ volt , $q = 4\mu C$

$$\Delta V = V_B - V_A = \frac{W}{q}$$

$$W = q\,(V_B - V_A)$$

$$= 4 \times 10^{-6}\,(60 - 40)$$

$$= 80 \times 10^{-6}\ J$$

$$= \frac{80 \times 10^{-6}}{1.6 \times 10^{-19}} = 5 \times 10^{14}\ e.V$$

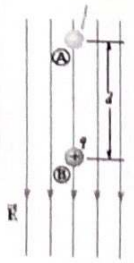
25.2 Potential difference in a uniform electric field:-

$$\Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{s}$$

$$= -\int_A^B E \cos(0)\, ds$$

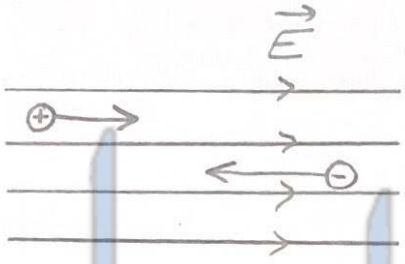
$$= -\int_A^B E\, ds = -E \int_A^B ds$$

$$= -Ed$$



-ve Potential difference $\Rightarrow V_A > V_B$

Electric field lines always point in the direction of decreasing electric potential.



$$\Delta U = q\Delta V = -qEd$$

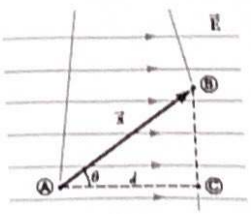
for +ve charge ΔU decreases

for -ve charge ΔU increases.

in general acharged particle moves between A and B in uniform $\vec{E}$ such that $\vec{s}$ is not parallel to the field lines

$$\Delta V = -\int_A^B \vec{E}\cdot d\vec{s}$$

$$= -\vec{E}\cdot\vec{s}$$



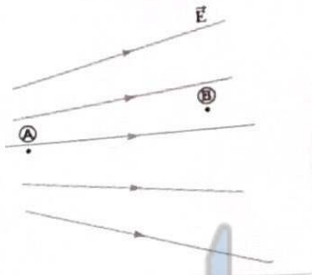
$$\Delta U = q_o\Delta V = -q_o\vec{E}\cdot\vec{s}$$

* All points in a plane ⊥ to $\vec{E}$ are at the same electric potential (equipotential surfaces).

Quick Quiz 25.1 In Figure 25.1, two points A and B are located within a region in which there is an electric field. **(i)** How would you describe the potential difference $\Delta V = V_B - V_A$? (a) It is positive. (b) It is negative. (c) It is zero. **(ii)** A negative charge is placed at A and then moved to B. How would you describe the change in potential energy of the charge–field system for this process? Choose from the same possibilities.

i-(b) it's negative.

ii-(a) it's positive.

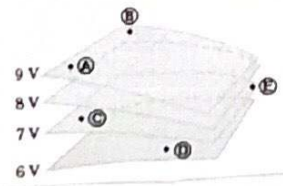


$$\Delta U = -q\vec{E}\cdot\vec{S}$$

$$= -(-q)\vec{E}\cdot\vec{S}$$

$$= q\vec{E}\cdot\vec{S} \quad +ve.$$

Quick Quiz 25.2 The labeled points in Figure 25.4 are on a series of equipotential surfaces associated with an electric field. Rank (from greatest to least) the work done by the electric field on a positively charged particle that moves from A to B, from B to C, from C to D, and from D to E.



from A to B

$W_{AB} = q\Delta V = q(9-9) =$ zero

$W_{BC} = q(7-9) = -2q$

$W_{CD} = q(6-7) = -q$

$W_{DE} = q(7-6) = q$

$W_{BC} > W_{CD} > W_{AB} > W_{DE}$

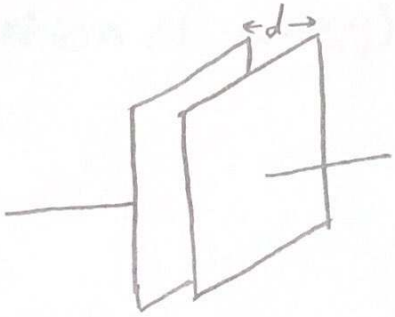
$-2q > -q > \text{zero} > q$

-6-

Ex(25.1):- $\vec{E}$ between two parallel plate:-

$\Delta V = 12$ volt

$d = 0.3$ cm



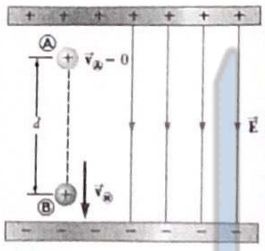
$$E = \frac{\Delta V}{d} = \frac{12}{0.3 \times 10^{-2}}$$

$$= 4 \times 10^{3} \text{ V/m}$$

Ex(25.2):- motion of a proton in a uniform $\vec{E}$:-

$E = 8 \times 10^{4}$ V/m

$d = 0.5$ m



$$\Delta V = -Ed$$

$$= -8 \times 10^{4} \times 0.5$$

$$\Delta V = -4 \times 10^{4} \text{ volt} \Rightarrow V_B < V_A$$

$$\Delta U = q_o \Delta V$$

$$= 1.6 \times 10^{-19} \times -4 \times 10^{4}$$

$$= -6.4 \times 10^{-15} \text{ J}$$

decreasing of potential energy corresponding to increasing of Kinetic energy of the charge.

For isolated system

$\Delta K + \Delta U = 0$ (conservation of energy).

$\left(\frac{1}{2}mv^2 - 0\right) + e\Delta V = 0$

$\frac{1}{2}mv^2 = -e\Delta V$

$v^2 = \frac{-2e\Delta V}{m} \Rightarrow v = \sqrt{\frac{-2e\Delta V}{m}}$

$v = \sqrt{\frac{-2e(-Ed)}{m}} = \sqrt{\frac{2eEd}{m}}$

$v = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 8 \times 10^{4} \times 0.5}{1.67 \times 10^{-27}}}$

$= 2.8 \times 10^{6}$ m/s

25.3 Electric Potential and potential energy due to Point charges:-

$$V_B - V_A = -\int_A^B E \cdot ds$$

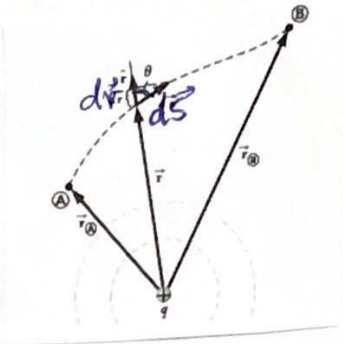
$$\vec{E} \cdot d\vec{s} = \frac{Kq}{r^2} dr$$

$$V_B - V_A = -Kq \int_{r_A}^{r_B} \frac{dr}{r^2}$$

$$= -Kq \left(-\frac{1}{r}\right)_{r_A}^{r_B}$$

$$\Delta V = Kq \left[\frac{1}{r_B} - \frac{1}{r_A}\right]$$

$$\boxed{V = \frac{Kq}{r}}$$ Potential due to point charge.



$dr = ds \cos\theta$

for a group of point charges:-

$$V = K \sum_i \frac{q_i}{r_i}$$

the work needed to bring q_2 from ∞ to P

$$W = q_2 V_P$$

$$= q_2 \left(\frac{Kq_1}{r}\right)$$

$$W = \frac{K q_1 q_2}{r} = U$$

V_P

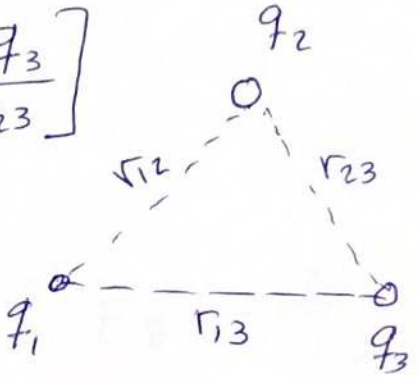
P $\quad \vec{r} \quad$ q_1

for charges of same sign, U is +ve

for charges of opposite sign, U is -ve.

if the system consist of more than 2 charges

$$U = K\left[\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right]$$



in general

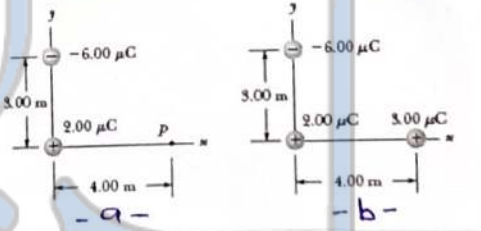
$$U = K \sum_i \sum_j \frac{q_i q_j}{r_{ij}}$$

Ex(25.3):- electric potential due to 2-point charges:-

a- $V_p = K\left(\frac{q_1}{r_1} + \frac{q_2}{r_2}\right)$

$$V_p = 9\times10^9\left(\frac{2\times10^{-6}}{4} - \frac{6\times10^{-6}}{5}\right)$$

$$= -6.3\times10^3 \text{ volt}$$



b- $\Delta U = U_f - U_i = U_f - \cancel{U_\infty}^{0} = U_f$

$$U_f = q_3 V_p = 3\times10^{-6} \times -6.3\times10^3$$

$$= -1.9\times10^{-2} \text{ J}$$

$W = -U = +1.9\times10^{-2}$ J to take q_3 back from P to ∞.

25.4 obtaining the value of $\vec{E}$ from the value of V

$$\Delta V = -\int \vec{E} \cdot d\vec{s}$$

$$dV = -\vec{E} \cdot d\vec{s}$$

$$d\vec{s} = d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$$

$$\vec{E} = -\frac{dV}{d\vec{s}}$$

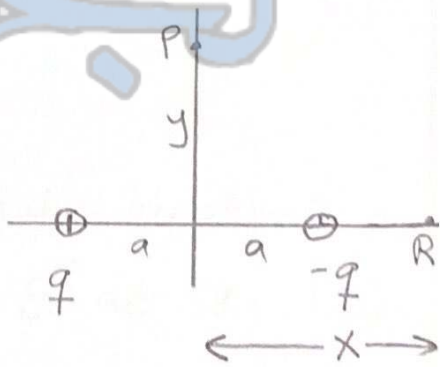
$$E_x = -\frac{dV}{dx} \quad , E_y = -\frac{dV}{dy} \quad , E_z = -\frac{dV}{dz}$$

$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$$= -\left[\frac{dV}{dx}\hat{i} + \frac{dV}{dy}\hat{j} + \frac{dV}{dz}\hat{k}\right]$$

Ex(25.4):- Electric potential due to a dipole:-

a- $V_p = K\sum_i \frac{q_i}{r_i}$



$$V_p = \left[\frac{Kq}{\sqrt{a^2+y^2}} - \frac{Kq}{\sqrt{a^2+y^2}}\right] = 0$$

b- $V_R = K\left[\frac{-q}{x-a} + \frac{q}{x+a}\right]$

$$= Kq\left[\frac{1}{x+a} - \frac{1}{x-a}\right]$$

$$V_R = Kq\left[\frac{x-a-x-a}{x^2-xa+xa-a^2}\right]$$

$$V_R = \frac{-2Kqa}{x^2-a^2}$$

C- for $x \gg a$

$$V = \lim_{x\gg a}\left(\frac{-2Kqa}{x^2-a^2}\right) \approx \frac{-2Kqa}{x^2}$$

$$E_x = -\frac{dv}{dx} = -\frac{d}{dx}\left(\frac{-2Kqa}{x^2}\right)$$

$$= 2Kqa\,\frac{d}{dx}\left(\frac{1}{x^2}\right)$$

$$= \frac{-4Kqa}{x^3}$$

Example:- let $V = a_0(5x^2y + 3xyz - 2yz^2)$

a_0 is constant $= 4\times10^{-2}$, find E at $(2,-1,1)$

$$E_x = -\frac{dv}{dx} = -a_0(10xy + 3yz) = -0.92\ V/m$$

$$E_y = -\frac{dv}{dy} = -a_0(5x^2 + 3xz - 2z^2) = -0.96\ V/m$$

$$E_z = -\frac{\partial V}{\partial z} = -a_0(3xy - 4yz) = 0.08 \text{ V/m}$$

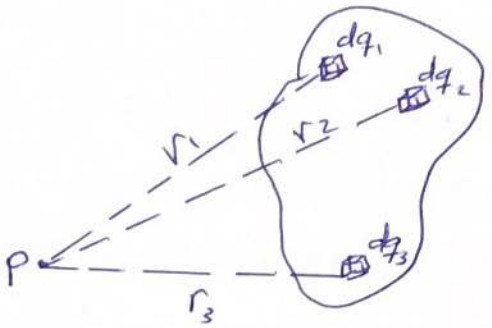
$$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$$

$$= -0.92\,\hat{i} - 0.96\,\hat{j} + 0.08\,\hat{k} \quad (\text{V/m})$$

25.5 Electric potential due to continous charge distribution:-

Electric potential due to charge element dq is

$$dv = \frac{K\,dq}{r} \Rightarrow$$



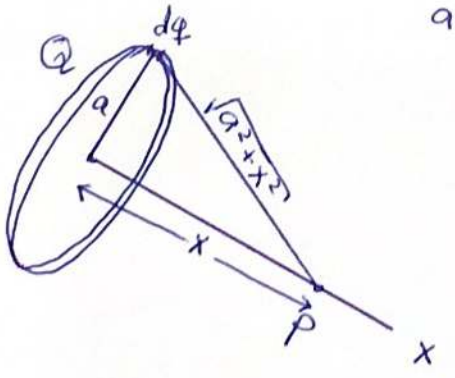
$$V_p = \int \frac{K\,dq}{r}$$

remember that $\lambda = \frac{Q}{L} \equiv C/m \Rightarrow dq = \lambda\,dL$

$$\sigma = \frac{Q}{A} \equiv C/m^2 \Rightarrow dq = \sigma\,dA$$

$$\rho = \frac{Q}{V} \equiv C/m^3 \Rightarrow dq = \rho\,dv$$

Ex(25.5):- Electric Potential due to a uniformly charged ring:-



a- $V_p = K\int \frac{dq}{r} = K\int_0^Q \frac{dq}{\sqrt{a^2+x^2}}$

$$= \frac{Kq}{\sqrt{a^2+x^2}}\Big]_0^Q$$

$$= \frac{KQ}{\sqrt{a^2+x^2}}$$

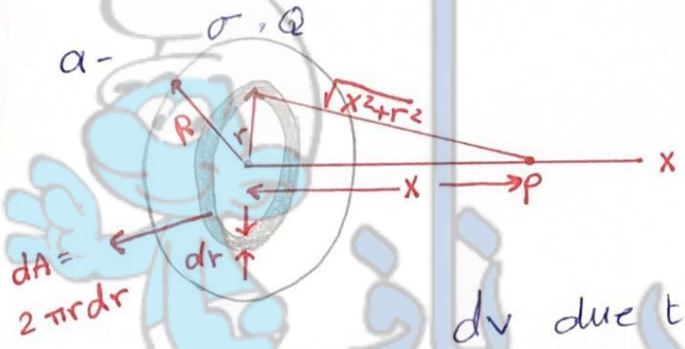
b- $E_x = -\frac{dV}{dx} = -KQ\frac{d}{dx}(a^2+x^2)^{-1/2}$

$= -KQ\left(-\frac{1}{2}\right)(a^2+x^2)^{-3/2}(2x)$

$= \frac{K \times Q}{(a^2+x^2)^{3/2}}$

Ex(25.6):- Electric potential due to a uniformly charged disk.

a-



$dq = \sigma\, dA$

$= \sigma(2\pi r\, dr)$

$= 2\pi\sigma r\, dr$

dV due to ring from previous example

$$dV = \frac{K\,dq}{\sqrt{r^2+x^2}} = \frac{K\,2\pi\sigma r\,dr}{\sqrt{r^2+x^2}}$$

$$V_P = K\pi\sigma \int_0^R \frac{2r\,dr}{\sqrt{r^2+x^2}}$$

let $U = r^2 + x^2 \Rightarrow dU = 2r\,dr$

$$\int \frac{2r\,dr}{\sqrt{r^2+x^2}} = \int \frac{dU}{\sqrt{U}} = 2\sqrt{U} = 2\sqrt{r^2+x^2}\Big]_0^R$$

$$= 2\left[\sqrt{R^2+x^2} - x\right]$$

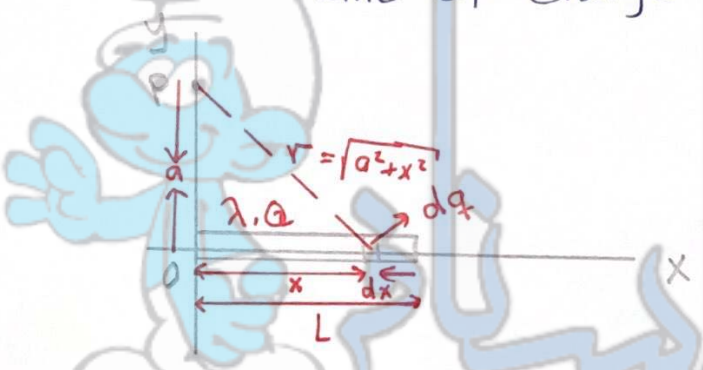
$$V_P = 2\pi\sigma K\left[\sqrt{R^2+x^2} - x\right]$$

b- $E_x = -\frac{dV}{dx} = -2\pi K\sigma \frac{d}{dx}\left[\sqrt{R^2+x^2} - x\right]$

$= -2\pi K\sigma \left[\frac{1}{2} \cdot \frac{2x}{\sqrt{R^2+x^2}} - 1\right]$

$= 2\pi K\sigma \left[1 - \frac{x}{\sqrt{R^2+x^2}}\right]$

Ex (25.7):- Electric Potential due to a finite Line of charge:-



$dV = \frac{K\,dq}{r} = \frac{K\lambda dx}{\sqrt{a^2+x^2}}$

$V_p = K\lambda \int_0^L \frac{dx}{\sqrt{a^2+x^2}}$

$\int \frac{dx}{\sqrt{a^2 \pm x^2}} = \ln\left|x \pm \sqrt{a^2 \pm x^2}\right|$

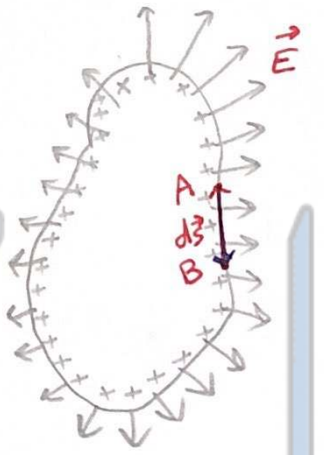
$V_p = K\lambda \ln\left|x + \sqrt{a^2+x^2}\right|\Big]_0^L$

$= K\lambda \left[\ln\left(L + \sqrt{a^2+L^2}\right) - \ln a\right]$

$= K\lambda \left[\ln\left(\frac{L + \sqrt{a^2+L^2}}{a}\right)\right]$

25.6 Electric Potential due to a charged conductor:-

we Know that for solid charged conductor in equalibrium charge resides on the outer surface of conductor, so Electric field inside it = zero and $\vec{E}$ just outside the conductor is ⊥ to the surface.



$\vec{E}$ always ⊥ $d\vec{s}$

$\therefore \vec{E} \cdot d\vec{s} = 0$

$$\Delta V = V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{s}$$

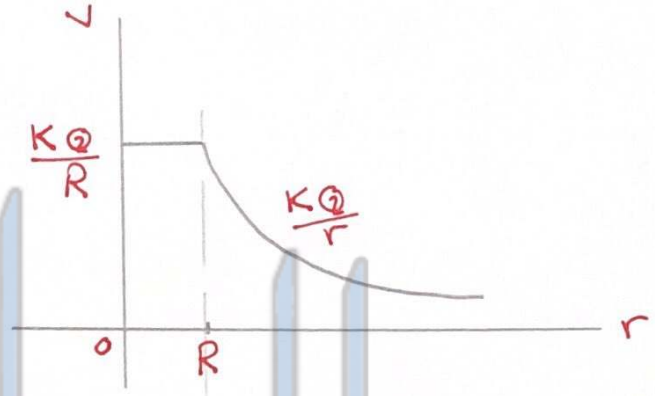
$$= 0$$

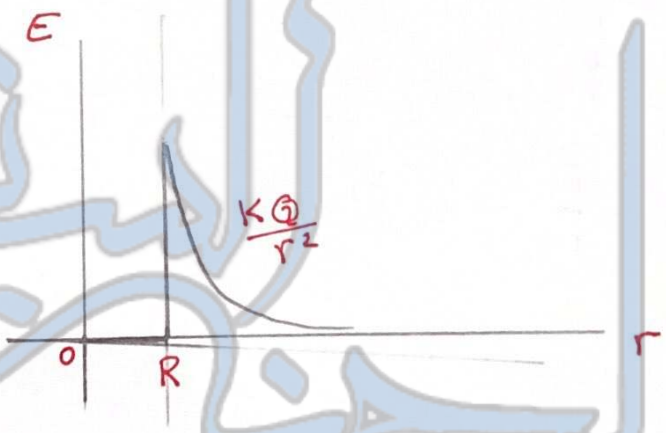
$\therefore V_B = V_A \Rightarrow$ ① V is constant every where on the surface of the charged conductor.

② the surface of any charged conductor in electro-static equalibrium is an equipotential surface.

③ because $\vec{E}$ = zero inside conductor; the electric Potential is constant inside it and equal it's value on the surface.

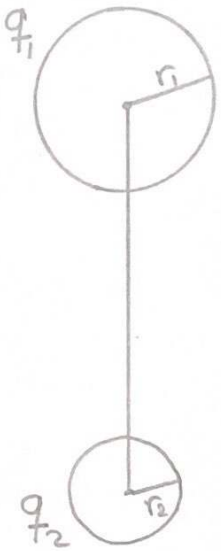
R
charged
conductor


V
KQ/R
KQ/r
0
R
r


E
KQ/r²
0
R
r

EX(25.8):- Two connected charged spheres:-

for connected spheres



$$V_1 = V_2$$

$$\frac{K q_1}{r_1} = \frac{K q_2}{r_2}$$

$$\frac{q_1}{q_2} = \frac{r_1}{r_2}$$

$$E_1 = \frac{K q_1}{r_1^2} \quad , \quad E_2 = \frac{K q_2}{r_2^2}$$

$$\frac{E_1}{E_2} = \frac{\frac{K q_1}{r_1^2}}{\frac{K q_2}{r_2^2}}$$

$$= \frac{\frac{q_1}{r_1^2}}{\frac{q_2}{r_2^2}} = \frac{q_1}{r_1^2} \cdot \frac{r_2^2}{q_2}$$

$$= \frac{q_1}{q_2} \cdot \frac{r_2^2}{r_1^2} = \frac{r_1}{r_2} \cdot \frac{r_2^2}{r_1^2}$$

$$\frac{E_1}{E_2} = \frac{r_2}{r_1} \Rightarrow \boxed{E_1 = E_2 \frac{r_2}{r_1}}$$

$\therefore V_1 = V_2$

$E_2 > E_1$

Solved Problem
chapter (25)

Problem (3) Page (769)

$v_i = 0$ $\quad \Delta V = 120$ volt $\quad m_p = 1.67 \times 10^{-27}$ kg

$\Delta V = 120V \rightarrow 0V$

a- $v_f = ??$

$$\Delta K = -\Delta U$$

$$K_f - K_i = -(U_f - U_i)$$

$$K_f + U_f = K_i + U_i$$

$$\frac{1}{2} m v_f^2 + q V_f = \frac{1}{2} m v_i^2 + q V_i$$

$$\frac{1}{2} \times 1.67 \times 10^{-27} \times v_f^2 + 0 = 0 + 1.6 \times 10^{-19} \times 120$$

$$\Rightarrow v_f = 1.52 \times 10^5 \text{ m/s}$$

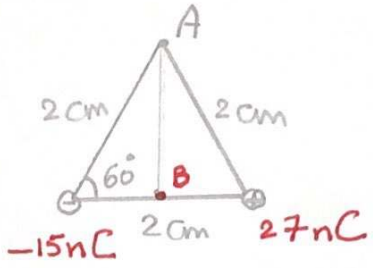
b- e^- will move from $v_i = 0$ volt to $v_f = 120$ volt

$$K_i + U_i = K_f + U_f$$

$$0 + 0 = \frac{1}{2} \times 9.11 \times 10^{-31} v_f^2 - 1.6 \times 10^{-19} \times 120$$

$$v_f = 6.49 \times 10^6 \text{ m/s}$$

Problem (14) Page (770)



a- $V_A = V_1 + V_2$

$= K\left[\frac{q_1}{r_1} + \frac{q_2}{r_2}\right]$

$= 9\times10^{9}\left[\frac{-15\times10^{-9}}{2\times10^{-2}} + \frac{27\times10^{-9}}{2\times10^{-2}}\right]$

$= 5.39\times10^{3}$ volt

$= 5.39$ Kvolt.

b- $V_B = V_1 + V_2$

$= 9\times10^{9}\left[\frac{-15\times10^{-9}}{1\times10^{-2}} + \frac{27\times10^{-9}}{1\times10^{-2}}\right]$

$= 10.8$ K volt.

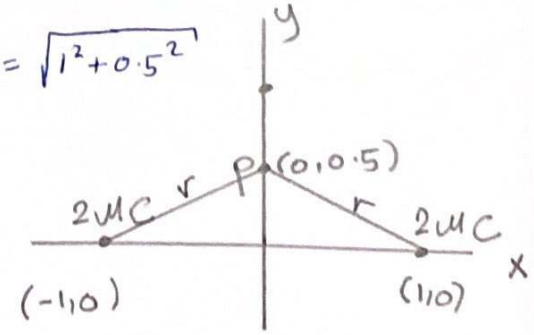
Problem (25) Page (771)

$q_1 = q_2 = +2\,\mu C$ $\quad \vec{x}_1 = \hat{i}m$ $\quad \vec{x}_2 = -\hat{i}m$

a- $V_p = \frac{K q_1}{r_1} + \frac{K q_2}{r_2}$, $r_1 = r_2 = \sqrt{1^2+0.5^2}$

$V_p = 9\times10^{9}\left[\frac{2\times10^{-6}}{\sqrt{1+0.5^2}} + \frac{2\times10^{-6}}{\sqrt{1+0.5^2}}\right]$

$= 32.2$ KV



b- $U = qV$

$= -3 \times 10^{-6} \times 3.22 \times 10^{4}$

$= -9.65 \times 10^{-2}$ J

Problem #36 Page (772)

$E_x = -\frac{dV}{dx} = -\frac{\Delta V}{\Delta x}$ = slope of the Line

$X = 0 \rightarrow X = 1$ cm

$E_x = -\frac{\Delta V}{\Delta x} = -\frac{20-0}{1\,cm} = -20$ v/cm

$X = 1$ cm to $x = 3$ cm

$E_x = -\frac{\Delta V}{\Delta x} = -\frac{0}{2} = 0$

$X = 3$cm to $X = 4$cm

$E_x = -\frac{\Delta V}{\Delta x} = -\frac{(0-20)}{4-3}$

$= 20$ v/cm

Problem (39) Page (772)

$V = 5x - 3x^2y + 2yz^2$

a- $E_x = -\frac{dV}{dx} = -5 + 6xy$

$E_y = -\frac{dV}{dy} = 3x^2 - 2z^2$

$E_z = -\frac{dV}{dz} = -4yz$

$\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$

$= (-5 + 6xy)\hat{i} + (3x^2 - 2z^2)\hat{j} - 4yz\hat{k}$

b- at $(x, y, z) = (1, 0, -2)$ m

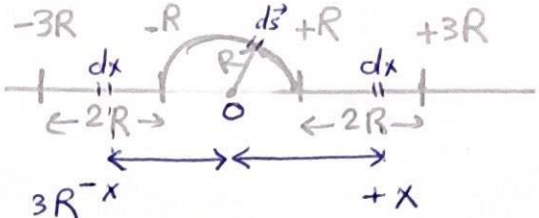
$\vec{E} = (-5 + 6 \times 1 \times 0)\hat{i} + (3 \times 1 - 2 \times 4)\hat{j} - 4 \times 0 \times -2\hat{k}$

$= -5\hat{i} - 5\hat{j}$

$|\vec{E}| = \sqrt{25 + 25} = 7.07$ N/C

Problem (47) Page (773)

$$V_o = \int_{\text{all charge}} \frac{K\,dq}{r}$$



$$V_o = K\int_{-3R}^{-R} \frac{\lambda\,dx}{-x} + \int_{\text{semi circle}} K\lambda\frac{ds}{R} + K\int_{R}^{3R} \frac{\lambda\,dx}{x}$$

$$= -K\lambda \ln(-x)\Big|_{-3R}^{-R} + \frac{K\lambda}{R}\left(\tfrac{1}{2}\cdot 2\pi R\right) + K\lambda \ln(x)\Big|_{R}^{3R}$$

$$= -K\lambda[\ln R - \ln 3R] + \pi K\lambda + K\lambda[\ln 3R - \ln R]$$

$$= K\lambda \ln\frac{3R}{R} + \pi K\lambda + K\lambda \ln\frac{3R}{R}$$

$$= K\lambda(\pi + 2\ln 3)$$

Problem(50) Page (773)

$R = 14\ cm \qquad Q = 26\ \mu C$

a- $r = 10\ cm$

$r < R$ inside conductor

$E = 0$

$$V = \frac{KQ}{R} = \frac{9\times10^{9}\times26\times10^{-6}}{0.14} = 1.67\times10^{6}\ V$$

b- $r = 20\ cm$ $\quad r > R$ outside conductor

$$E = \frac{KQ}{r^2} = \frac{9\times10^{9}\times26\times10^{-6}}{(0.2)^2} = 5.84\times10^{6}\ N/C$$ outward

$$V = \frac{KQ}{r} = \frac{9\times10^{9}\times26\times10^{-6}}{0.2} = 1.17\times10^{6}\ volt$$

c- $r = 14\ cm = R$ on the surface of conductor

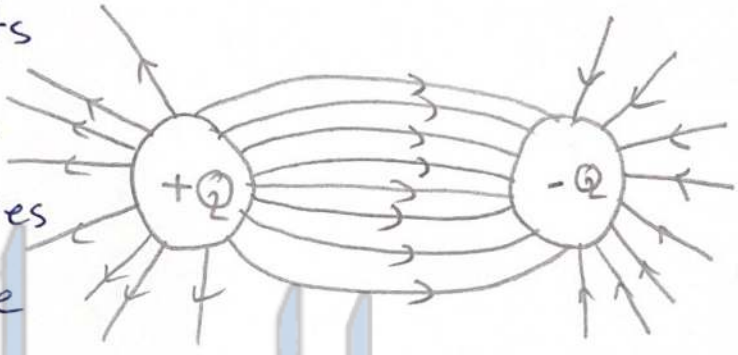
$$E = \frac{KQ}{r^2} = \frac{9\times10^{9}\times26\times10^{-6}}{(0.14)^2} = 11.9\times10^{6}\ N/C$$ outward

$$V = \frac{KQ}{R} = 1.67\times10^{6}\ volt$$

Chapter (26):- Capacitance and Dielectrics

26.1 Definition of capacitance:-

Capacitor consists of two conductors which carry charges of equal magnitude and opposite sign.

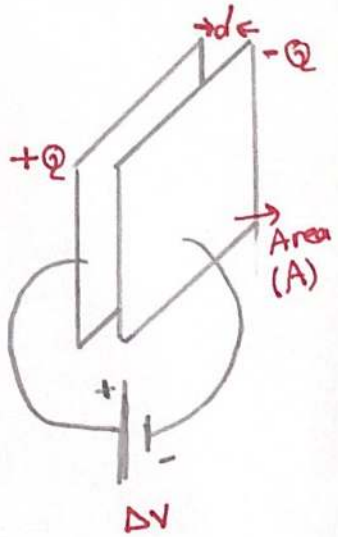


each conductor called plates. and for charged capacitors a potential difference exists between them (ΔV).

Capacitance:- the ratio of magnitude of the charge on either conductor to the magnitude of the potential difference between the conductors.

$$C = \left| \frac{Q}{\Delta V} \right| \text{ , } C \text{ is always +ve}$$

Q and ΔV expressed as +ve quantities.



$$[C] \equiv \frac{1\,C}{1\,V} \equiv 1 \text{ Farad}$$

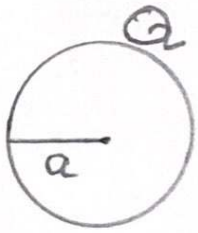
$$1\,\mu F = 10^{-6}\,F \quad , \quad 1\,pF = 10^{-12}\,F$$

26.2 Calculating Capacitance.

a single conductor has capacitance

① Capacitance of single spherical charged conductor :-

$$C = \frac{Q}{\Delta V}$$



$V = \frac{KQ}{a}$ for sphere of radius a.

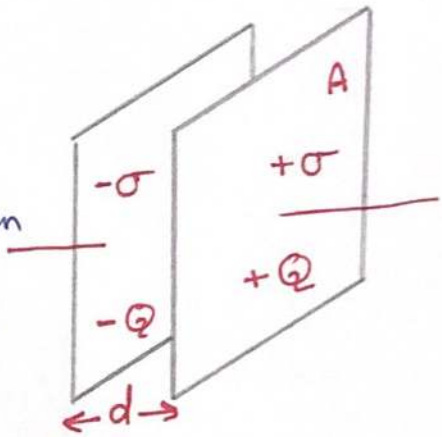
to find ΔV, imagin that a spherical shell of radius ∞ surrounds the spherical conductor and concentric with it, it's potential is zero. $\Rightarrow \Delta V = \frac{KQ}{a} - 0 = \frac{KQ}{a}$

$$\therefore C = \frac{Q}{KQ/a} = \frac{a}{K} = 4\pi\epsilon_0 a$$

$\therefore$ for isolated charged sphere C depends on the radius directly propotional and is independent on the charge and potential.

② parallel plate capacitor:-

$$\sigma = \frac{Q}{A}$$



Electric field is uniform between the plates and zero elsewhere.

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

$$\Delta V = E\,d = \frac{Q\,d}{\epsilon_0 A}$$

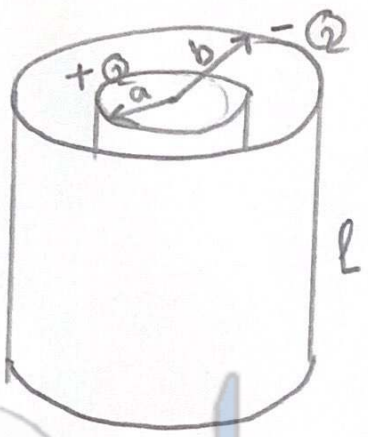
$$C = \frac{Q}{\Delta V} = \frac{Q}{Qd/\epsilon_0 A}$$

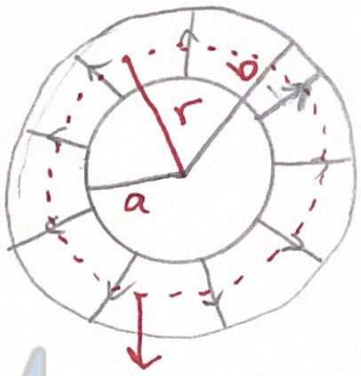
$$C = \frac{\epsilon_0 A}{d}$$

Capacitance of parallel plate capacitor is directly proportional to the area of it's plates and inversely proportional to it's separation.

$$C \propto A \qquad C \propto \frac{1}{d}$$

Ex(26.1):- The Cylindrical Capaeitor.





$$C = \frac{Q}{\Delta V}$$

$$\Delta V = V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{s} = -\int_a^b E_r \, dr$$

for cylindrically charge distribution

$$E = \frac{2K\lambda}{r}$$

$$\Delta V = -2K\lambda \int_a^b \frac{dr}{r} = -2K\lambda \ln r \Big]_a^b$$

$$= -2K\lambda \ln\left(\frac{b}{a}\right)$$

$$C = \frac{Q}{2K\frac{Q}{L}\ln(\frac{b}{a})} = \frac{L}{2K\ln(b/a)}$$

the capaeitance of cylindrical capaeitor depend on the radii a and b and proportional to the length of cylinders.

What if ?? suppose $b = 2a$, to increase the capacitance we increase a by 10%.

$b = 2a$ $\qquad a' = 1.1a$

$$C' = \frac{L}{2K \ln(b/a')} \quad , \quad C = \frac{L}{2K \ln(b/a)}$$

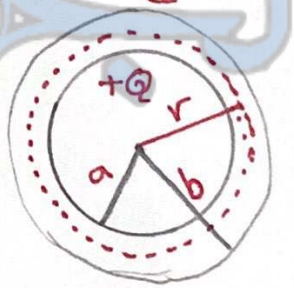
$$\frac{C'}{C} = \frac{L/2K \ln(b/a')}{L/2K \ln(b/a)} = \frac{\ln(b/a)}{\ln(b/a')}$$

$$= \frac{\ln(2a/a)}{\ln(2a/1.1a)} = \frac{\ln 2}{\ln 1.82} = 1.16$$

16 % increase in capacitance.

Ex(26.2):- The spherical capacitor :-

$$\Delta V = V_b - V_a = -\int_a^b \vec{E} \cdot d\vec{s} = -\int_a^b E_r dr \quad -①$$



for spherical conductor

$$E = \frac{KQ}{r^2}$$

$$\Delta V = -\int_a^b KQ \frac{dr}{r^2} = \frac{KQ}{r}\Big]_a^b$$

$$= KQ\left(\frac{1}{b} - \frac{1}{a}\right) = KQ\,\frac{(a-b)}{ab}$$

$$C = \frac{Q}{\Delta V} = \frac{Q\,ab}{KQ(a-b)}$$

$$= \frac{ab}{K(a-b)}$$, C depends on a and b

what if the radius $b \to \infty$

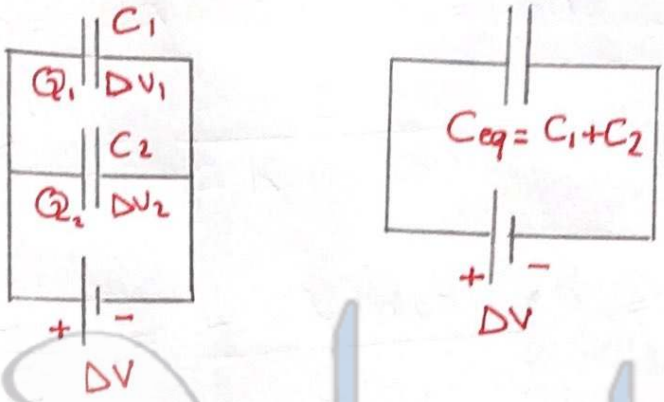
$$C = \lim_{b\to\infty} \frac{ab}{K(a-b)} = \frac{ab}{Kb}$$

$$C = \frac{a}{K} = 4\pi\epsilon_0 a$$

which is the capacitance of isolated conducting sphere.

26.3 Combinations of Capacitors

1- Parallel combination:-



the potential difference across capacitors connected in parallel are the same and equal to potential difference applied across the combination.

$$\Delta V_1 = \Delta V_2 = \Delta V$$

$$Q_{tot} = Q_1 + Q_2$$

$$C_{eq}\,\Delta V = C_1 \Delta V_1 + C_2 \Delta V_2$$

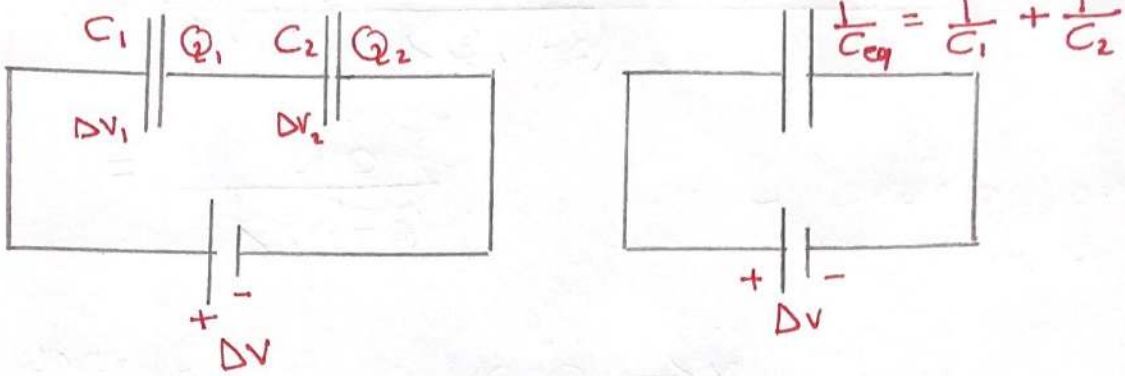
$$C_{eq} = C_1 + C_2$$

for three or more capacitors connected in Parallel $C_{eq} = C_1 + C_2 + C_3 + \cdots$

C_{eq} is greater than any of individual capacitances.

$$C_{eq} > C_1 \quad , \quad C_{eq} > C_2$$

2- Series Combination :-



the charges on capacitors connected in series are same.

$$Q_1 = Q_2 = Q$$

$$\Delta V_{tot} = \Delta V_1 + \Delta V_2$$

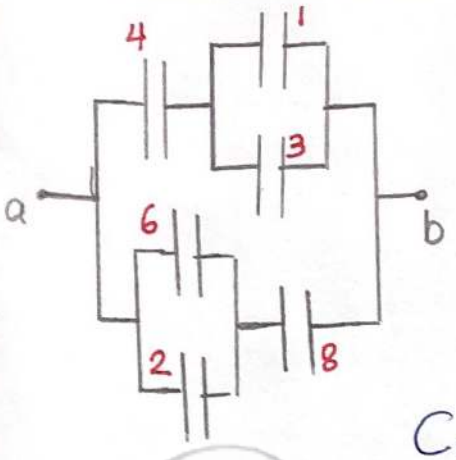
$$\frac{Q}{C_{eq}} = \frac{Q_1}{C_1} + \frac{Q_2}{C_2}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2}$$

For 3 or more capacitors connected in series

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$$

Ex (26.3):- Equivalent Capacitance.



all capacitors capacitance in μf.

capacitors 1 and 3 are connected in parallel.

$C_{eq} = 1 + 3 = 4\ \mu f$

Capacitors 6 and 2 are connected in parallel

$C_{eq} = 2 + 6 = 8\ \mu f$

capacitors 4 and 4 are connected in series

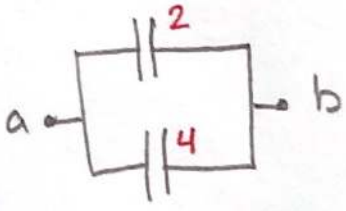
$$\frac{1}{C_{eq}} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$\therefore C_{eq} = 2\ \mu f$

Capacitors 8 and 8 are connected in series

$$\frac{1}{C_{eq}} = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$C_{eq} = 4\ \mu f$

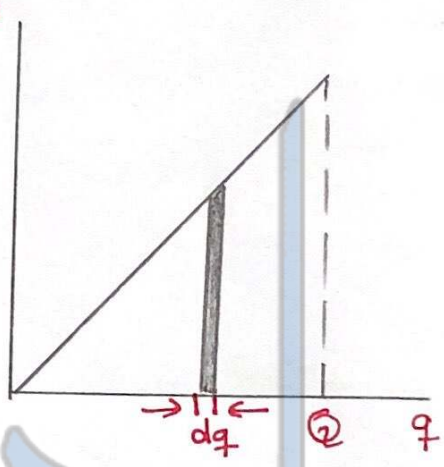


capacitors 2 and 4 are connected in parallel

$C_{eq} = 2 + 4 = 6\ \mu f$

26.4 Energy stored in a charged capacitor:-

Electric potential energy is stored in the capacitor. as it's transformed from the chemical energy in the battery. to calculate this energy suppose that +ve charge on one plate will move to the other plate under a force effect; work must be done on a charge to move it between two plates.



$W \simeq$ area of rectangular

$$dW = dq\,\Delta V = \frac{q}{C}\,dq$$

$$W = \int_0^Q \frac{1}{C}\,q\,dq = \frac{Q^2}{2C}$$

$$W = \Delta V \Rightarrow U = \frac{Q^2}{2C}$$

$$U = \frac{Q^2}{2C} = \frac{1}{2}Q\,\Delta V = \frac{1}{2}C(\Delta V)^2$$

$\equiv$ Energy stored in a charged capacitor

For parallel plate capacitor

$$C = \frac{\epsilon_0 A}{d} \quad , \quad \Delta V = Ed$$

$$U = \frac{1}{2} C (\Delta V)^2$$

$$= \frac{1}{2} \left(\frac{\epsilon_0 A}{d} \right) (Ed)^2$$

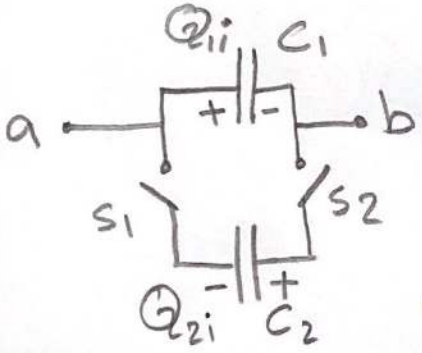
$$U = \frac{1}{2} \epsilon_0 A d E^2$$

the energy per unit volume U_E Known as energy density

$$U_E = \frac{U}{\text{Volume}} = \frac{U}{Ad} = \frac{1}{2} \epsilon_0 E^2$$

$$[U_E] \equiv J/m^3$$

Ex (26.4):- Rewiring two charged capacitors :-

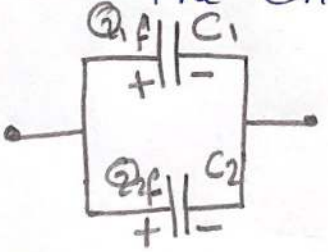


$C_1 > C_2$

$Q_{1i} > Q_{2i}$

when the switches are closed the charges redistribute. and both capacitors will have the same ΔV.

a-



the left hand plates of the capacitors form an isolated system because they are not connected to the right hand side.

Q_i :- is the initial charge on the left hand plates.

$$Q_i = Q_{1i} + Q_{2i}$$
$$= C_1 \Delta V_i - C_2 \Delta V_i$$
$$= (C_1 - C_2) \Delta V_i$$

Q_f :- is the final charge on the left hand Plates.

$$Q_f = Q_{1f} + Q_{2f} = C_1 \Delta V_f + C_2 \Delta V_f$$
$$= (C_1 + C_2) \Delta V_f$$

because the system is isolated $Q_f = Q_i$

$$(C_1 - C_2) \Delta V_i = (C_1 + C_2) \Delta V_f$$
$$\Delta V_f = \frac{(C_1 - C_2)}{(C_1 + C_2)} \Delta V_i$$

b- $U_i = \frac{1}{2} C_1 (\Delta V_i)^2 + \frac{1}{2} C_2 (\Delta V_i)^2$

$$= \frac{1}{2} (C_1 + C_2) \Delta V_i^2$$

$$U_f = \frac{1}{2} C_1 (\Delta V_f)^2 + \frac{1}{2} C_2 (\Delta V_f)^2$$

$$= \frac{1}{2} (C_1 + C_2) (\Delta V_f)^2$$

$$= \frac{1}{2} (C_1 + C_2) \left[\frac{C_1 - C_2}{C_1 + C_2} \right]^2 \Delta V_i^2$$

$$= \frac{1}{2} \frac{(C_1 - C_2)^2}{C_1 + C_2} \Delta V_i^2$$

$$\frac{U_f}{U_i} = \frac{\frac{1}{2} \frac{(C_1 - C_2)^2}{C_1 + C_2} \Delta V_i^2}{\frac{1}{2} (C_1 + C_2) \Delta V_i^2}$$

$$\frac{U_f}{U_i} = \frac{(C_1 - C_2)^2}{(C_1 + C_2)^2} < 1$$

$U_f < U_i$; the missing energy transfered out of the system by the mechanism of electromagnetic waves.

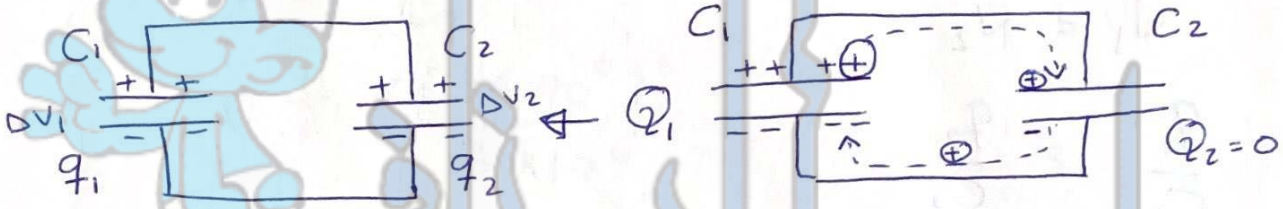
what if ? if $C_1 = C_2$

$Q_{i1} = Q_{2i}$ and

$Q_{1f} = Q_{f2} = 0$ in opposite Polarity.

Rewiring 2 charged capacitors:-

1- Charged capacitor with charge $Q_{1\,initial}$ and capacity C_1 connected to second capacitor C_2 which is not charged, $Q_{2\,initial}$ = zero, find final charges on each capacitor?? if $C_1 = C_2$



$\Delta V_1 = \Delta V_2$

$\frac{q_1}{C_1} = \frac{q_2}{C_2}$, but $C_1 = C_2$

$q_1 = q_2$, $q_1 + q_2 = Q_1 + Q_2^{\,0}$

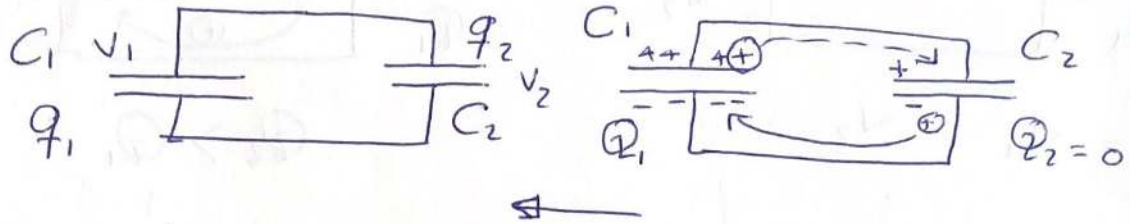
$q_1 = Q_1 - q_1$

$2q_1 = Q_1$

$q_1 = \frac{1}{2} Q_1 = q_2$

Ex:- $C_1 = C_2 = 2\mu f$

$Q_1 = 40\mu C$ $\quad$ $Q_2 = 0$



$V_1 = V_2$

$\frac{q_1}{C_1} = \frac{q_2}{C_2} \Longrightarrow q_1 = q_2$

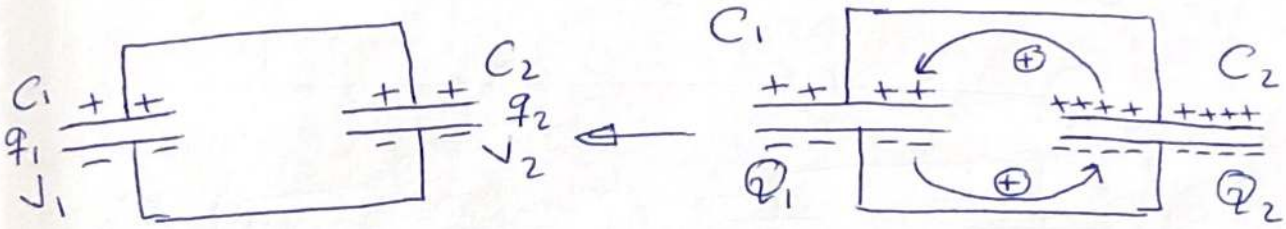
$q_1 + q_2 = Q_1 + Q_2 = 40\mu C + 0 = 40\mu C$

$q_1 + q_2 = 40\mu C$

$2q_1 = 40\mu C \Longrightarrow q_1 = q_2 = 20\mu C$

---

[2]- 2 charged capacitor $C_1 = 2\mu f$ and $C_2 = 3\mu f$ with initial values of charges $Q_1 = 40\mu C$ and $Q_2 = 100\mu C$ if they are connected with the same Polarity i.e +ve plates are connected together and negative plates are connected to gether find final charges on each one of them??



$V_1 = V_2$ $\qquad$ $Q_2 > Q_1$

$$q_1 = Q_1 + (Q_2 - q_2)$$

$$q_1 = 40\mu C + (100\mu C - q_2)$$

$$q_2 = 140\mu C - q_1$$

$$V_1 = V_2$$

$$\frac{q_1}{C_1} = \frac{q_2}{C_2}$$

$$\frac{q_1}{2\mu f} = \frac{140\mu C - q_1}{3\mu f}$$

$$3q_1 = 280\mu C - 2q_1$$

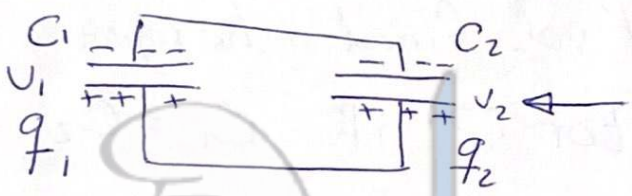
$$5q_1 = 280\,\mu C \Rightarrow q_1 = 56\,\mu C$$

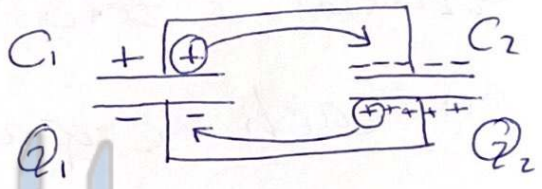
$$q_2 = 140\,\mu C - q_1 = 140 - 56 = 84\,\mu C$$

3 if we solve the preceding example but with opposite polarity i.e -ve plate connected to +ve plate.

$C_1 = 2\mu f$ $\quad C_2 = 3\mu f$

$Q_1 = 40\mu C$ $\quad Q_2 = 100\mu C$





$V_1 = V_2$

$$\frac{q_1}{C_1} = \frac{q_2}{C_2}$$

$$\frac{q_1}{2} = \frac{q_2}{3}$$

but $q_2 = |Q_2 - Q_1| - q_1 = (100-40) - q_1$

$$q_2 = 60 - q_1$$

$$\frac{q_1}{2} = \frac{60-q_1}{3} \Rightarrow 3q_1 = 120 - 2q_1$$

$$5q_1 = 120 \Rightarrow q_1 = 24\mu C$$

$$q_2 = 60 - 24 = 36\ \mu C$$

in example [3] find stored energy in each capacitor after they acheive equalibrium?

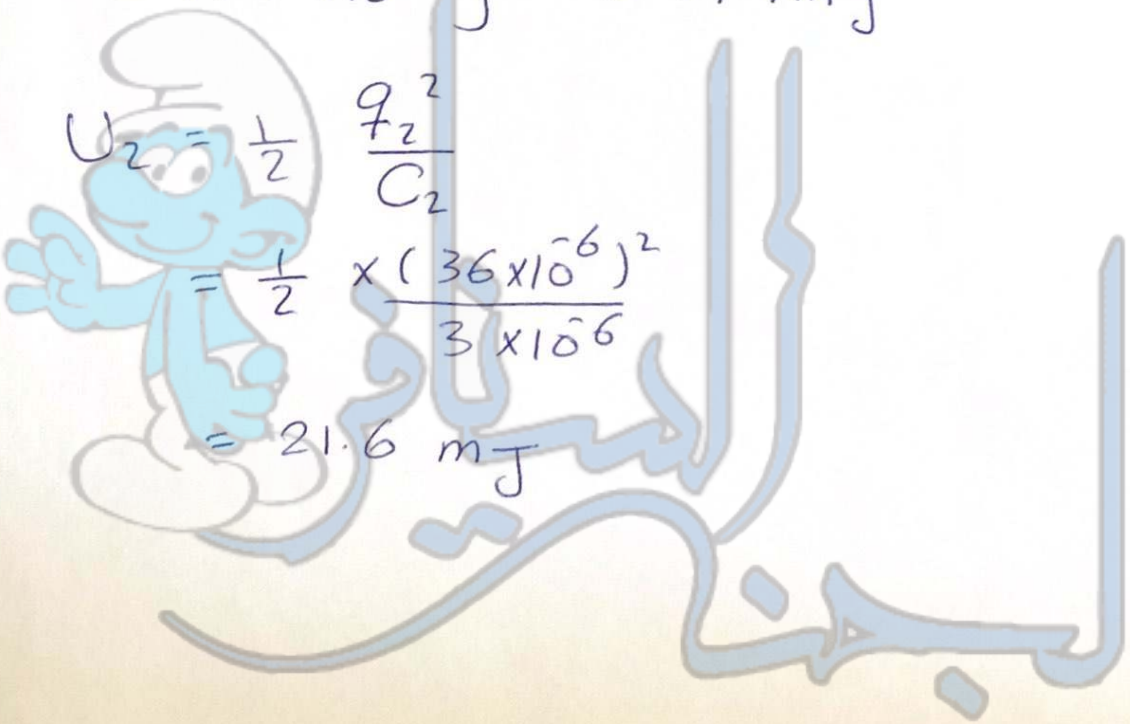
$$U_1 = \frac{1}{2}\frac{q_1^2}{C_1} = \frac{1}{2} \times \frac{(24\times10^{-6})^2}{2\times10^{-6}}$$

$$= 14.4\times10^{-3}\ J = 14.4\ mJ$$

$$U_2 = \frac{1}{2}\frac{q_2^2}{C_2}$$

$$= \frac{1}{2} \times \frac{(36\times10^{-6})^2}{3\times10^{-6}}$$

$$= 21.6\ mJ$$



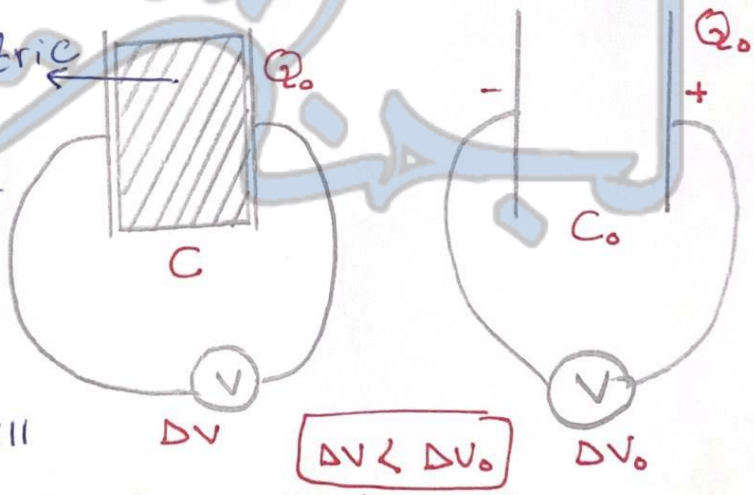
26.5 Capacitors with dielectrics:-

dielectric is anonconducting material.

by experiments they found that potential difference across capacitor decreases by inserting dielectric between capacitor terminals.

for capacitor without dielectric $\Delta V_o = \frac{Q_o}{C_o}$

where ΔV_o is the potential difference between capacitor terminals, Q_o is the charge on capacitor and C_o is the capacitor.



as NO charge can flow through voltmeter and NO battery in the circuits; this means that Q_o will not change with and without dielectric.

the voltage with and without dielectric are related by dielectric constant (K).

$$\Delta V = \frac{\Delta V_o}{K}$$

$\Delta V < \Delta V_o \Rightarrow K > 1$, K is dimensionless constant.

$$C = \frac{Q_o}{\Delta V} = \frac{Q_o}{\Delta V_o / K} = K \frac{Q_o}{C_o}$$

$$C = K C_o$$

∴ after the dielectric is inserted between the plates:- 1- the charge remains the same.

2- potential difference decreases.

3- capacitance increases.

for parallel plate capacitor :-

$$C = K C_o = K \frac{\epsilon_o A}{d}$$

we can increase the capacitance of parallel-plate capacitor by inserting dielectric and decreasing (d) between the plates.

the lowest value of (d) is limited by the electric discharge that could occur through the dielectric.

max voltage can be applied to the capacitor

depends on dielectric strength.

dielectric strength :- is the max electric field of the dielectric. and can be measured in (V/m)

if the magnitude of $\vec{E}$ in the dielectric exceeds dielectric strength the dielectric begins to conduct!

working voltage
breakdown voltage
rated voltage
] 3 names for the same Parameter.

this parameter represents the largest voltage that can be applied to the capacitor without exceeding the dielectric strength.

the value of K depends on the material of dielectric. see Table (26.1) page 791 in the text book, Please.

advanges of dielectric between capacitor terminals :-

1- an increase in capacitance.

2- an increase in the max operating voltage.

3- mechanical support between the plates which decrease (d) and increase (C).

Ex(26.5):- Energy stored before and after.

U_o :- energy stored in capacitor without dielectric

$$U_o = \frac{Q_o^2}{2C_o}$$

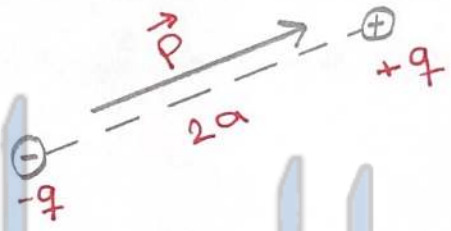
U :- energy stored in capacitour with dielectric

$$U = \frac{Q_o}{2C} = \frac{Q_o}{2C_oK} = \frac{U_o}{K}$$

$\therefore U < U_o$; energy stored in the capacitor with dielectric is less than without dielectric.

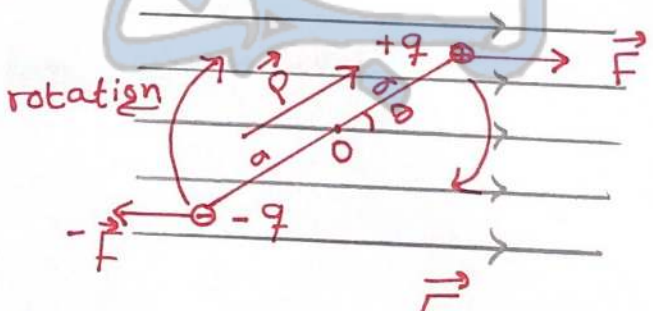
26.6 Electric dipole in an electric field :-

The electric dipole consists of 2 charges of equal magnitude and opposite sign seperated by a distance (2a).



$\vec{P}$ is the electric dipole moment which is defined as a vector directed from -ve charge to +ve charge along (2a).

For electric dipole in a uniform electric field $\vec{E}$:-



$$P = 2aq \equiv [C \cdot m]$$

the electric force acting on the two charges are equal in magnitude and opposite in direction $\Rightarrow$ the net force on the dipole = 0.

the two forces produce anet torque on the dipole.

$$\vec{\tau} = \vec{F} \times \vec{r} \equiv [N.m]$$

$$= F a \sin\theta + F a \sin\theta \quad \text{clockwise rotation}$$

$$= 2 F a \sin\theta$$

but $F = qE$, $P = 2aq$

$$\therefore \tau = 2qEa\sin\theta$$

$$= PE\sin\theta$$

$$\Rightarrow \vec{\tau} = \vec{P} \times \vec{E} = [C.m \cdot \frac{N}{C}] = [N.m]$$

the work done by external force to rotate the dipole, the work done is stored as electric potential in the system.

the work required to rotate the dipole through an angle ($d\theta$) is:-

$$dw = \tau \, d\theta$$

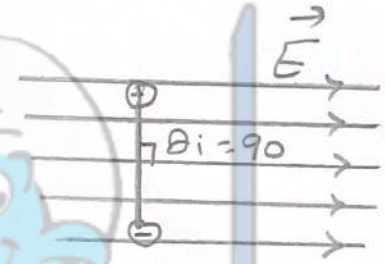
$$w = \int_{\theta_i}^{\theta_f} \tau \, d\theta = \Delta U$$

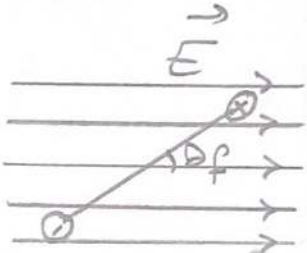
$$U_f - U_i = \int_{\theta_i}^{\theta_f} PE \sin\theta \, d\theta$$

$$= PE \int_{\theta_i}^{\theta_f} \sin\theta \, d\theta$$

$$= PE \left(-\cos\theta\right)_{\theta_i}^{\theta_f}$$

$$= PE \left(\cos\theta_i - \cos\theta_f\right) \equiv [J]$$





let $\theta_i = 90^\circ \Rightarrow U_i = 0$

$$U_f - 0 = PE\left(0 - \cos\theta_f\right)$$

$$U_f = -PE\cos\theta_f$$

$$U_E = -\vec{P}\cdot\vec{E}$$

$U_g = mgy$
$U_E = -PE\cos\theta$

1- both expressions contain parameters of the entity placed in the field mass and dipole moment

2- both expressions contain field g and E.

3- both expressions contain configuration description translational (y) and rotational (θ).

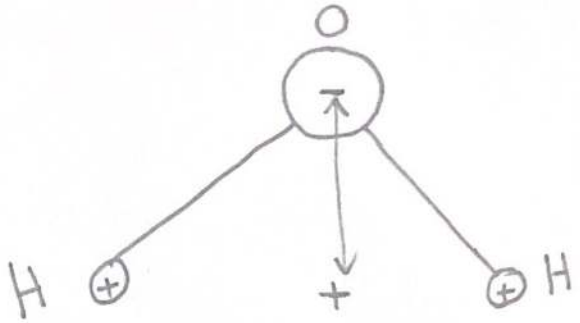
Ex(26.6):- The H_2O molecule:-

for H_2O

$P = 6.3 \times 10^{-30}$ C.m

$E = 2.5 \times 10^{5}$ N/C

$N = 10^{21}$ molecules



$\theta_i = 0$, $\theta_f = 90^\circ$

$W = \Delta U_E = U_{90} - U_0$

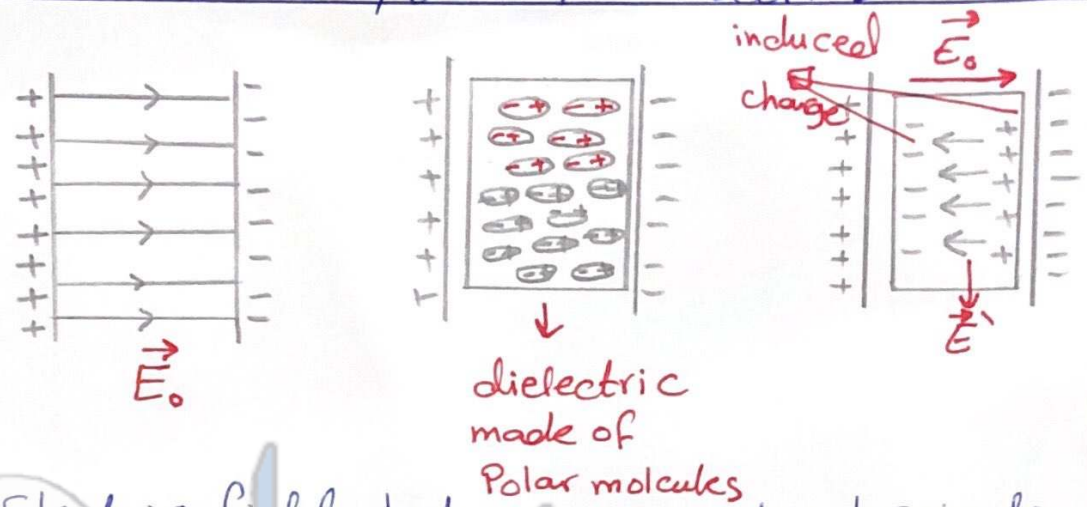
$= [-NPE\cos 90] - [-NPE\cos 0]$

$= NPE$

$= 10^{21} \times 6.3 \times 10^{-30} \times 2.5 \times 10^{5}$

$= 1.6 \times 10^{-3}$ J

26.7 An atomic description of dielectrics:-



$\vec{E}_0$:- Electric field between capacitor terminals without dielectric.

dielectric made of polar molcules, when an electric field is applied, the molcules partially align with the field.

$\vec{E}'$:- induced electric field due to induced charge on dielectric.

$\vec{E}_{tot} = \vec{E}_0 - \vec{E}'$, Electric field is reduced by inserting dielectric between capacitor terminals.

$E = \frac{E_0}{K}$ and this explain why ΔV reduced also.

$\Delta V = Ed$

$= \frac{E_0 d}{K} = \frac{\Delta V_0}{K}$

surface charge density $\sigma = \frac{Q}{A} \equiv [C/m^2]$

$$E = \frac{\sigma}{\epsilon_0}$$

$$E' \equiv E_{ind} = \frac{\sigma_{ind}}{\epsilon_0} \quad , \sigma_{ind} = \frac{Q_{ind}}{\epsilon_0}$$

$$E = E_0 - E_{ind}$$

$$\frac{\sigma}{K\epsilon_0} = \frac{\sigma}{\epsilon_0} - \frac{\sigma_{ind}}{\epsilon_0}$$

$$\sigma_{ind} = \sigma - \frac{\sigma}{K} = \sigma\left(1 - \frac{1}{K}\right)$$

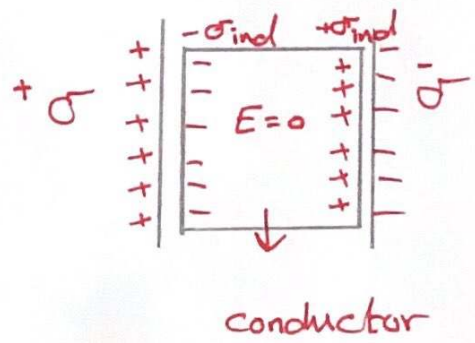
$$\sigma_{ind} = \sigma\left(\frac{K-1}{K}\right) \Rightarrow \sigma_{ind} < \sigma$$

for $K = 1$; NO dielectric between terminals $\Rightarrow$

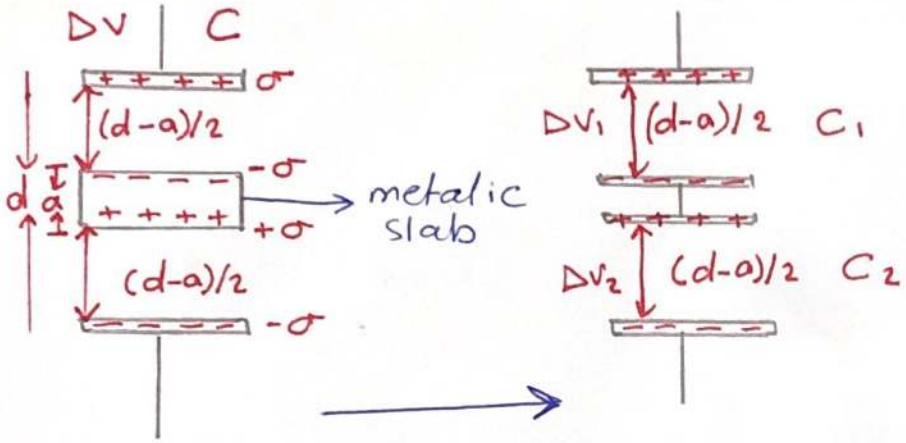
$$\sigma_{ind} = 0$$

if we put electrical conductor between conductor terminals $\Rightarrow E = 0$ inside conductor $\Rightarrow$

$$E_0 = E_{ind} \Rightarrow \sigma_{ind} = \sigma$$



Ex(26.7):- Effect of metalic slab:-



inserting metalic slab between terminals of capacitor change it from one capacitor into 2 capacitors that are connected in series.

why in series not parallel ??

because $\Delta V = \Delta V_1 + \Delta V_2$

$$E \cdot d = E\frac{(d-a)}{2} + E\frac{(d-a)}{2}$$

a- $$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$C_1 = \frac{\epsilon_o A}{(d-a)/2} \quad , \quad C_2 = \frac{\epsilon_o A}{(d-a)/2}$$

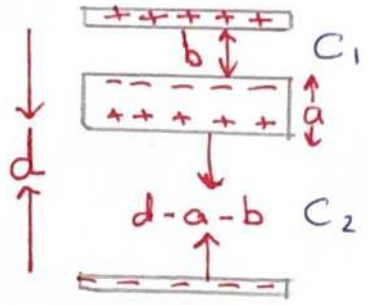
$$\frac{1}{C} = \frac{d-a}{2\epsilon_o A} + \frac{d-a}{2\epsilon_o A} = \frac{d-a}{\epsilon_o A}$$

$$\boxed{C = \frac{\epsilon_o A}{d-a}}$$

b- for thin slab, $a \to 0$

$$C = \frac{\epsilon_o A}{d-0} = \frac{\epsilon_o A}{d}$$

C- what if metalic slab not midway between the plates?



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

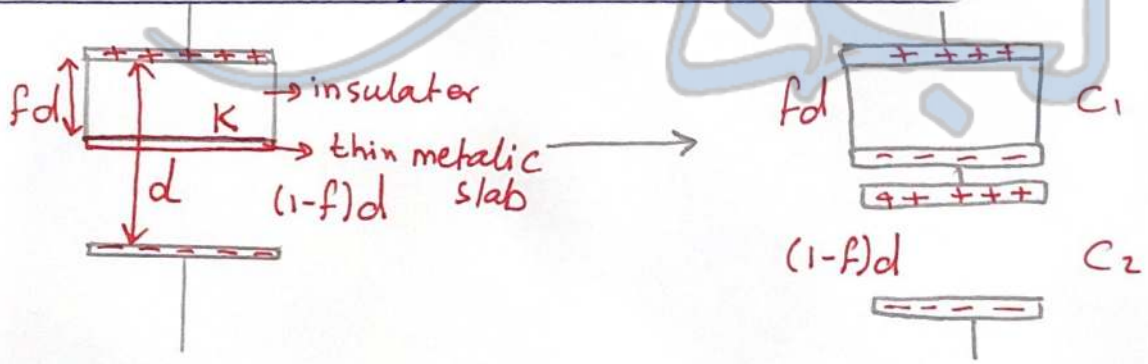
$$= \frac{1}{\frac{\epsilon_0 A}{b}} + \frac{1}{\frac{\epsilon_0 A}{d-a-b}}$$

$$= \frac{b}{\epsilon_0 A} + \frac{d-a-b}{\epsilon_0 A}$$

$$\frac{1}{C} = \frac{d-a}{\epsilon_0 A} \Rightarrow C = \frac{\epsilon_0 A}{d-a}$$

∴ total capacitance does not affect by the position of metalic slab.

Ex(26.8):- Partially filled capacitor:-



f is fraction between (0-1)

$$C_1 = \frac{K\epsilon_0 A}{fd}$$

$$C_2 = \frac{\epsilon_0 A}{(1-f)d}$$

$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$= \frac{fd}{K\epsilon_0 A} + \frac{(1-f)d}{\epsilon_0 A}$$

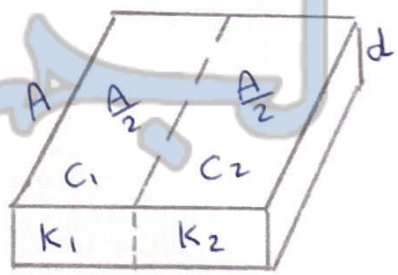
$$= \frac{fd}{K\epsilon_0 A} + \frac{K(1-f)d}{K\epsilon_0 A}$$

$$= \frac{d}{\epsilon_0 A}\left[\frac{f + K(1-f)}{K}\right]$$

$$C = \frac{\epsilon_0 A}{d}\left[\frac{K}{f + K(1-f)}\right]$$

$$C = \frac{K}{f + K(1-f)} C_0$$

Ex:- ΔV for each half is the same, we may treat the system as (2) capacitors in parallel



$$C_1 = \frac{K_1 \epsilon_0 (A/2)}{d} \quad , C_2 = \frac{K_2 \epsilon_0 (A/2)}{d}$$

$$C = C_1 + C_2 = \frac{K_1 \epsilon_0 (A/2)}{d} + \frac{K_2 \epsilon_0 (A/2)}{d}$$

$$C = \frac{\epsilon_0 A}{2d}(K_1 + K_2)$$

$$C = \frac{(K_1 + K_2)}{2} C_0$$

Solved problems chapter (26)

Q3:- a- $C = 4\,\mu f$ $\quad \Delta V = 12$ volt

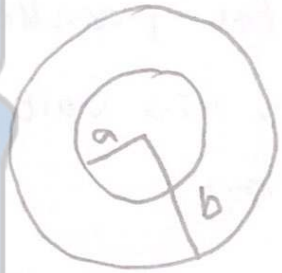
$$Q = C \cdot \Delta V = 4 \times 10^{-6} \times 12 = 48\,\mu C$$

b- $\Delta V = 1.5$ volt

$$Q = C \cdot \Delta V = 4 \times 10^{-6} \times 1.5 = 6\,\mu C$$

Q4:- a- spherical capacitor

$a = 7$ cm $\quad b = 14$ cm



$$C = \frac{ab}{K(b-a)}$$

$$= \frac{0.07 \times 0.14}{9 \times 10^{9}(0.14 - 0.07)}$$

$$= 15.6\ PF$$

b- $Q = 4\,\mu C$ $\quad \Delta V = ??$

$$\Delta V = \frac{Q}{C} = \frac{4 \times 10^{-6}}{15.6 \times 10^{-12}} = 2.57 \times 10^{5} \text{ volt}$$

Q5:- cylindrical capacitor $\quad L = 50$ m $\quad Q = 8.1\,\mu C$

$2a = 2.58$ mm $\quad 2b = 7.27$ mm

$$a- C = \frac{l}{2K \ln(b/a)} = \frac{50}{2 \times 9 \times 10^{9} \ln(3.635/1.29)}$$

$$= 2.68 \times 10^{-9}\ F = 2.68\ nf$$

b- For cylindrical charge distribution

$$\Delta V = 2K\lambda \ln\left(\frac{b}{a}\right) \quad OR \quad \Delta V = \frac{Q}{C}$$

$$\Delta V = 2\times 9\times 10^{9} \times \frac{8.1\times 10^{-6}}{50} \ln\left(\frac{3.635}{1.29}\right)$$

$$= 3.02 \times 10^{3} \text{ volt}$$

OR $$\Delta V = \frac{8.1\times 10^{-6}}{2.68\times 10^{-9}}$$

$$= 3.02\times 10^{3} \text{ volt}$$

Q7:- for parallel plate capacitor

$\Delta V = 150$ volt $\quad \sigma = 30\ nC/cm^2$

$d = ??$

$$C = \frac{\epsilon_0 A}{d} = \frac{Q}{\Delta V} \Rightarrow \frac{Q}{A} = \frac{\epsilon_0 \Delta V}{d} = \sigma$$

$$d = \frac{\epsilon_0 \Delta V}{\sigma} = \frac{8.85\times 10^{-12} \times 150}{30\times 10^{-9} \times 10^{4}}$$

$$= 4.425\times 10^{-6} \text{ m}$$

$$= 4.425\ \mu m$$

$$\sigma = \frac{30\times 10^{-9}\ C}{10^{-4}\ m^2} = 30\times 10^{-9} \times 10^{4}\ C/m^2$$

Q8:- for parallel plate capacitor

$A = 2.3\ cm^2 = 2.3 \times 10^{-4}\ m^2$

$d = 1.5\ mm = 1.5 \times 10^{-3}\ m$ $\quad \Delta V = 12\ volt$

a- $C = \frac{\epsilon_o A}{d} = \frac{8.85 \times 10^{-12} \times 2.3 \times 10^{-4}}{1.5 \times 10^{-3}}$

$= 1.36 \times 10^{-12}\ f$

$= 1.36\ pf$

b- $Q = C \cdot \Delta V$

$= 1.36 \times 10^{-12} \times 12$

$= 16.3 \times 10^{-12}\ C$

$= 16.3\ pC$

c- $E = \frac{\Delta V}{d} = \frac{12}{1.5 \times 10^{-3}} = 8 \times 10^{3}\ V/m$

Q16:- $C_1 = 2.5\ \mu f$ $\quad C_2 = 6.25\ \mu f$ $\quad \Delta V = 6\ volt$

a- when connected in series:-

$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2.5 \times 10^{-6}} + \frac{1}{6.25 \times 10^{-6}}$

$C_{eq} = 1.79\ \mu f$

$Q_1 = Q_2 = Q_{tot}$ in series combination

$Q_{tot} = C_{eq} \Delta V = 1.79 \times 10^{-6} \times 6$

$= 10.7\ \mu C = Q_1 = Q_2$

b- in parallel combination

$C_{eq} = C_1 + C_2$

$= 2.5 + 6.25 = 8.75\ \mu f$

$Q_1 = C_1 \Delta V = 2.5 \times 10^{-6} \times 6 = 15\ \mu C$

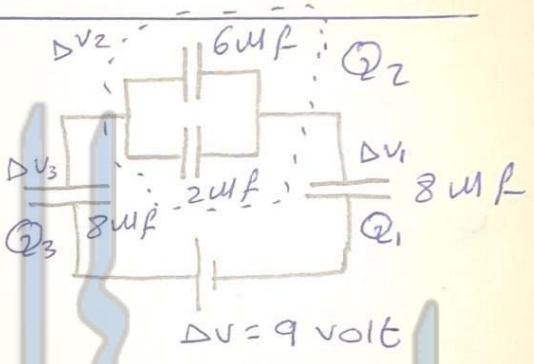
$Q_2 = C_2 \Delta V = 6.25 \times 10^{-6} \times 6 = 37.5\ \mu C$

Q26:- a- $C_{parallel} = 6 + 2$

$= 8\ \mu f$

$\frac{1}{C_{series}} = \frac{1}{8} + \frac{1}{8} + \frac{1}{8}$

$C_{series} = \frac{8}{3} = 2.67\ \mu f$



b- $Q_1 = Q_2 = Q_3$ as they are in series

+C- $= Q_{tot}$

$Q_{tot} = C_{tot} \times \Delta V = 2.67 \times 10^{-6} \times 9 = 24\ \mu C$

$\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3$

$\Delta V_1 = \frac{Q_1}{C_1} = \frac{24 \times 10^{-6}}{8 \times 10^{-6}} = 3 volt$

$\Delta V_2 = \frac{Q_2}{C_{parallel}} = \frac{24 \times 10^{-6}}{8 \times 10^{-6}} = 3 volt$

$\Delta V_3 = \frac{Q_3}{C_3} = \frac{24 \times 10^{-6}}{8 \times 10^{-6}} = 3 volt$

$\Delta V_{6\mu f} = \Delta V_{2\mu f} = 3 volt$

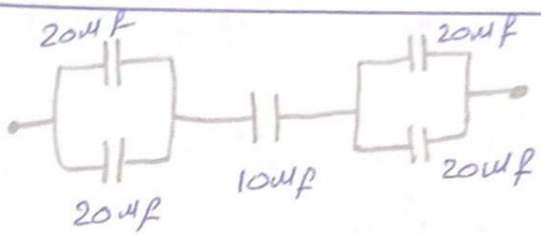
$Q_{6\mu f} = 6 \times 10^{-6} \times 3 = 18\ \mu C$

$Q_{2\mu f} = 2 \times 10^{-6} \times 3 = 12\ \mu C$

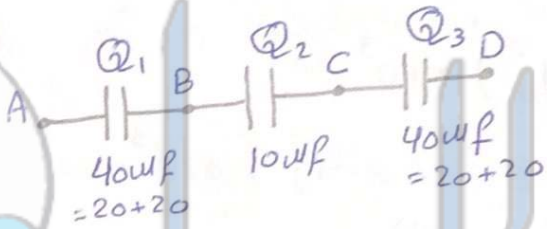
Q31:- $\Delta V = 12$ volt $\quad Q = 54 \mu C$

$U = \frac{1}{2} Q \Delta V = \frac{1}{2} \times 54 \times 10^{-6} \times 12 =$

Q48:-



break down voltage = 15 volt for each capacitor.



$Q_1 = Q_2 = Q_3$ in series.

$40 \times 10^{-6} \Delta V_{AB} = 10 \times 10^{-6} \Delta V_{BC} = 40 \times 10^{-6} \Delta V_{CD}$

$4 \Delta V_{AB} = \Delta V_{BC} = 4 \Delta V_{CD}$

when $\Delta V_{BC} = 15$ volt $\Rightarrow$ 10μf will break down first

$\Rightarrow \Delta V_{AB} = \frac{1}{4} \Delta V_{BC} = \frac{15}{4}$ volt

$\Delta V_{CD} = \frac{1}{4} \Delta V_{BC} = \frac{15}{4}$ volt

$\Delta V_{AD} = \Delta V_{AB} + \Delta V_{BC} + \Delta V_{CD}$

$= \frac{15}{4} + 15 + \frac{15}{4}$

$= 22.5$ volt.

Q50:- $q = \mp 3.5\,nC$ for dipole in $\vec{E}$

$\vec{E} = 7.8\times10^{3}\,\hat{i} - 4.9\times10^{3}\,\hat{j}$ (N/C)

$\vec{r}_{+q} = (-1.2\,\hat{i} + 1.1\,\hat{j})$ mm $\qquad \vec{r}_{-q} = (1.4\,\hat{i} - 1.3\,\hat{j})$ mm

a- $\vec{p} = 2\vec{a}\,q$

$2\vec{a} = \Delta\vec{r} = \vec{r}_{-q} - \vec{r}_{+q}$

$= -(1.4\,\hat{i} - 1.3\,\hat{j}) + (-1.2\,\hat{i} + 1.1\,\hat{j}) \times 10^{-3}$ m

$= (-2.6\,\hat{i} + 2.4\,\hat{j}) \times 10^{-3}$ m

$\vec{p} = 3.5\times10^{-9}\,(-2.6\,\hat{i} + 2.4\,\hat{j}) \times 10^{-3}$

$= (-9.1\,\hat{i} + 8.4\,\hat{j}) \times 10^{-12}$ C·m

b- $\vec{\tau} = \vec{p} \times \vec{E} = [(-9.1\,\hat{i} + 8.4\,\hat{j}) \times 10^{-12}] \times [7.8\,\hat{i} - 4.9\,\hat{j}] \times 10^{3}$

$= 10^{-9} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -9.1 & 8.4 & 0 \\ 7.8 & -4.9 & 0 \end{vmatrix} = (44.6\,\hat{k} - 65.5\,\hat{k}) \times 10^{-9}$

$= -2.09\times10^{-8}\,\hat{k}$ N·m

C- $U = -\vec{p}\cdot\vec{E}$

$U = -[(-9.1\,\hat{i} + 8.4\,\hat{j}) \times 10^{-12}] \cdot [(7.8\,\hat{i} - 4.9\,\hat{j}) \times 10^{3}]$

$= (71\times10^{-9} + 41.2\times10^{-9})$

$= 112$ nJ

d- U_{min} ; when the dipole aligned with $\vec{E}$

θ = zero.

$$U_{min} = -\vec{P}\cdot\vec{E} = -PE\cos(0)$$

$$= -\left(\sqrt{(9.1)^2+(8.4)^2}\times10^{-12}\right)\left(\sqrt{(7.8)^2+(4.9)^2}\times10^{3}\right)(1)$$

$$= -114\,n\,J$$

U_{max} ; when $\vec{P}$ is opposite to $\vec{E} \Leftrightarrow \theta = 180^\circ$

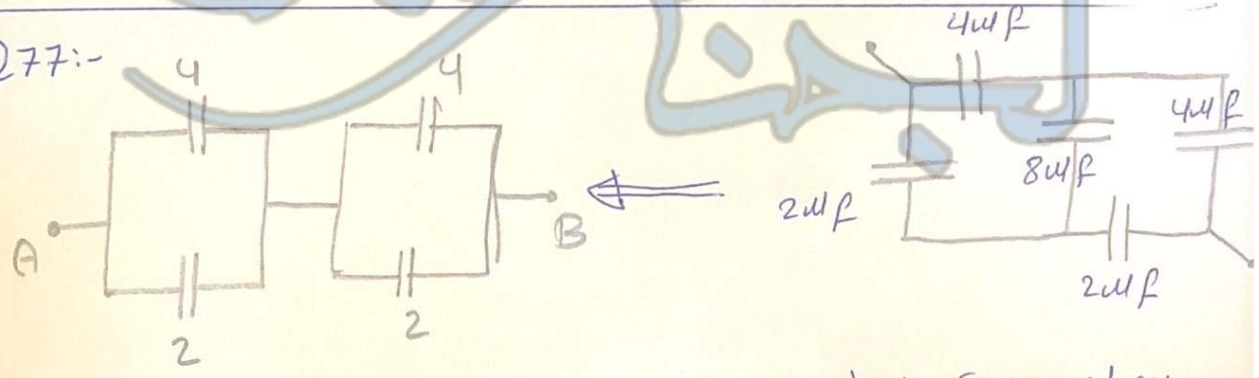
$$U_{max} = -\left[12.4\times10^{-12}\right]\left[9.21\times10^{3}\right](-1)$$

$$= +114\,n\,J$$

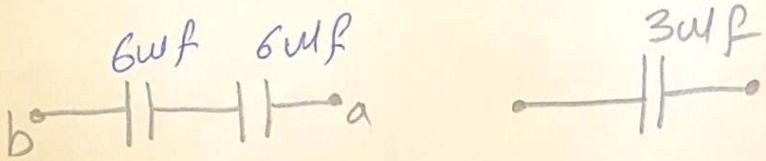
$$\Delta U = U_{max} - U_{min}$$

$$= 114 - -114 = 228\,n\,J$$

Q77:-



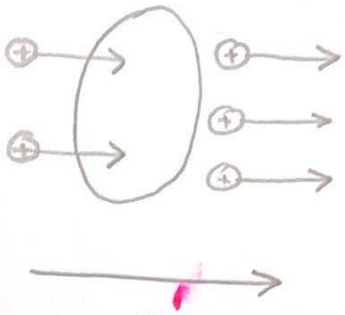
ΔV across 8 μf must be zero by symmetry



chapter (27):- Current and Resistance:-

27.1 Electric current:-

flow of electric charge through a piece of material makes current.



I

Current:- the rate at which charge flows through a surface.

average current : the charge that passes through cross sectional area (A) per unit time.

$$I_{avg} = \frac{\Delta Q}{\Delta t} \equiv [C/s] \equiv \text{ampere (A)}$$

as $\Delta t \to 0$ we define instantanous current

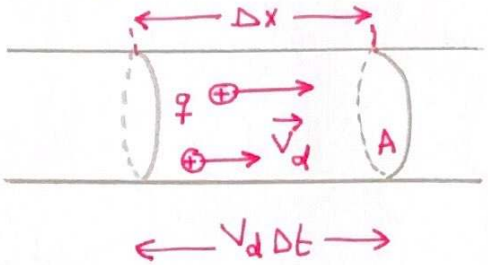
$$I = \frac{dQ}{dt} \equiv [A]$$

the direction of the current is the same direction of +ve charges, and opposite the direction of electrons flow. Applying potential difference across wire creating electric field affects on the electrons

by electric force that moves electrons to make current.

Microscopic Model of Current:-

segment volume $V = A\Delta x$



Charge carrier:- moving charge (positive or negative).

charge carriers density (n) = $\frac{\text{number of charge carriers}}{\text{volume of the segment}}$

the total charge in the segment ΔQ

$\Delta Q = (nA\Delta x)q$, q: is the charge on each carrier.

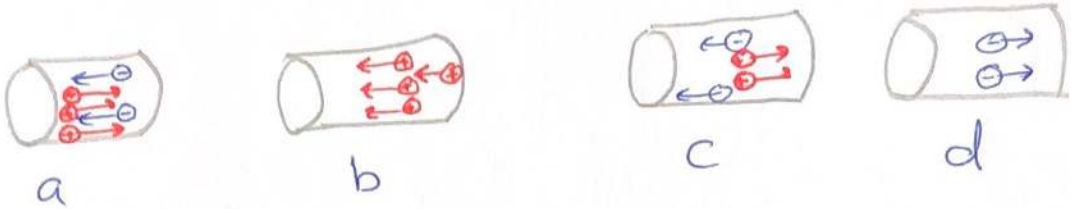
v_d :- velocity of the charge carriers. (drift speed)

$v_d = \frac{\Delta x}{\Delta t} \Rightarrow \Delta x = v_d \Delta t$

$\frac{\Delta Q}{\Delta t} = \frac{(nA v_d \Delta t)q}{\Delta t}$

$I_{avg} = \frac{\Delta Q}{\Delta t} = nA v_d q$

Quick quiz (27.1):- Rank the current from highest to lowest.



a, b and c, d

$a > b = c > d$.

Ex (27.1):- Drift speed in a copper wire:-

$A = 3.31 \times 10^{-6}\ m^2$ $\quad I = 10\ A$

each copper atom contributes one free electron

$\rho_{Cu} = 8.92\ g/cm^3$ $\quad$ molar mass of $Cu = 63.5\ g/mol$

One mole of substance = Avogadro's number of atoms (N_A)

$N_A = 6.02 \times 10^{23}\ mol^{-1}$

Volume of 1 mole of $Cu = \frac{\text{molar mass}}{\text{density}} = V$

electron density in $Cu = \frac{N_A}{V} = \frac{N_A \rho}{M}$

$I_{avg} = n q v_d A \Rightarrow v_d = \frac{I_{avg}}{n q A}$

$$V_d = \frac{IM}{N_A \rho q A} = \frac{10 \times 63.5 \times 10^{-3}}{6.02 \times 10^{23} \times 8.92 \times 10^{3} \times 1.6 \times 10^{-19} \times 3.31 \times 10^{-6}}$$

$$= 2.23 \times 10^{-4} \text{ m/s}$$

27.2 Resistance:-

$\vec{E} = 0$ inside a conductor only when it in electrostatic equalibrium.

current density ($\vec{J}$): is the current per unit area.

$$|\vec{J}| = \frac{I}{A} = \frac{nqv_d A}{A} = nqv_d \equiv [A/m^2]$$

$$\vec{J} \propto \vec{E}$$

$\vec{J} = \sigma \vec{E}$, where σ is the conductivity of the conductor.

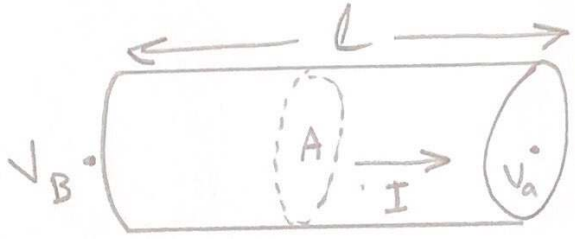
general formula of Ohm's Law, material that obays this formula are called Ohmic materials and those don't obay that formula are nonohmic materials.

$$\Delta V = V_B - V_A$$

$$\Delta V = E L$$

$$J = \sigma E$$

$$J = \frac{\sigma \Delta V}{L} \Rightarrow \Delta V = \frac{L J}{\sigma}$$



$$\Delta V = \frac{L}{\sigma} \frac{I}{A} = \frac{L}{\sigma A} I$$

$$\Delta V = R I \quad , R = \frac{L}{\sigma A}$$

R :- resistance of conductor

$$R = \frac{\Delta V}{I} \equiv [\text{volt}/A] \equiv \text{ohm}(\Omega)$$

$$R = \frac{L}{\sigma A} \Rightarrow \sigma = \frac{L}{R A} \equiv \left[\frac{m}{\Omega \cdot m^2}\right] = (\Omega \cdot m)^{-1}$$

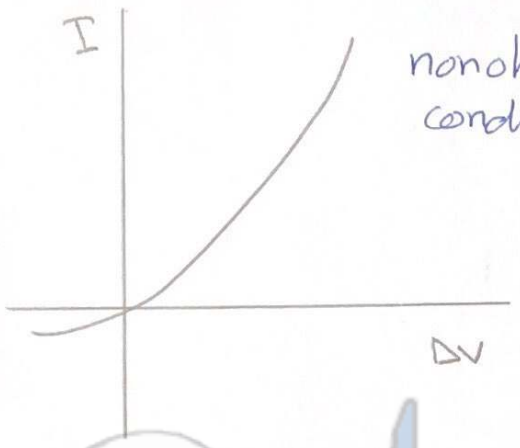
the inverse of conductivity is resistivity (ρ)

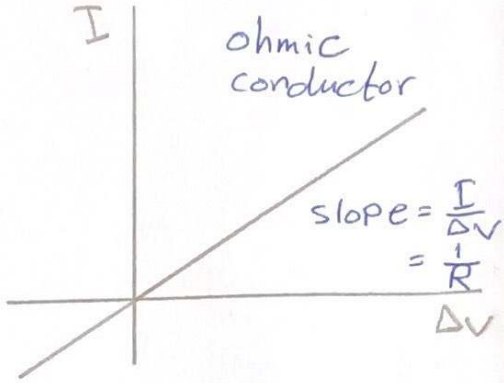
$$\rho = \frac{1}{\sigma} \equiv (\Omega \cdot m)$$

$$R = \frac{L}{\sigma A} = \frac{\rho L}{A}$$

ρ depends on the properties of material and temperature.

$R = \rho \frac{L}{A} \Rightarrow R \propto L \quad , R \propto \frac{1}{A}$





the relationship between ΔV and I is Linear when conductor is ohmic and nonlinear for nonohmic conductors

Quiz (27.2):- a conductor has radius r and length L, if both of them are doubled then the resistance:-

a- increase (b)- deerease c- remain the same.

Quiz (27.3):- for nonohmic conductor if we increase ΔV then R:-

a- increase (b)- decrease c- remain the same.

Ex (27.2):- The Resistance of Nichrome wire:-

$r = 0.32\ mm$ $\qquad \rho = 1 \times 10^{-6}\ \Omega \cdot m$

a- $R = \dfrac{\rho L}{A}$

Resistance per unit length $\dfrac{R}{L} = \dfrac{\rho}{A}$

$A = \pi r^2$

$$\frac{R}{L} = \frac{\rho}{\pi r^2} = \frac{1 \times 10^{-6}}{\pi (0.32 \times 10^{-3})^2} = 3.1\ \Omega/m$$

b- $\Delta V = 10$ volt $\qquad L = 1\ m$

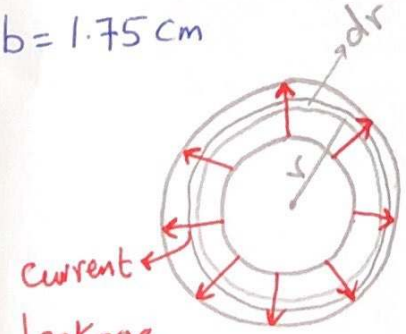
$$\frac{R}{L} = 3.1 \Rightarrow R = L \times 3.1 = 1 \times 3.1 = 3.1\ \Omega$$

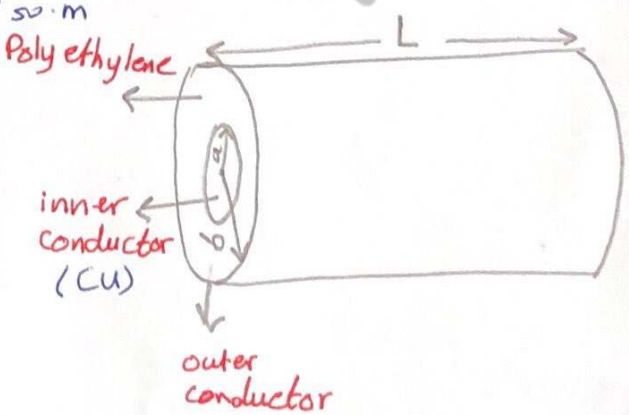
$$\Delta V = IR \Rightarrow I = \frac{\Delta V}{R} = \frac{10}{3.1} = 3.2\ A$$

Ex(27.3):- The radial Resistance of a Coaxial cable:-

$a = 0.5\ cm$ $\quad L = 15\ cm$ $\quad \rho = 1 \times 10^{13}\ \Omega \cdot m$

$b = 1.75\ cm$





the resistance of plastic shell $dR = \frac{\rho\, dr}{A}$

$$dR = \frac{\rho\, dr}{2\pi r L}$$

$$R = \int dR = \frac{\rho}{2\pi L} \int_a^b \frac{dr}{r} = \frac{\rho}{2\pi L} \ln\left(\frac{b}{a}\right)$$

$$R = \frac{1 \times 10^{13}}{(2\pi \times 0.15)} \ln\left(\frac{1.75}{0.5}\right) = 1.33 \times 10^{13}\ \Omega$$

$$R_{Cu} = \frac{\rho L}{A} = \frac{1.7 \times 10^{-8} \times 0.15}{\pi (0.5 \times 10^{-2})^2} = 3.2 \times 10^{-5}\ \Omega$$

$R_{Polyethylene} >>> R_{Cu}$; to prevent current leakage.

27.3 A Model for Electrical Conduction:-

electrical conduction has the following Properties:-

1- array of atoms plus free electrons (conduction electrons) make system of electrical conduction. where conducton electrons bound to their atoms.

2- in the absence of electric field, the conduction electrons move in random directions throug conductor

3- when the electric field is applied, the free electrons drift slowly in the direction that opposite to $\vec{E}$. $v_d \ll v_{avg}$

4- the electronismotion after collision is independent of it's motion before collision. the work done on $\bar{e}$'s by $\vec{E}$ stored as energy that transfered to atoms due to collisions, which increase the internal energy and temperature of the system.

for electron in an electric field, it will experience a force

$$\vec{F} = q\vec{E}$$

$$\vec{a} = \frac{\sum \vec{F}}{m_e} = \frac{q\vec{E}}{m_e} = \text{Constant}$$

$$\vec{v}_f = \vec{v}_i + \vec{a}t = \vec{v}_i + \frac{q\vec{E}}{m_e}t$$

$$\vec{v}_{f\,avg} = \vec{v}_d = 0 + \frac{q\vec{E}}{m_e}\tau$$

where τ is the average time interval between successive collision's. τ depends on n and the atom size.

$$\vec{v}_d = \frac{q\vec{E}}{m_e}\tau$$

$$I_{avg} = nqv_dA$$

$$= nq\left(\frac{qE\tau}{m_e}\right)A$$

$$= \frac{nq^2E}{m_e}\tau A$$

$$J = \frac{I}{A} = \frac{nq^2E}{m_e}\tau \quad \text{and} \quad J = \sigma E$$

$$\Rightarrow \sigma = \frac{nq^2\tau}{m_e} \qquad \rho = \frac{1}{\sigma} = \frac{m_e}{nq^2\tau}$$

$\therefore$ σ and ρ don't depend on the strength of $\vec{E}$.

v_{avg} :- average speed between collisions.

mean free path (ℓ_{avg}) :- average distance between collisions.

$$\tau = \frac{\ell_{avg}}{v_{avg}}$$

27.4 Resistance and Temperature

the resistivity varies with temperature according to :-

$$\rho = \rho_0 [1 + \alpha (T - T_0)]$$

ρ :- resistivity at some temperature T.

ρ_0 :- " " " reference temperature

$(T_0 = 20^\circ C)$

α :- temperature coefficient of resistivity.

$$\Delta\rho = \rho - \rho_0 \qquad \Delta T = T - T_0$$

$$\alpha = \frac{1}{\rho_0} \frac{\Delta\rho}{\Delta T} \equiv {}^\circ C^{-1}$$

R is proportional to ρ

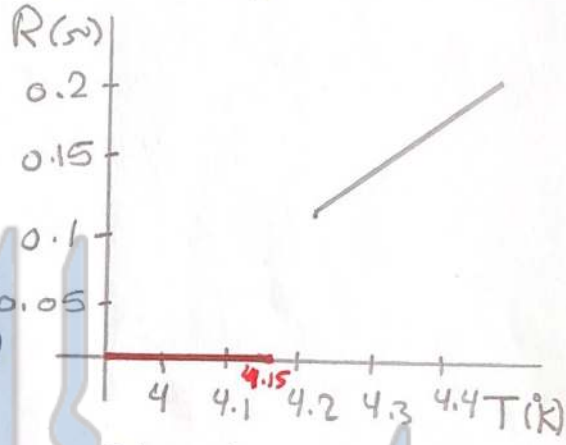
$R = R_0 [1 + \alpha (T - T_0)]$

- at Low temperature the relationship between ρ and T is nonlinear due to collisions of electrons with impurities and imperfections in the metal.
- at higher temperature the relationship between ρ and T is linear where the collisions are between electrons and metal atoms.

27.5 Superconductors:-

Superconductors are:- metals whose resistance decreases to zero when they are below a certain temperature T_c (critical temperature).

for mercury (Hg) $T_c = 4.15\,^{\circ}K$



below that temperature $R = 0$.

because $R = 0$; current is persists without any applied ΔV. One of the important application of superconductors is in the development of super conducting magnets which is used in medical magnetic resonance imaging (MRI).

Ex:- Resistance dependence thermometer made of Platinium, it's resistance was 50Ω at 20°C. if you put it in melt indium the resistance increases to (76.8)Ω, determine indium melting temperature? where $\alpha = 3.92 \times 10^{-3}\ ^{\circ}C^{-1}$

$$R = R_0[1 + \alpha(T - T_0)]$$

$$R = R_0 + R_0\alpha(T - T_0)$$

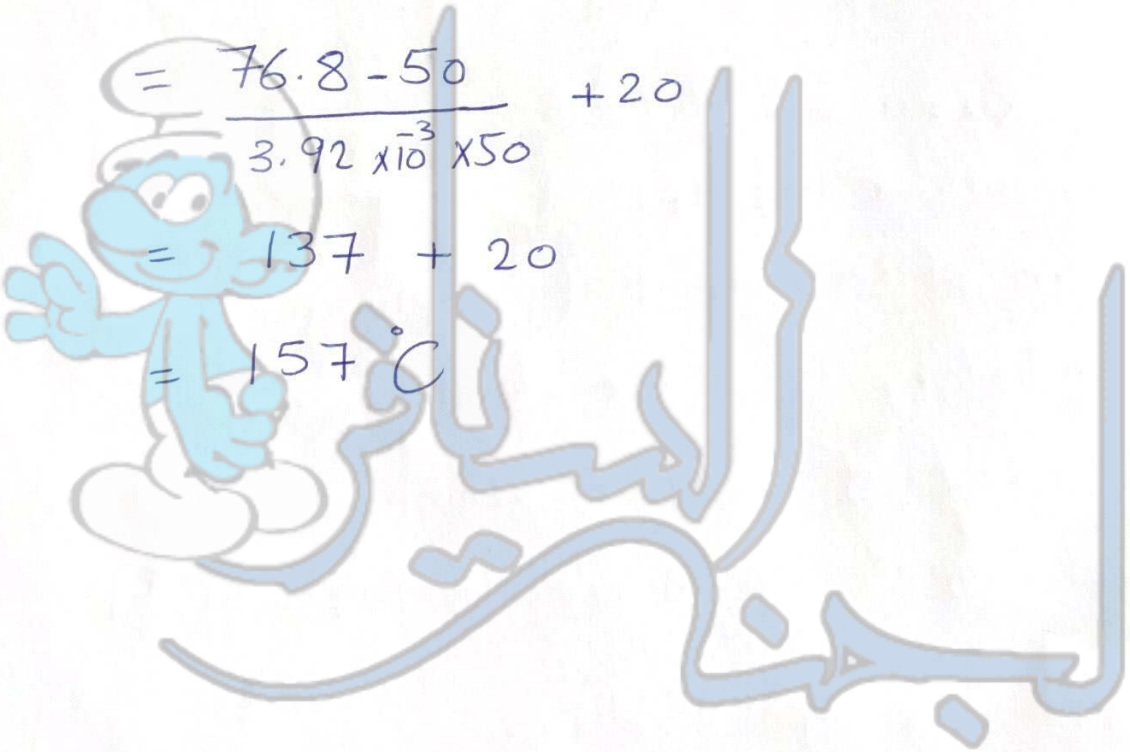
$$R - R_0 = R_0\alpha(T - T_0)$$

$$T = \frac{R - R_0}{R_0\alpha} + T_0$$

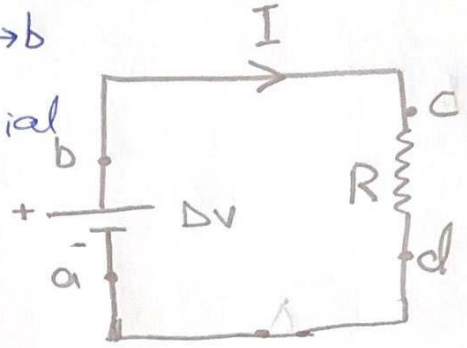
$$= \frac{76.8 - 50}{3.92 \times 10^{-3} \times 50} + 20$$

$$= 137 + 20$$

$$= 157\,^{\circ}C$$



26.7 Electrical Power:-



for charge @ moving from a → b

from low potential to high potential

⇒ electric potential increases

by $U = @\,\Delta V$ and chemical

energy in the battery decreases

by $@\,\Delta V$.

as the charge moves from c to d ⇒ U decreases

because it's transfered to the resistor as heat.

the rate of loses electrical potential energy is

Power (P), where

$$P = \frac{dU}{dt} = \frac{d}{dt}(@\,\Delta V) = I\,\Delta V \equiv [\text{watt (w)}]$$

$$P = I\,\Delta V = I^2 R = \frac{(\Delta V)^2}{R}$$

the process by which power is lost as internal energy in a conductor of resistance R is called Joule heating.

Ex(27.4):- Power in an Electric heater:-

$\Delta V = 120$ volt $\quad R = 8\ \Omega$

a- $I = \frac{\Delta V}{R} = \frac{120}{8} = 15\ A$

b- $P = I^2 R$

$= (15)^2 (8)$

$= 1.8 \times 10^3$ watt

$= 1.8$ Kwatt

Ex(27.5):- Linking electricity and thermodynamics:-

$m_{water} = 1.5\ kg \quad T_1 = 10^\circ C \quad T_2 = 50^\circ C$

$t = 10$ minutes $\quad \Delta V = 110$ volt $\quad c = 4186\ J/kg.^\circ C$

a- Power $= \frac{\text{thermal energy } (Q)}{\text{time}} = \frac{Q}{\Delta t} = \frac{(\Delta V)^2}{R}$

$Q = mc\Delta T$, C is the specific heat for water

$Q = 1.5 \times 4186 \times (50 - 10) = 251160\ J$

$\frac{(\Delta V)^2}{R} = \frac{Q}{\Delta t} \Rightarrow R = \frac{(\Delta V)^2 \Delta t}{Q}$

$R = \frac{(110)^2 \times 10 \times 60}{251160} = 28.9\ \Omega$

b- Energy transferred (T) = $P \Delta t$

$$T = \frac{(\Delta V)^2}{R} \Delta t$$

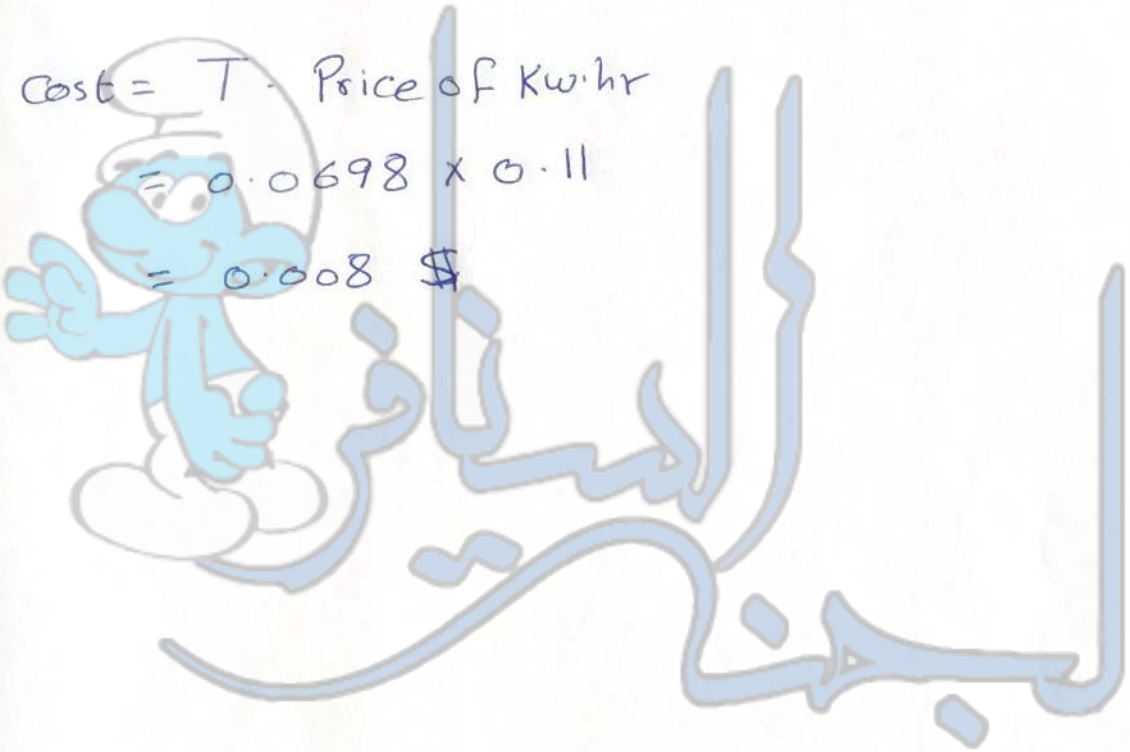
$$= \frac{(110)^2}{28.9} \times \frac{10}{60} = 69.8 \text{ W·hr}$$

$$= 0.0698 \text{ Kw·hr}$$

Cost = T · Price of Kw·hr

$$= 0.0698 \times 0.11$$

$$= 0.008 \$$$



Solved Problem's chapter (27)

Q3:- $A = 4 \times 10^{-6}\ m^2$ $I = 5A$ $\rho_{AL} = 2.7 g/cm^3$

1 free electron per atom of AL

molar mass = 27g/mole $N_A = 6.02 \times 10^{23}$ atom/mole

$$I = n v_d q A \Rightarrow v_d = \frac{I}{n q A}$$

$$n = \frac{N_A}{V} = \frac{N_A \rho}{m}$$

$$= \frac{6.02 \times 10^{23} \times 2.7 g / 10^{-6} m^3}{27 g} = 6.02 \times 10^{28} \text{ atom}/m^3$$

$$v_d = \frac{5}{6.02 \times 10^{28} \times 1.6 \times 10^{-19} \times 4 \times 10^{-6}}$$

$$= 1.3 \times 10^{-4} \ m/s$$

Q5:- $I = 125 \mu A$ $t = 23$ sec

$$I = \frac{\Delta Q}{\Delta t} \quad , \quad \Delta Q = N q_p$$

$$I = \frac{N q_p}{\Delta t} \Rightarrow N = \frac{I \Delta t}{q_p} = \frac{125 \times 10^{-6} \times 23}{1.6 \times 10^{-19}}$$

$$N = 1.8 \times 10^{16} \text{ Proton.}$$

Q_7:- $I(t) = I_o e^{-t/\tau}$

$I = \frac{dQ}{dt} \Rightarrow dQ = I\,dt$

$$Q = \int dQ = \int_0^T I_o e^{-t/\tau}\, dt$$

$$= -I_o \tau\, e^{-t/\tau}\Big]_0^T$$

$$= -I_o \tau\,(e^{-T/\tau} - 1)$$

$$= I_o \tau\,(1 - e^{-T/\tau})$$

a- if $T = \tau$

$Q(\tau) = I_o \tau\,(1 - e^{-1}) = 0.632\, I_o \tau$

b- $T = 10\tau$

$Q(10\tau) = I_o \tau\,(1 - e^{-10}) = 0.99995\, I_o \tau$

c- $T = \infty$

$Q(\infty) = I_o \tau\,(1 - e^{-\infty})$

$= I_o \tau$

Q9:- $q = 4t^3 + 5t + 6$ $\quad A = 2\,cm^2 = 2\times10^{-4}\,m^2$

a- $I = \frac{dq}{dt} = 12t^2 + 5\,|_{t=1}$

$= 17\,A$

b- $J = \frac{I}{A} = \frac{17}{2\times10^{-4}} = 85\times10^3\,A/m^2$

Q16:- $\Delta V = 0.9$ $\quad L = 1.5\,m$ $\quad A = 0.6\,mm^2$ $\quad \rho = 5.6\times10^{-8}\,\Omega\cdot m$

$\Delta V = I R \Rightarrow I = \frac{\Delta V}{R}$

$R = \frac{\rho L}{A} = \frac{5.6\times10^{-8}\times1.5}{0.6\times10^{-6}} = 0.14\,\Omega$

$I = \frac{0.9}{0.14} = 6.43\,A$

Q22:- a- if current doubled (n) is constant; it will not be affected.

b- $|\vec{J}| = \frac{I}{A}$, if I doubled $\Rightarrow$ J doubles.

c- $I = nqv_dA$, if I doubled $\Rightarrow$ v_d doubles

d- $\tau = \frac{m\sigma}{nq^2}$; τ is not changed.

Q23:- $J = 6 \times 10^{-13}\ A/m^2$ $\quad E = 100\ V/m$

$$J = \sigma E \Rightarrow \sigma = \frac{J}{E} = \frac{6 \times 10^{-13}}{100} = 6 \times 10^{-15}\ (\Omega \cdot m)^{-1}$$

Q26:- $R_o = 19\ \Omega$ $\quad T_o = 20\ ^\circ C$ $\quad \alpha = 4.5 \times 10^{-3}\ ^\circ C^{-1}$

$R = 140\ \Omega$ $\quad T = ??$

$$R = R_o[1 + \alpha(T - T_o)]$$

$$140 = 19[1 + 4.5 \times 10^{-3}(T - 20)]$$

$$T = 1.44 \times 10^{3}\ ^\circ C$$

Q33:- $d = 0.1\ mm$ $\quad E = 0.2\ V/m$ $\quad T = 50^\circ C$

$\rho_o = 2.82 \times 10^{-8}\ \Omega \cdot m$ $\quad \alpha = 3.9 \times 10^{-3}\ ^\circ C^{-1}$

a- $\rho = \rho_o[1 + \alpha(T - T_o)]$

$= 2.82 \times 10^{-8}[1 + 3.9 \times 10^{-3}(50 - 20)]$

$= 3.15 \times 10^{-8}\ \Omega \cdot m$

b- $J = \sigma E = \frac{E}{\rho} = \frac{0.2}{3.15 \times 10^{-8}}$

$= 6.35 \times 10^{6}\ A/m^2$

c- $I = JA = J\left(\pi \frac{d^2}{4}\right)$

$= 6.35 \times 10^{6} \left(\pi \times \frac{1 \times 10^{-4}}{4}\right)$

$= 49.9 \times 10^{-3}\ A$

d- $n = \frac{6.02 \times 10^{23}}{26.98\,g / 2.7 \times 10^{6}\,g/m^3}$ ~~$= 49.9 \times 10^{-3}$~~

$= 6.02 \times 10^{28}$ electron/m^3

$V_d = \frac{J}{nq} = \frac{6.35 \times 10^{6}}{6.02 \times 10^{28} \times -1.6 \times 10^{-19}}$

$= -6.59 \times 10^{-4}$ m/s

e- $\Delta V = EL = 0.2 \times 2 = 0.4$ volt

---

② 37:- $P = 1.5\ hp \times 0.8$ $\Delta V = 2000$

$= 1.5 \times 746 \times 0.8$

$= 8.95 \times 10^{5}$ watt

$P = I \Delta V \Rightarrow I = \frac{P}{\Delta V}$

$I = \frac{8.95 \times 10^{5}}{2000} = 448\ A$

Q67:- $L = 0.5m$ $d = 0.2mm$ $\rho = 4\times10^{-8}\ \Omega.m$

$V = 4$ volt at $x=0$ $V=0$ at $x=0.5m$

a- $E_x = -\frac{dV(x)}{dx} = -\frac{\Delta V(x)}{\Delta x}$

$= -\frac{(0-4)}{(0.5-0)} = 8\hat{i}\ V/m$

b- $R = \frac{\rho L}{A} = \frac{4\times10^{-8}\times0.5}{\pi\times(1\times10^{-4})^2} = 0.637\ \Omega$

c- $I = \frac{\Delta V}{R} = \frac{4}{0.637} = 6.28\ A$

d- $J = \frac{I}{A} = \frac{6.28}{\pi(1\times10^{-4})^2} = 2\times10^{8}\ A/m^2$

e- $J = \frac{I}{A}, I = \frac{\Delta V}{R} = \frac{EL}{R} = \frac{EA}{\rho}$

$J = \frac{I}{A} = \frac{EA/\rho}{A} = \frac{E}{\rho}$

$E = \rho J = 4\times10^{-8}\times2\times10^{8}$

$= 8\ V/m$

Chapter (28):-

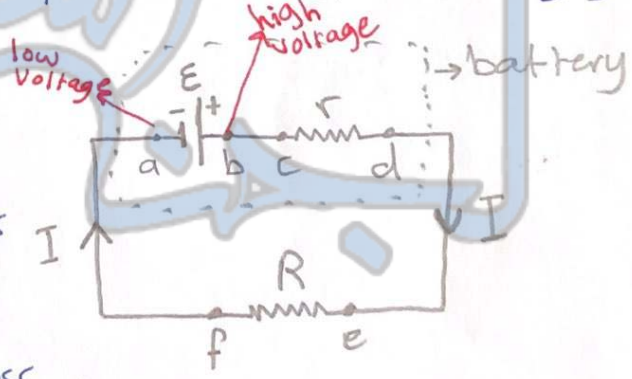
Direct-current Circuits:-

28.1 Electromotive force:-

battery produces a current in a circuit so it's called a source of electromotive force (emf) ($\mathcal{E}$).

as emf is constant the current is constant too which is called direct current (DC).

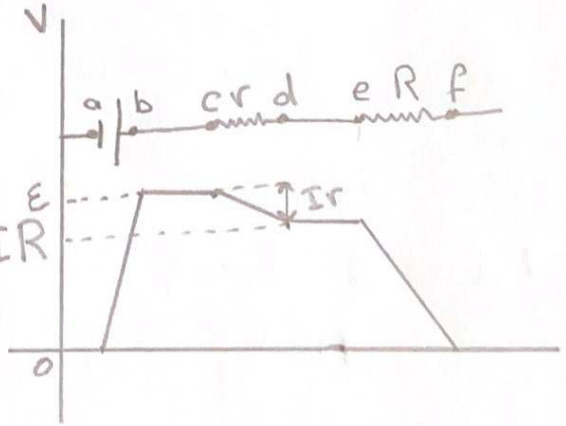
electromotive force ($\mathcal{E}$) is :- the maximum possible voltage the battery can provide between it's terminals.



battery moves charges from Low to high voltage across it's terminals, and from high to Low voltage across the circuit.

r:- internal resistance of the battery.

as the charge moving from a→b Potential difference increases by $\mathcal{E}$.



moving through r reduces the potential difference by Ir. $\Rightarrow$ terminal voltage for the battery (ΔV_{ad})

$$\Delta V_{ad} = \mathcal{E} - Ir$$

R :- is the Load resistance

the potential difference across $R = IR = \Delta V_{ad}$

$$\mathcal{E} - Ir = IR$$

$$\mathcal{E} = Ir + IR$$

$$\boxed{I = \frac{\mathcal{E}}{r+R}}$$

$$\left[\mathcal{E} = Ir + IR\right] * I$$

$$I\mathcal{E} = I^2 r + I^2 R$$

the total power associated by emf ($I\mathcal{E}$) is delivered to the load resistance by I^2R and battery internal resistance by I^2r.

$\Delta V_{ad} = \mathcal{E}$ if, 1- $I=0$, open circuit

2- ideal battery ($r=0$)

Ex(28.1):- Terminal Voltage of a battery:-

emf = 12 V $r = 0.05\ \Omega$ $R = 3\ \Omega$

a- $I = \frac{\mathcal{E}}{r+R} = \frac{12}{3+0.05} = 3.93\ A$

terminal voltage of the battery $\Delta V = \mathcal{E} - Ir$

$\Delta V = 12 - 3.93 \times 0.05$

$= 11.8$ volt

$\Delta V = IR = 3.93 \times 3 = 11.8\ V$

b- $P_R = I^2 R = (3.93)^2 \times 3 = 46.3$ watt

$P_r = I^2 r = (3.93)^2 \times 0.05 = 0.772$ watt

$P = P_R + P_r = 46.3 + 0.772$

$= 47.1$ watt.

what if?? $r = 2\ \Omega$ for old battery

$I = \frac{\mathcal{E}}{r+R} = \frac{12}{3+2} = 2.4\ A$

$\Delta V = \mathcal{E} - Ir = 12 - 2.4 \times 2 = 7.2$ volt

$P_R = I^2 R = 2.4^2 \times 3 = 17.3$ watt

$P_r = I^2 r = 2.4^2 \times 2 = 11.5$ watt

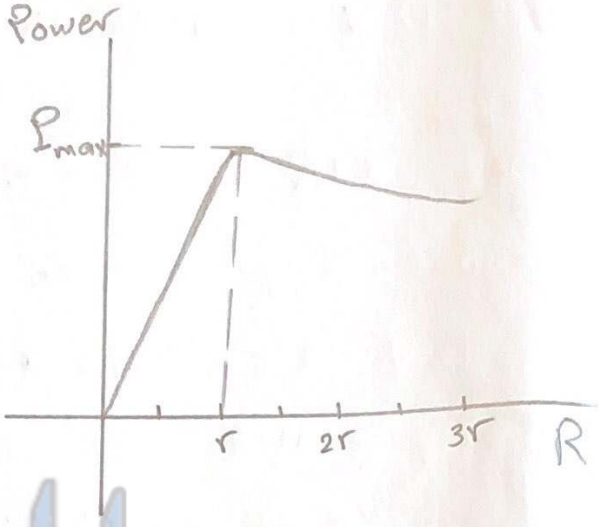
∴ increasing (r) reduces battery ability to deliver energy to R.

Ex(28.2):- Matching the Load:-

$$P_R = I^2 R$$

$$I = \frac{\mathcal{E}}{r+R} \Rightarrow I^2 = \frac{\mathcal{E}^2}{(r+R)^2}$$

$$P_R = \frac{\mathcal{E}^2 R}{(R+r)^2}$$



For maximum power $\frac{dP}{dR} = 0$

$$\frac{d}{dR}\left[\frac{\mathcal{E}^2 R}{(R+r)^2}\right] = 0$$

$$\frac{\mathcal{E}^2(R+r)^2 - 2R\mathcal{E}^2(R+r)}{(R+r)^4} = 0$$

$$\frac{1}{(R+r)^2} - \frac{2R}{(R+r)^3} = 0$$

$$\frac{1}{(R+r)^2} = \frac{2R}{(R+r)^3} \Rightarrow R+r = 2R$$

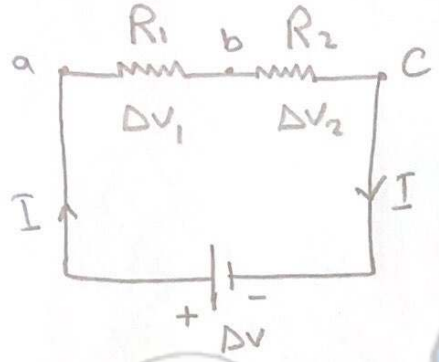
$$\Rightarrow R = r$$

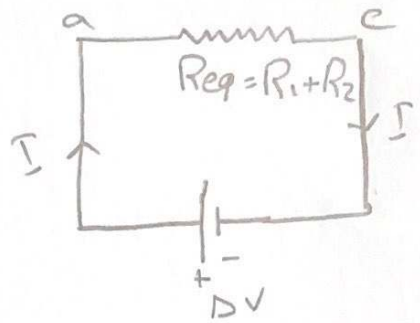
$$P_{max} = \frac{\mathcal{E}^2 r}{(r+r)^2} = \frac{\mathcal{E}^2 r}{4r^2}$$

$$\boxed{P_{max} = \frac{\mathcal{E}^2}{4r}}$$

28.2 Resistors in Series and Parallel:-

1) Series combination:-





$$I = I_1 + I_2$$

$$\Delta V = \Delta V_1 + \Delta V_2$$

$$I R_{eq} = I_1 R_1 + I_2 R_2$$

$$R_{eq} = R_1 + R_2$$

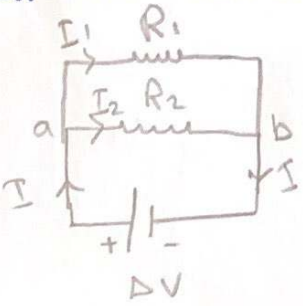
for more than 2 resistors connected in series

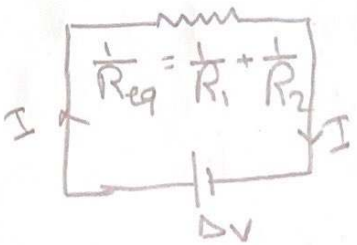
$$R_{eq} = R_1 + R_2 + R_3 + \cdots$$

R_{eq} > individual resistors.

if one resistor of those connected in series creates an open circuit, all resistors are inoperative.

2) Parallel combination





for parallel combination

$$\Delta V = \Delta V_1 = \Delta V_2$$

$$I = I_1 + I_2$$

$$\frac{\Delta V}{R_{eq}} = \frac{\Delta V_1}{R_1} + \frac{\Delta V_2}{R_2}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2}$$

for more than 2 resistance in parallel combination

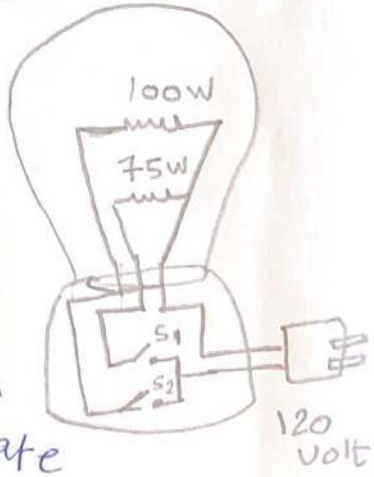
$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots$$

R_{eq} < smallest resistance in the group.

in house hold circuits the appliances are connected in parallel; if one device switched off, the others remain on.

Examples of practical applications of series and Parallel circuits:-

1- this figure shows 3- levels of light intensity, using Lamp with 2- filaments. When S_1 closed only current will operate 75 watt- filament, when S_2 closed only current will operate



100-watt filament and when both switches are closed, the current will operate both filaments with 175 watt Power as they connected in parallel. but if they connected in series and one of them breaks the another one will not work.

2- strings of light that are used in decorating. what if we use parallel or series combination for those strings?

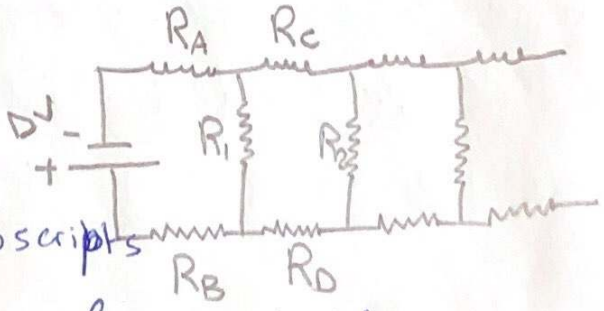
Series combination	Parallel combination
- less operating energy per bulb.	- higher operating energy per bulb.
- less glowing bulbs	- brighter bulbs.
- lower temperature bulbs	- higher temperature bulbs
- more safe inside home	- more dangours.
- if single filament failed all the rest will go out	- single filament failed will not affect the rest.

To use benifits of both ways of combination, anew design was developed (miniature lights in series).

as one bulb breaks it's represent higher resistance, must of ΔV is applied across it. inside each light bulb in this connection a small jumper loop covered by an isulator, when the filament fails an arc burns the insulator and connects the filament leads which is complete the circuits and light the rest of light bulbs. This reduces the R_{eq} which increases current in each bulb and makes the light bulbs glower with higher temperature, which will reduce the lifetime of the bulb. You should ~~reduce~~ change the break bulb to give the rest longer life.

EX (28.3):- land scape Lights:-

the resistance with letter subscripts represent the resistance of wires, and resistors with number subscripts represents the resistance of the light.

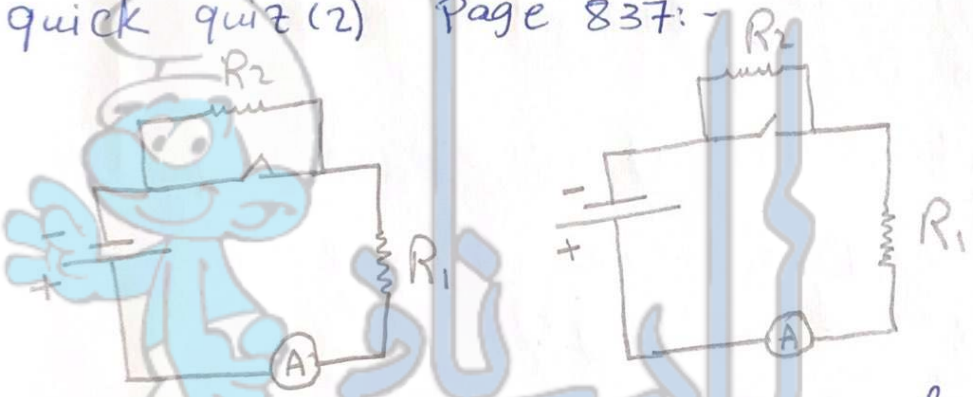


Part of ΔV is dropped due to R_A and R_B, so the voltage across R_1 is less than ΔV, more voltage drop across R_C and R_D $\Rightarrow$ ΔV across R_2 < ΔV across R_1

this make each successive light has a smaller voltage across it and glow less brightly than the one before.

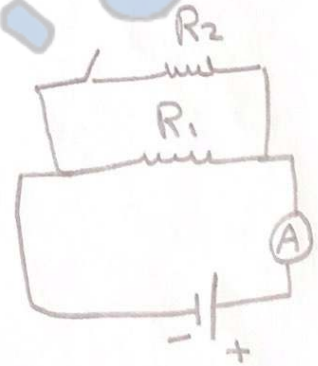
quick quiz (1) Page 834:-
to maximize power that delivered to R, the internal resistance should be (a) as low as Possible

quick quiz (2) Page 837:-



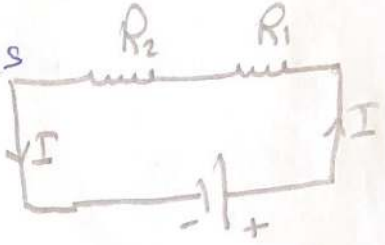
what happens to the reading of the ammeter when switch is opened?
(b)- the reading goes down.

quick quiz (3) Page (840):-
if the switch is closed, what happens to the reading of ammeter?
(a) - the reading increases.



quick quiz (4) page (840):-

i- a third resistor is added in series what happens to the current?

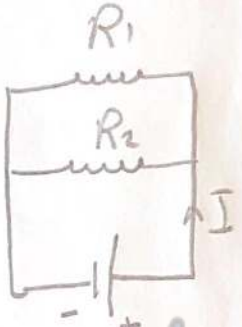


(b)- current decreases

ii- what happens to the terminal voltage?

(a)- terminal voltage increases.

iii- a third resistor is added in parallel, what happens to the current in battery?



(a)- current in battery increases.

iv- what happens to the terminal voltage of battery?

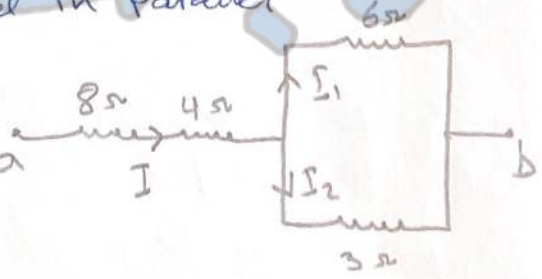
(b)- terminal voltage decreases.

Ex(28.4):- Find the equivalent resistance:-

---

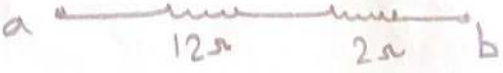
a- 6Ω and 3Ω are connected in parallel

$\frac{1}{R_p} = \frac{1}{6} + \frac{1}{3} \Rightarrow R_p = 2\ \Omega$



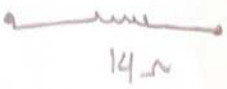
8Ω and 4Ω are connected in series

$R_s = 8 + 4 = 12\ \Omega$



2Ω and 12Ω are in series

$R_{eq} = 2 + 12 = 14\ \Omega$



b - $\Delta V_{ab} = 42$ volt

I passes through 8Ω and 4Ω as they connected in series which is = I passes through $R_{eq} = 14\,\Omega$

$$I = \frac{\Delta V_{ab}}{R_{eq}} = \frac{42}{14} = 3A$$

6Ω and 3Ω are connected in parallel so they have the same voltage difference

$$\Delta V_{6\Omega} = \Delta V_{3\Omega}$$

$$6I_1 = 3I_2 \Rightarrow I_2 = 2I_1$$

but $I = 3A = I_1 + I_2$

$$3 = I_1 + 2I_1 \Rightarrow I_1 = 1A$$

$$I_2 = 2A$$

C- $\Delta V_{8\Omega} = IR = 3 \times 8 = 24$ volt

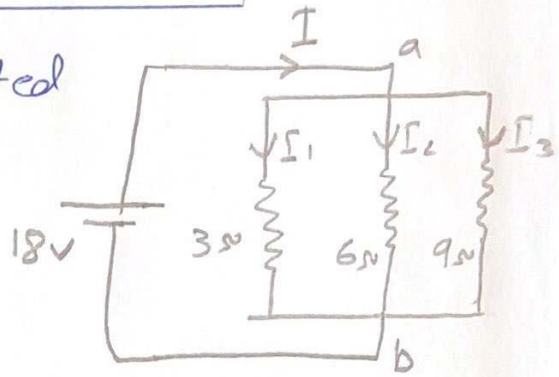
$\Delta V_{4\Omega} = 3 \times 4 = 12$ volt

$\Delta V_{6\Omega} = \Delta V_{3\Omega} = 42 - 24 - 12$

$= 6$ volt

Ex(28.5):- Three resistors in parallel:-

a- 3Ω, 6Ω and 9Ω are connected in parallel



$$\frac{1}{R_{eq}} = \frac{1}{3} + \frac{1}{6} + \frac{1}{9}$$

$$= \frac{6 + 3 + 2}{18}$$

$$R_{eq} = \frac{18}{11} = 1.64\ \Omega$$

b- $I_1 = \frac{\Delta V}{R_1} = \frac{18}{3} = 6A$

$I_2 = \frac{\Delta V}{R_2} = \frac{18}{6} = 3A$

$I_3 = \frac{\Delta V}{R_3} = \frac{18}{9} = 2A$

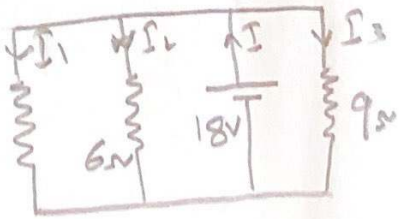
c- $P_1 = I_1^2 R_1 = 6^2 \times 3 = 108$ watt

$P_2 = I_2^2 R_2 = 3^2 \times 6 = 54\ W$

$P_3 = I_3^2 R_3 = 2^2 \times 9 = 36\ W$

$P = P_1 + P_2 + P_3 = 198\ W = \frac{(\Delta V)^2}{R_{eq}} = \frac{18^2}{1.64}$

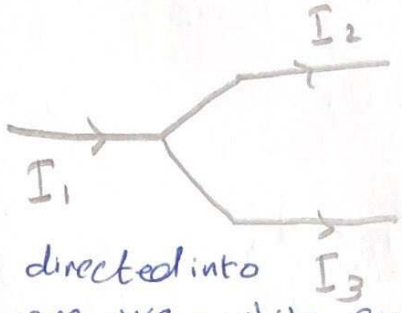
what if ?? there is no effect to the calculations as $\Delta V = 18V$ and resistors are still in parallel.



28.3 Kirchhoff's Rules:-

1- Junction rule:- at any junction, the sum of currents must = zero.

$$\sum_{\text{junction}} I = 0$$



$I_1 - I_2 - I_3 = 0$

$I_1 = I_2 + I_3$

currents directed into junction are +ve, while currents directed out of junction are -ve.

2- Loop rule:- the sum of potential differences across all elements around any closed loop must be zero.

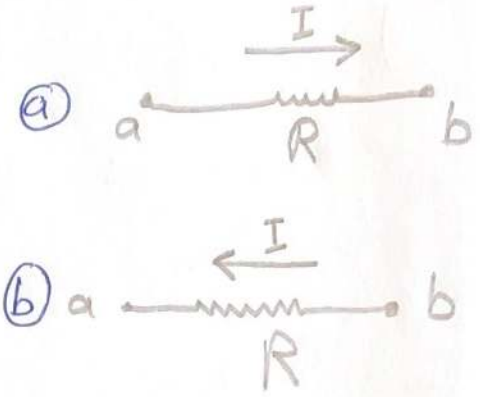
$$\sum_{\text{closed loop}} \Delta V = 0$$

from junction rule:- the electric charge is conserved

$$\sum Q_{in} = \sum Q_{out}$$

from loop rule: the energy is conserved, for moving charge in a closed loop, the sum of increases of energy as a charge passes through elements = the sum of decreases of energy as a charge passes through another elements.

in the figures to the right, the traveling through the circuit elements to the right.



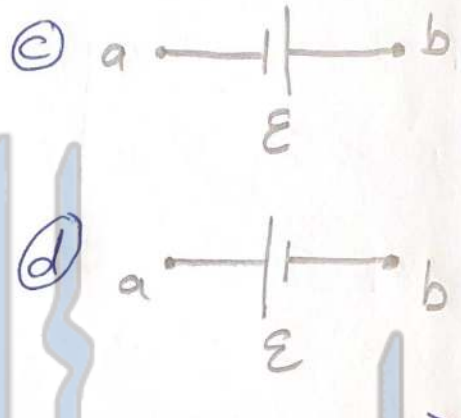
$\Delta V = V_b - V_a$.

ⓐ $\Delta V = -IR$, $V_b < V_a$

ⓑ $\Delta V = IR$, $V_b > V_a$

ⓒ $\Delta V = \mathcal{E}$, $V_b > V_a$

ⓓ $\Delta V = -\mathcal{E}$, $V_b < V_a$



Problem-solving strategy:-

1- determine all elements in the circuit, identify the polarity of each battery and imagine the directions of current in each battery.

2- reduce the circuit by means of combination in parallel and series.

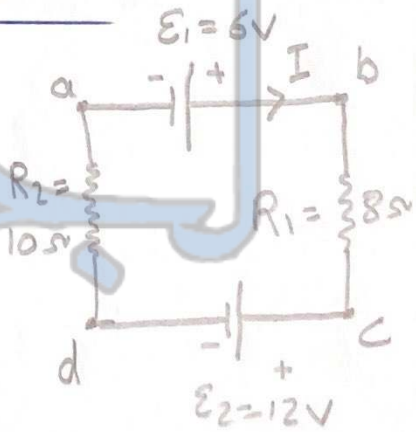
3- assign directions to the current in each Part of the circuit.

4- apply the junction rule to the junction in the circuit, and apply the loop rule to all loops in the circuit after choosing the direction of travel around the loops.

5 - solve equations for unknown quantities.

6- if any of the resulting currents have a -ve value, that means you have guessed incorrect direction for the current but the value is correct.

Ex(28.6):- A single-loop circuit:-



$\sum \Delta V = 0$

closed loop

$\mathcal{E}_1 - I R_1 - \mathcal{E}_2 - I R_2 = 0$

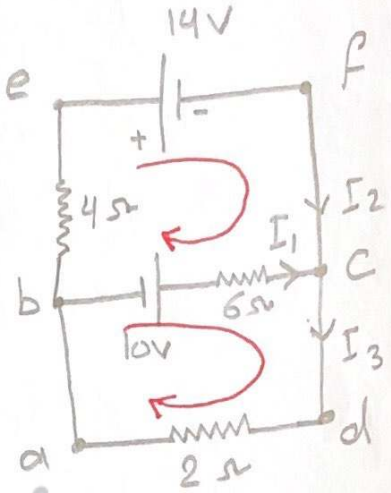
$6 - 8I - 12 - 10I = 0$

$-6 - 18I = 0$

$I = -0.33\,A$, the direction of the current is opposite to the assumed one.

Ex(28.7):- A multiloop Circuit:-

$\sum_{\text{junction c}} I = 0$



$I_1 + I_2 - I_3 = 0$ ----①

$\sum \Delta V = 0$ for closed loop (a b c d a):-

$10 - 6I_1 - 2I_3 = 0$ ---②

for closed loop (b e f c b):-

$-4I_2 - 14 + 6I_1 - 10 = 0$

$-24 + 6I_1 - 4I_2 = 0$ ---③

from eq ① $I_3 = I_1 + I_2$, substitute I_3 into eq (2)

$10 - 6I_1 - 2(I_1 + I_2) = 0$

$10 - 8I_1 - 2I_2 = 0$ ---②'

multiply eq(3) by 4 and eq(2)' by 3 then add the results

$$\begin{array}{r} -96 + 24I_1 - 16I_2 = 0 \\ + \quad 30 - 24I_1 + 6I_2 = 0 \\ \hline -66 - 22I_2 = 0 \end{array}$$

$\Rightarrow I_2 = -3\,A$ (in opposite dir)

Put I_2 in eq (3) $\Rightarrow$

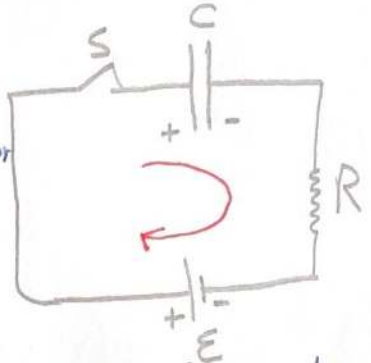
$-24 + 6I_1 + 12 = 0 \Rightarrow I_1 = 2\,A$

$I_3 = I_1 + I_2 = 2 - 3 = -1\,A$ (in opposite dir)

28.4 RC Circuits:-

a circuit containing a series combination of a resistor and a capacitor is called an RC circuit.

1- Charging a capacitor:-



when S is open, Q on the capacitor equall zero, closing S makes C starts to charge increasing it's Potential difference until C completely full charged it's potential difference = $\mathcal{E}$ then current in R becomes zero.

$$\mathcal{E} - \frac{q}{C} - IR = 0$$

at $t = 0$, $q = 0 \Rightarrow \mathcal{E} = IR$

$$I_{o_{max}} = \frac{\mathcal{E}}{R}$$ (max current when $t = 0$)

when the capacitor is charged to the max Q the current is zero

$$\mathcal{E} - \frac{q}{C} = 0 \Rightarrow q_{max} = C\mathcal{E}$$

to determine analytical expressions for time dependence of the charge and current we solve eq $\mathcal{E} = \frac{q}{C} + IR$

$$IR = \varepsilon - \frac{q}{C}$$

$$I = \frac{\varepsilon}{R} - \frac{q}{RC}$$

$$\frac{dq}{dt} = \frac{\varepsilon}{R} - \frac{q}{RC}$$

$$= \frac{\varepsilon C - q}{RC} = -\frac{(q - \varepsilon C)}{RC}$$

$$\frac{dq}{q - \varepsilon C} = -\frac{dt}{RC}$$

$$\int_0^q \frac{dq}{q - \varepsilon C} = -\int_0^t \frac{dt}{RC}$$

$$\ln\left(\frac{q - \varepsilon C}{-\varepsilon C}\right) = -\frac{t}{RC}$$

$$\frac{q - \varepsilon C}{-\varepsilon C} = e^{-t/RC}$$

$$q - \varepsilon C = -\varepsilon C\, e^{-t/RC}$$

$$\boxed{q(t) = \varepsilon C\left[1 - e^{-t/RC}\right]}$$

Charging current $I(t) = \frac{dq}{dt}$

$$I(t) = \frac{Q}{RC} e^{-t/RC}$$

$$= \frac{\mathcal{E}}{R} e^{-t/RC}$$

$$\boxed{I(t) = I_0 e^{-t/RC}}$$

the quantity RC is called the time constant (τ)

$$\tau = RC = \left[\frac{\Delta V}{I} \cdot \frac{Q}{\Delta V}\right] = \left[\frac{Q}{\frac{Q}{t}}\right] = [\text{time}] \equiv [\text{sec}]$$

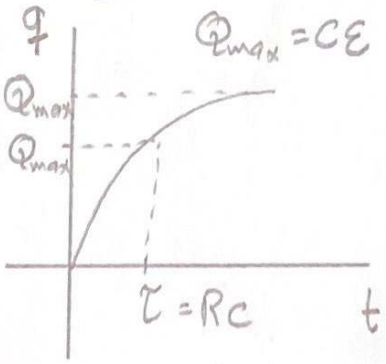
at $t = \tau$

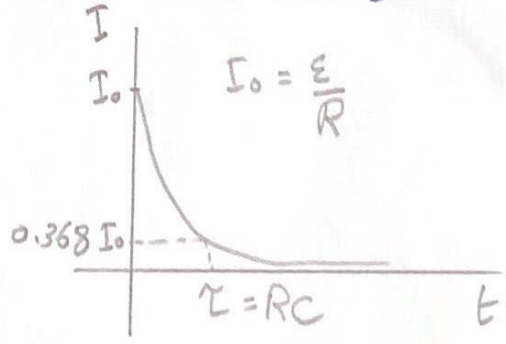
$$q(\tau) = Q_{max}\left(1 - e^{-1}\right) = 0.632\, Q_{max} = 0.632\, C\mathcal{E}$$

$$I(\tau) = I_0 e^{-1} = 0.368\, I_0 = 0.368\, \frac{\mathcal{E}}{R}$$

τ represent the time interval during which current decreasing to e^{-1} of it's initial value.

after τ passed, the current is 0.632 Q_{max} 36.8 % of I_0 and the charge is 36.2 of Q_{max}.





the energy of battery during charging time

$$E = Q_{max}\,\mathcal{E} = C\mathcal{E}^2$$

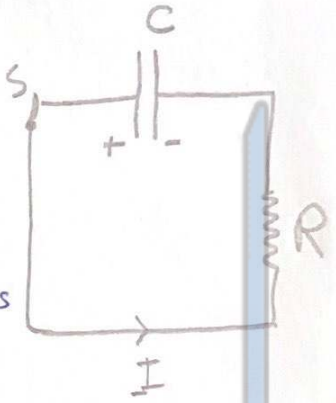
half of this energy is stored in the capacitor and the remaining half stored in the resistance as internal energy.

$$E_C = \frac{1}{2} Q_{max}\,\mathcal{E} = \frac{1}{2} C\mathcal{E}^2 = E_R$$

2- Discharging a capacitor:-

for charged C , $\Delta V_c = \frac{Q_{max}}{C}$

$\Delta V_R = 0$ because $I = 0$



if S closed, at $t=0$, Q becomes discharges through R

$$-\frac{q}{C} - IR = 0$$

$$-R\frac{dq}{dt} = \frac{q}{C}$$

$$\frac{dq}{q} = -\frac{dt}{RC}$$

$$\int_{Q_{max}}^{q} \frac{dq}{q} = \frac{-1}{RC}\int_0^t dt$$

$$\ln\left(\frac{q}{Q_{max}}\right) = -\frac{t}{RC}$$

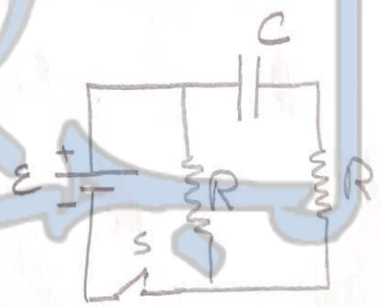
$$\frac{q(t)}{Q_{max}} = e^{-t/RC}$$

$$\boxed{q(t) = Q_{max}\, e^{-t/RC}}$$

$$I = \frac{dq}{dt} \Rightarrow I(t) = \frac{-Q}{RC}\, e^{-t/RC}$$

$$\boxed{I(t) = -I_o\, e^{-t/RC}}$$ the -ve sign indicates that the direction of charging current is opposite to the direction of discharging current.

Quick quiz (5) Page 849 :-



i- just after S is closed.

$$I = \frac{\varepsilon}{R_{eq}} \quad , \frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{R}$$

$$= \frac{2}{R}$$

$$R_{eq} = \frac{R}{2} \Rightarrow I_o = \frac{\varepsilon}{\frac{R}{2}} = \frac{2\varepsilon}{R} \quad (c)$$

ii- after very long time, C is full charge

$$I = \frac{\varepsilon}{R} \quad (d)$$

Ex(28.9):- Charging a capacitor in an RC circuit

$\mathcal{E} = 12\ V \qquad C = 5\ \mu f \qquad R = 8\times10^{5}\ \Omega$

a- $\tau = RC = 8\times10^{5}\times5\times10^{-6} = 4\ sec$

b- $Q_{max} = C\mathcal{E} = 5\times10^{-6}\times12 = 60\ \mu C$

c- $I_{max} = I_o = \frac{\mathcal{E}}{R} = \frac{12}{8\times10^{5}} = 15\ \mu A$

d- $q(t) = Q_{max}(1-e^{-t/\tau})$

$= 60\times10^{-6}(1-e^{-t/4})$

$I(t) = I_o e^{-t/\tau}$

$= 15\times10^{-6}\ e^{-t/4}$

Ex(28.10):- Discharging a capacitor in an RC circuit

a- $q(t) = \frac{Q_{max}}{4}$

$\frac{Q_{max}}{4} = Q_{max}\, e^{-t/RC}$

$\frac{1}{4} = e^{-t/\tau}$

$\ln(\frac{1}{4}) = \frac{-t}{\tau}$

$-\ln 4 = -\frac{t}{\tau}$

$t = \tau \ln 4 = 1.39\,\tau$

b- $U(t) = \frac{q^2(t)}{2C}$

$= \frac{Q_{max}^2}{2C} e^{-2t/\tau}$

$U(t) = \frac{1}{4} \frac{Q_{max}^2}{2C} = \frac{Q_{max}^2}{2C} e^{-2t/\tau}$

$\ln(\frac{1}{4}) = \frac{-2t}{\tau}$

$-\ln 4 = \frac{-2t}{\tau} \Rightarrow t = \frac{\tau}{2} \ln 4$

$t = 0.693\,\tau$

what if ?? half life $t_{1/2}$: is the time required for the charge $q(t)$ to become $\frac{1}{2} Q_{max}$

$\frac{Q_{max}}{2} = Q_{max} e^{-t_{1/2}/\tau}$

$-\ln 2 = -\frac{t_{1/2}}{\tau}$

$t_{1/2} = \tau \ln 2 = 0.693\,\tau$

Ex(28.11):- Energy delivered to a resistor :-

$C = 5\,\mu f \qquad \Delta V = 800 V$

$E_R = \frac{1}{2} C \mathcal{E}^2$

$= \frac{1}{2} \times 5 \times 10^{-6} \times (800)^2$

$= 1.6\ J$

Solved Problems chapter (28):-

1 $\mathcal{E} = 15$ Volt $\quad \Delta V = 11.6$ Volt $\quad P_R = 20$ watt

a- $P_R = I\Delta V = \frac{(\Delta V)^2}{R}$

$R = \frac{(\Delta V)^2}{P_R} = \frac{(11.6)^2}{20} = 6.73\ \Omega$

b- $\mathcal{E} = IR + Ir$

$I = \frac{\Delta V}{R} = \frac{11.6}{6.73} = 1.72\ A$

$r = \frac{\mathcal{E} - IR}{I} = \frac{15 - (1.72)(6.73)}{1.72}$

$= 1.97\ \Omega$

5

$P_{max} = 25$ watt

$P_{max} = I_{max}^2 R \Rightarrow I_{max} = \sqrt{\frac{P_{max}}{R}}$

$I_{max} = \sqrt{\frac{25}{100}} = 0.5\ A$

$\frac{1}{R_p} = \frac{1}{100} + \frac{1}{100} \Rightarrow R_p = 50\ \Omega$

$R_{eq} = 100 + 50 = 150\ \Omega$

$\Delta V_{max} = R_{eq} I_{max}$

$= 150 \times 0.5$

$= 75$ volt

b- $P_{series} = I^2 R = 0.25 \times 100$

$= 25$ watt

$P_{parallel} = (0.25)^2 \times 100$

$= 6.25$ watt for each of the parallel resistors.

c- $P_{total} = 25 + 6.25 + 6.25$

$= 37.5$ watt

$P_{total} = I^2 R_{eq}$

$= (0.5)^2 (150) = 37.5$ watt

[9] a- $R_s = 5 + 20 = 25\ \Omega$

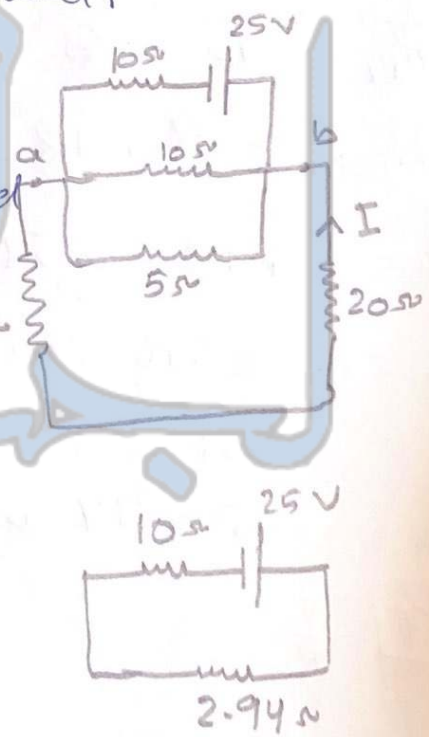
10 Ω, 5 Ω and 25 Ω in Parallel

$$\frac{1}{R_p} = \frac{1}{10} + \frac{1}{5} + \frac{1}{25}$$

$$= \frac{5 + 10 + 2}{50}$$

$$R_{eq} = \frac{50}{17} = 2.94\ \Omega$$

$$I = \frac{\Delta V}{R} = \frac{25}{(10 + 2.94)} = 1.93\ A$$

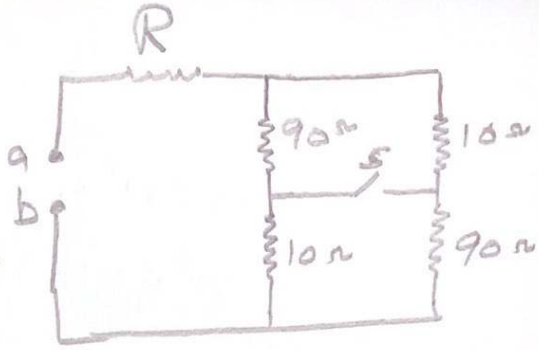


b- $\Delta V_{ab} = I R_{eq}$

$= 1.93 \times 2.94$

$= 5.68$ Volt

14 a- R_{tot} without closing S



$R_{S1} = 10 + 90 = 100\,\Omega$

$R_{S2} = 10 + 90 = 100\,\Omega$

$\frac{1}{R_p} = \frac{1}{100} + \frac{1}{100} \Rightarrow R_p = 50\,\Omega$

$R_{tot} = R + 50\,\Omega$

Closing the switch S

$\frac{1}{R_{P1}} = \frac{1}{10} + \frac{1}{90} \Rightarrow R_{P1} = 9\,\Omega$

$\frac{1}{R_{P2}} = \frac{1}{10} + \frac{1}{90} \Rightarrow R_{P2} = 9\,\Omega$

$R_{tot} = R + 18\,\Omega$

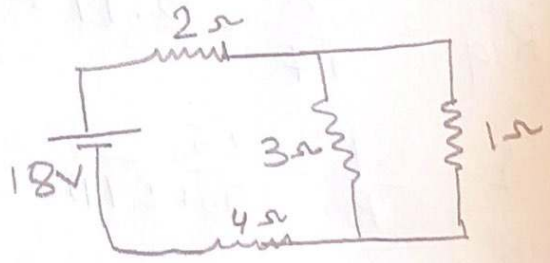
b- $R + 18 = 0.5(R + 50)$

$R - 0.5R = 25 - 18$

$0.5R = 7$

$R = 14\,\Omega$

19 $\frac{1}{R_p} = \frac{1}{3} + \frac{1}{1}$



$R_p = 0.75\,\Omega$

$R_{eq} = 2 + 0.75 + 4 = 6.75\,\Omega$

$I = \frac{\Delta V}{R_{eq}} = \frac{18}{6.75} = 2.67\,A$

$P_{2\Omega} = I^2R = (2.67)^2 \times 2 = 14.2$ watt

$P_{4\Omega} = I^2R = (2.67)^2 \times 4 = 28.4$ watt

$\Delta V_{Parallel} = IR = 2.67 \times 0.75 = 2$ Volt

$I_{3\Omega} = \frac{\Delta V}{R} = \frac{2}{3} = 0.66A$

$P_{3\Omega} = I\Delta V = 0.66 \times 2 = 1.33$ watt

$I_{1\Omega} = \frac{2}{1} = 2A$

$P_{1\Omega} = 2 \times 2 = 4$ watt

24 a- $I_1 = I_2 + I_3$ ---①

$12 - 2I_3 - 4I_1 = 0$ ---②

$8 - 6I_2 + 2I_3 = 0$ ---③

$12 - 2I_3 - 4I_2 - 4I_3 = 0$

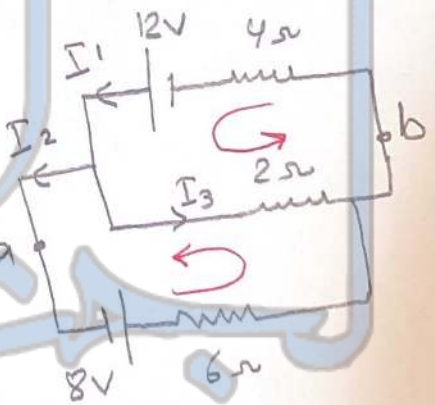
$12 - 6I_3 - 4I_2 = 0$ ---④

$8 - 6I_2 + 2I_3 = 0$

$36 - 22I_2 = 0 \Rightarrow I_2 = 1.63\ A$

$8 - 6(1.63) + 2I_3 = 0 \Rightarrow I_3 = 0.909\ A$

$I_1 = 1.63 + 0.909 = 2.539\ A$

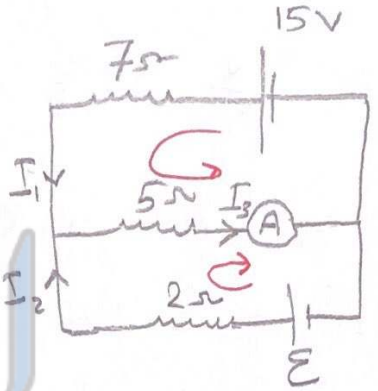


b- $V_a - 0.909 \times 2 = V_b$

$V_b - V_a = -1.82\ V$

$V_a - V_b = +1.82\ V$

29 $I_3 = 2A$



a- for upper loop

$\sum \Delta V = 0$

$15 - 7I_1 - 5 \times 2 = 0$

$I_1 = 0.714\ A$

b- $I_1 + I_2 = 2$

$I_2 = 2 - 0.714 = 1.29\ A$

c- $\varepsilon - 2I_2 - 5 \times 2 = 0$ from lower loop

$\varepsilon = 2 + 2 \times 1.29 + 10 = 12.6$ Volt

39 $C = 2nf$ $Q_o = 5.1\ \mu C$ $R = 1.3\ K\Omega$

$t = 8\ \mu s$

a- $I(t) = -I_o e^{-t/RC}$

$I_o = \frac{Q}{RC} = \frac{5.1 \times 10^{-6}}{1.3 \times 10^{3} \times 2 \times 10^{-9}} = 1.96\ A$

$$I(t) = -1.96\, e^{-t/RC}$$

$$= -1.96 \exp\left[\frac{-8\times10^{-6}}{1300\times2\times10^{-9}}\right]$$

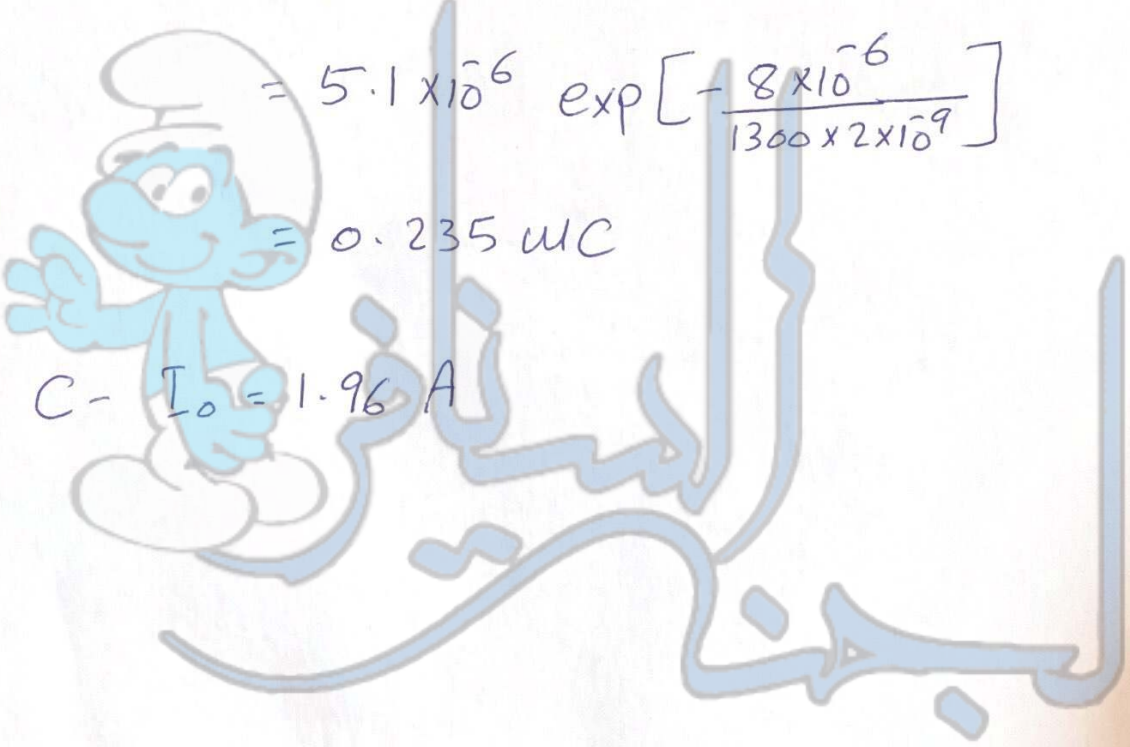
$$= -61.6 \text{ mA}$$

b - $q(t) = Q_0\, e^{-t/RC}$

$$= 5.1\times10^{-6} \exp\left[-\frac{8\times10^{-6}}{1300\times2\times10^{-9}}\right]$$

$$= 0.235\ \mu C$$

C - $I_0 = 1.96$ A

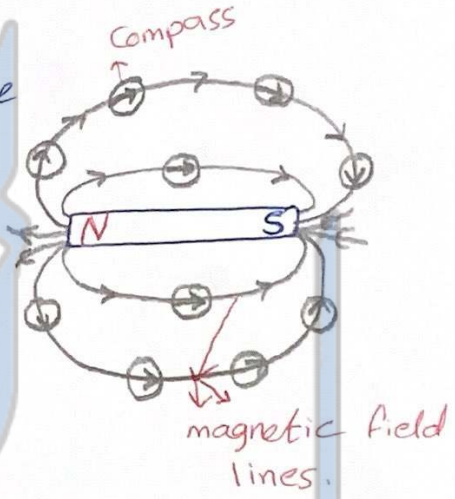


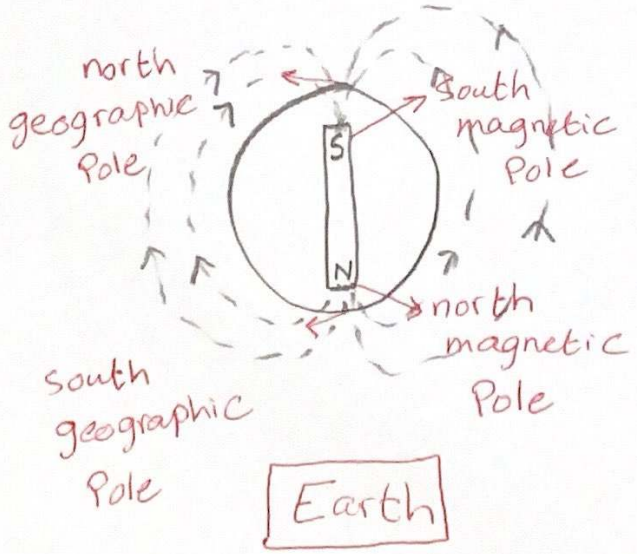
Chapter (29)
Magnetic Fields

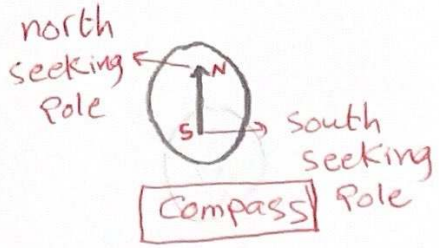
29.1 Analysis Model:- Particle in a magnetic field.

Any moving charge surrounding by magnetic field and magnetic substance making up magnetic field.

- the compass needles directed to the direction of magnetic field lines.
- the magnetic field lines directed out from north pole toward South Pole.







for moving charge (q) in a magnetic field ($\vec{B}$), it's affected by magnetic force ($\vec{F}_B$), where :-

1- the magnetic force is proportional to the charge (q)

2- " ~ ~ on a negative charge is directed opposite to the force on a positive charge moving in the same direction.

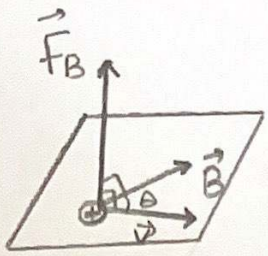
3 - the magnetic force is proportional to the magnetic field ($\vec{B}$).

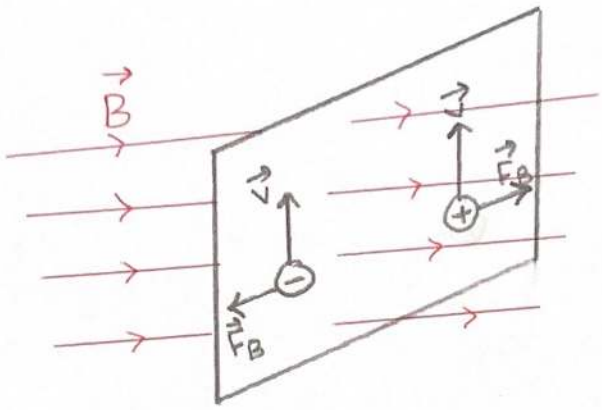
4- the magnetic force is propotional to the speed of Particle ($\vec{v}$).

5 - $\vec{v}$ makes angle θ with $\vec{B}$, $\vec{F}_B$ is propotional to $\sin\theta$.

6 - if q moves parallel to $\vec{B}$ then $\vec{F}_B$ = zero.

7 - $\vec{F}_B$ acts in a direction perpendicular to both $\vec{v}$ and $\vec{B}$; $\vec{F}_B$ ⊥ to the plane formed by $\vec{v}$ and $\vec{B}$.

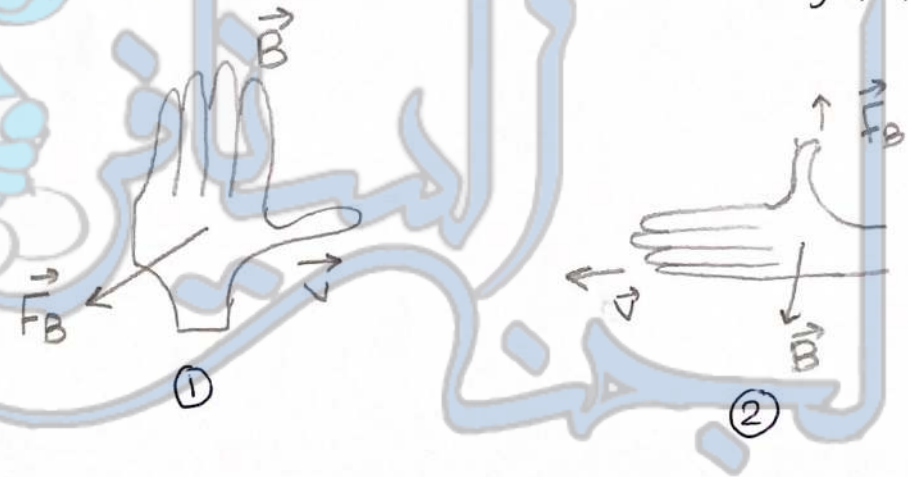




$\vec{F}_B = q\vec{v} \times \vec{B}$

$F_B = |q| vB \sin\theta$ (magnitude of $\vec{F}_B$)

to determine direction of $\vec{F}_B$ we use right hand rules :-



when $\vec{v}$ is parallel or antiparallel to $\vec{B}$, $\underline{\theta = 0}$ or $\underline{\theta = 180^\circ}$, $\vec{F}_B = 0$. when $\vec{v}$ is perpendicular to B, $\underline{\theta = 90}$, $\vec{F}_B$ is maximum.

Comparison between Electric and magnetic forces for charged particle moving in a field ($\vec{E}$ and $\vec{B}$)

1- $\vec{F}_E$ is along the direction of $\vec{E}$ while $\vec{F}_B$ is Perpendicular to $\vec{B}$.

2- $\vec{F}_E$ acts on (q) regardless if (q) is moving but $\vec{F}_B$ acts only on moving charge (q).

3- $\vec{F}_E$ does work in displacing a charged particle, $\vec{F}_B$ does not work in displaying a charged particle in steady magnetic field because $\vec{F}_B$ is $\perp$ to the displacement of (q).

$$F_B = qvB\sin\theta$$

$$[B] = \left[\frac{F_B}{qv}\right] = \left[\frac{N}{C.m/s}\right]$$

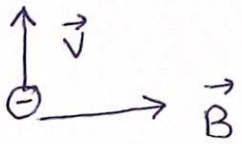
$$[B] = \left[\frac{N}{A.m}\right] = \text{tesla } (T) \quad (SI \text{ units})$$

$$[B] = \text{gauss } (G) \quad (\text{non-SI units})$$

$1\ T = 10^4$ gauss.

Quick quiz (29.1), Page (873):-

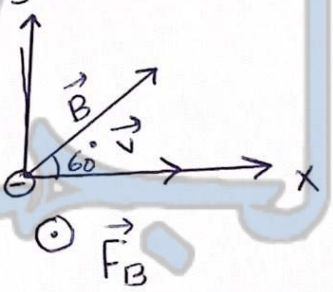
what is the direction of magnetic force $\vec{F}_B$ on the electron? ⓔ upward out of the page.



Ex(29.1):- An electron moving in a magnetic field:-

$q_e = -1.6 \times 10^{-19}\ C$ $\quad \vec{v} = 8 \times 10^6\ \hat{i}\ m/s$

$\vec{B} = 0.025\ T, \theta = 60^\circ$ with x-axis.



$|\vec{F}_B| = |q| v B \sin\theta$

$= 1.6 \times 10^{-19} \times 8 \times 10^6 \times 0.025 \times \sin 60$

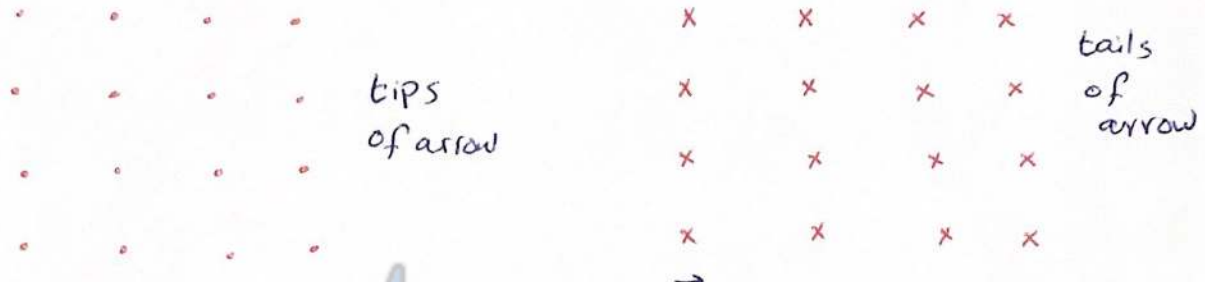
$= 2.8 \times 10^{-14}\ N$

$\vec{F}_B = 2.8 \times 10^{-14}\ \hat{k}\ N$

⊙ :- denotes the direction of vector out of the Page (+ Z-axis)

⊗ :- denotes the direction of vector inward the Page (- Z)

29.2 motion of a charged particle in a uniform magnetic field:-



$\vec{B}_{out}$:- magnetic field lines coming out of Paper.

$\vec{B}_{in}$:- magnetic field lines going into the Paper.

$\vec{F}_B \perp \vec{v}, \vec{B}$

as $\vec{F}_B$ always $\perp$ to $\vec{v}$ and $\vec{B}$; the path of a particle is a circle.

$|\vec{F}_B|$ is constant $= qvB$

for +ve charge the rotation is counterclockwise when $\vec{B}$ is directed into the page.

For -ve charge the rotation is clockwise when $\vec{B}$ is directed outward from the page.

$$\Sigma F = F_B = ma$$

$$F_B = qvB = \frac{mv^2}{r}$$

$$\boxed{r = \frac{mv}{qB}}$$ radius of circular path.

$$r \propto (p = mv) \quad , \quad r \propto \frac{1}{qB}$$

w :- angular speed or cyclotron frequency.

$$w = \frac{v}{r} = \frac{v}{\frac{mv}{qB}}$$

$$\boxed{w = \frac{qB}{m}} \equiv [\text{rad/sec}]$$

Period of the motion (T) :- time interval the particle requires to complete one revolution.

$$T = \frac{2\pi r}{V} = \frac{2\pi}{\omega}$$

$$T = \frac{2\pi}{\frac{qB}{m}}$$

$$\boxed{T = \frac{2\pi m}{qB}}$$

(W) and (T) don't depend on (V) and (r)

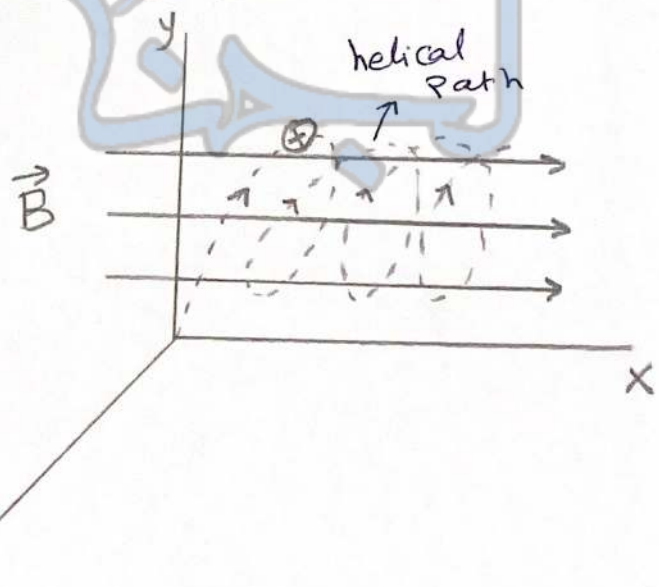
if q moves in uniform $\vec{B}$ with velocity $\vec{v}$, where $\vec{v}$ makes an angle with respect to $\vec{B}$, its path is a helix.

$\vec{B} = B\hat{i}$

$F_x = 0 \Rightarrow a_x = 0$

V_x = constant

$\vec{F}_B = q\vec{v} \times \vec{B}$



(V_y) and (V_z) are change with time. $\Rightarrow V_\perp = \sqrt{V_y^2 + V_z^2}$

$$r = \frac{m v_\perp}{qB}$$

$$\omega = \frac{qB}{m}$$

$$T = \frac{2\pi m}{qB}$$

quick quiz (29.2):-

i- 2- particles moving in uniform $\vec{B}$ with $\vec{v} \perp \vec{B}$

if $v_2 > v_1$ then r_2 smaller, larger or equal to r_1 ??

$$r = \frac{mv}{qB} \quad , \quad r \propto v$$

as $v_2 > v_1$ then $r_2 > r_1$

ii- for the particle 1 if B increased

$B_{final} > B_{initial}$ then r_f smaller, larger or equal to r_i ??

$$r = \frac{mv}{qB} \quad , \quad r \propto \frac{1}{B} \quad , \text{ as } B_f > B_i$$

ther $r_f < r_i$

Ex (29.2):- a proton moving ⊥ to a uniform $\vec{B}$:-

Proton $r = 14\,cm$ $B = 0.35\,T$ $m_p = 1.67 \times 10^{-27}$ kg

$\vec{B} \perp \vec{V}$ $V = ??$

$$V = \frac{qBr}{m_p} = \frac{1.6 \times 10^{-19} \times 0.35 \times 0.14}{1.67 \times 10^{-27}}$$

$$= 4.7 \times 10^{6}\ m/s$$

what if ?? electron not proton

$m_e << m_p \Rightarrow r = \frac{m_e V}{qB}$

$r_e < r_p$ for the same values of V, q and B.

Ex (29.3):- Bending an electron beam:-

$\Delta V = 350$ volt $r = 7.5$ cm $\vec{V} \perp \vec{B}$

a- $|\vec{B}| = ??$ $V_i = 0$

electron – Electric field (isolated system $\Rightarrow$ energy is conserved).

$$\Delta K + \Delta U = 0$$

$$\frac{1}{2} m V_f^2 - \frac{1}{2} m \cancel{V_i^2}^{\,0} + q\Delta V = 0$$

$$V_f = \sqrt{\frac{-2q\Delta V}{m_e}}$$

$$V = \sqrt{\frac{-2(-1.6\times10^{-19})(350)}{9.11\times10^{-31}}} = 1.11\times10^{7}\ m/s$$

$$B = \frac{m_e v}{q_e r} = \frac{9.11\times10^{-31}\times1.11\times10^{7}}{1.6\times10^{-19}\times0.075}$$

$$= 8.4\times10^{-4}\ T$$

B- what is w for electrons?

$$w = \frac{v}{r} = \frac{1.11\times10^{7}}{0.075} = 1.5\times10^{8}\ rad/sec$$

1 revolution $= 2\pi$ rad

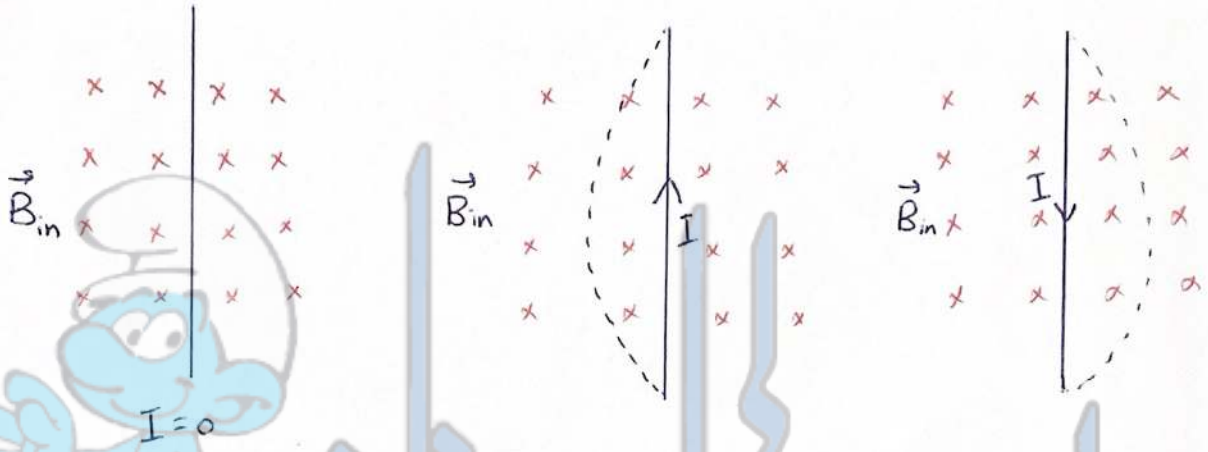
$$w = \frac{1.5\times10^{8}}{2\pi} = 2.4\times10^{7}\ rev/s$$

What if ?? ΔVoltage increase to 400 volt

ΔV increases $\Rightarrow$ velocity increases $\Rightarrow$

r increases, but $w = \frac{v}{r}$ remains constant.

29.4 Magnetic Force acting on a current Carrying conductor:-



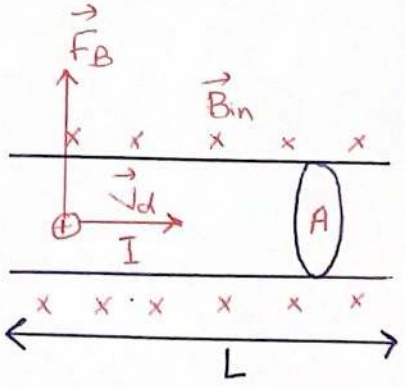
when current is zero; the wire remains vertical.

" ~ directed upward, the wire deflects to the left.

when current directed downward, the wire deflects to the right.

$$\vec{F}_B = q\vec{v}_d \times \vec{B}$$

Volume = AL



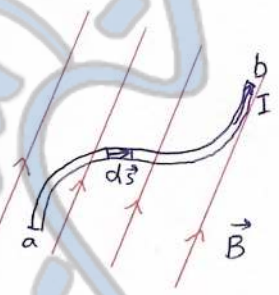
the number of charges in the segment = nAL , n : charge carrier density

$$\vec{F}_B = (q\vec{V}_d \times \vec{B})\, nAL$$

but $I = nqV_d A$

$\vec{F}_B = I\vec{L} \times \vec{B}$, where $\vec{L}$ is a vector points in the direction of I and has a magnitude of wire length in the magnetic field.

for arbitrary shaped wire in uniform $\vec{B}$ with uniform cross section.



$d\vec{F}_B = I\, d\vec{s} \times \vec{B}$

$\vec{F}_B \perp d\vec{s}$ and $\vec{B}$, so it's directed out of the page.

$$\vec{F}_B = \int_a^b I\, d\vec{s} \times \vec{B}$$

quick quiz (29.3):-

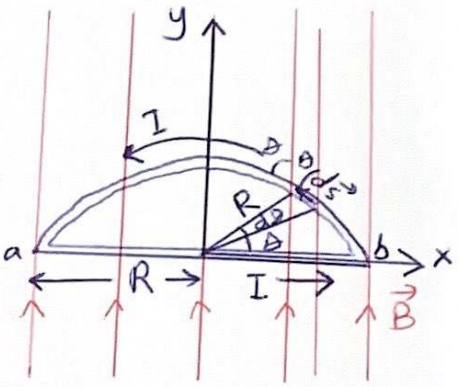
the direction of B??

C- upward out of the page. using right hand rule.

for closed loop carrying current I placed in $\vec{B}$

$$F = I \oint d\vec{s} \times \vec{B} = 0$$

Ex (29.4) :- force on a semicircular conductor :-



on the straigt part $\vec{F}_1$

$\vec{F}_1 \perp I, \vec{B} \Rightarrow$ along +ve z-axis

$$\vec{F}_1 = I \int_a^b d\vec{s} \times \vec{B}$$

$$= I \int_a^b B\, dx\, \hat{k}$$

$$\vec{F}_1 = 2IRB\hat{k}$$

on the curved part $\vec{F}_2 \perp I, \vec{B} \Rightarrow$ $\vec{F}_2$ along -ve z-axis

$$d\vec{F}_2 = I\, d\vec{s} \times \vec{B} = -IB \sin\theta\, ds\, \hat{k}$$

$\sin(d\theta) = d\theta$ (small angle)

$$d\theta = \frac{ds}{R} \Rightarrow ds = R\, d\theta$$

$$d\vec{F}_2 = -IBR \sin\theta\, d\theta\, \hat{k}$$

$$\vec{F}_2 = -\int_0^{\pi} IBR \sin\theta\, d\theta\, \hat{k}$$

$$= -IBR[-\cos\theta]_0^{\pi}\, \hat{k}$$

$$= +IBR[\cos\pi - \cos 0]\, k = -2IBR\hat{k}$$

29.5 Torque on a current Loop in a uniform magnetic field:-

a torque is exerted on a current loop placed in a magnetic field.

$$\vec{F}_1 = I\vec{L}\times\vec{B} = ILB\sin 180 = 0$$

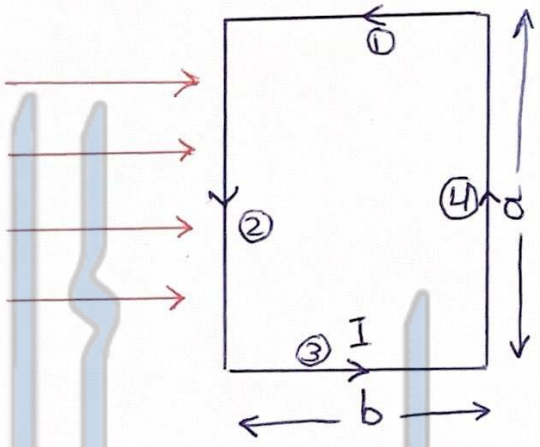
$$\vec{F}_2 = I\vec{L}\times\vec{B} = IaB\sin 90\,\hat{k}$$

$$= IaB\,\hat{k}$$

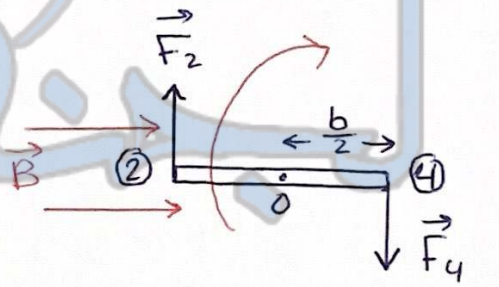
$$\vec{F}_3 = I\vec{L}\times\vec{B} = ILB\sin(0) = 0$$

$$\vec{F}_4 = I\vec{L}\times\vec{B} = -IaB\sin 90\,\hat{k}$$

$$= -IaB\,\hat{k}$$



the magnetic forces $\vec{F}_2$ and $\vec{F}_4$ creat a torque that rotates the Loop clockwise; the two forces Point in opposite directions but not along the same line of aetion.



$$\tau_{max} = F_2\left(\frac{b}{2}\right) + F_4\left(\frac{b}{2}\right) = IaB\frac{b}{2} + IaB\frac{b}{2}$$

$$= Iab\,B$$

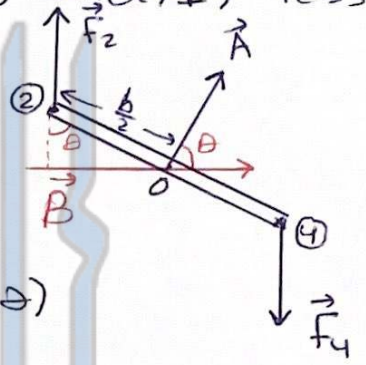
$$\tau_{max} = I a b B$$

$$= I A B \quad , A = ab$$

τ_{max} is valid when $\theta = 90^\circ$ between (B) and (A)

i.e $\vec{B}$ // Plane of the Loop.

but if θ between (B and A) less than 90°.



$$\tau = F_2 \frac{b}{2} \sin\theta + F_4 \frac{b}{2} \sin\theta$$

$$= I a B \left(\frac{b}{2} \sin\theta\right) + I a B \left(\frac{b}{2} \sin\theta\right)$$

$$= I a b B \sin\theta$$

$$\tau = I A B \sin\theta$$

for coil of (N) turns

$$\tau = N I A B \sin\theta$$

$$\vec{\tau} = I \vec{A} \times \vec{B}$$

$\vec{A}$:. vector $\perp$ to the plane of the Loop and has a magnitude = the area of the loop.

to determine the direction of $\vec{A}$, use right hand rule :-

1- curl your fingers in the direction of current around the loop.

2- your thumb points in the direction of $\vec{A}$.

magnetic dipole moment ($\vec{\mu}$) of the Loop is

$$\vec{\mu} = I\vec{A} \equiv [\text{ampere} \cdot m^2] \equiv [A \cdot m^2]$$

for coil contains N loops :-

$$\vec{\mu}_{coil} = NI\vec{A}$$

the direction of $\vec{\mu}$ is the same direction of $\vec{A}$.

$$\vec{\tau} = I\vec{A} \times \vec{B}$$
$$= \vec{\mu} \times \vec{B}$$

Potential energy of a system of magnetic dipole moment in a magnetic field is

$$U = -\vec{\mu} \cdot \vec{B} = -\mu B \cos\theta$$

$U_{min} = -\mu B \cos(0) = -\mu B$, $\vec{\mu}$ in the same direction of $\vec{B}$

$U_{max} = -\mu B \cos(180)$

$= \mu B$, $\vec{\mu}$ in the opposite direction of $\vec{B}$.

Ex(29.5):- the magnetic dipole of a coil

coil of dimensions 5.4cm x 8.5cm

N = 25 turns I = 15 mA

B = 0.35 T Parallel to the plane of the coil.

a- μ_{coil} = ??

$\mu_{coil} = NIA$

$$\mu_{coil} = 25 \times 15 \times 10^{-3} \times (0.054 \times 0.085)$$

$$= 1.72 \times 10^{-3} \ A \cdot m^2$$

b- $\tau = ??$

$$\tau = \mu_{coil} B$$

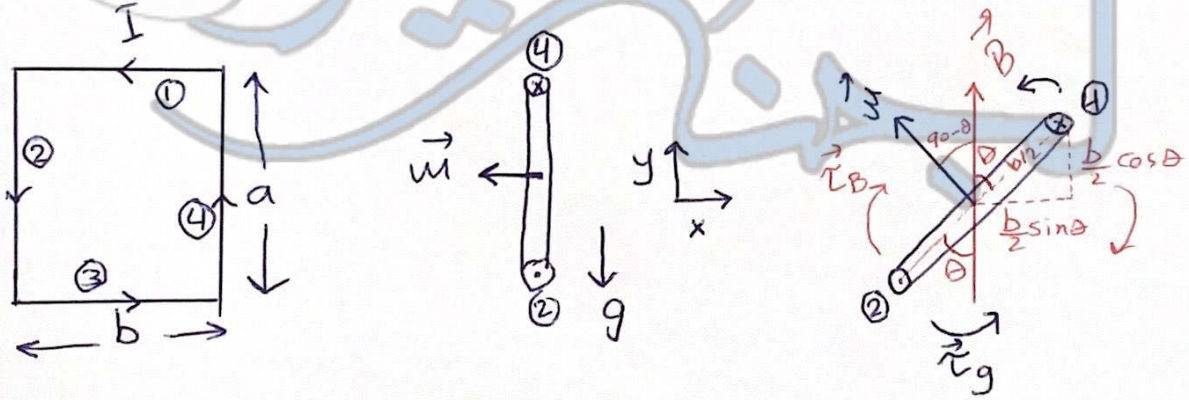
$$= 1.72 \times 10^{-3} \times 0.35$$

$$= 6.02 \times 10^{-4} \ N \cdot m$$

Ex (29.6):- Rotating a coil

$m = 50$ gram $\quad a = 0.2$ m $\quad b = 0.1$ m

$I = 3.5$ A $\quad \vec{B} = 0.01 \hat{j}$ T



$$\vec{\tau}_B = \vec{\mu} \times \vec{B} = -\mu B \sin(90-\theta)\hat{k}$$

$$= -IAB\cos\theta \hat{k}$$

$$= -Iab B\cos\theta \hat{k}$$

$$\vec{\tau}_g = \vec{r} \times m\vec{g} = mg\frac{b}{2}\sin\theta\,\hat{k}$$

rigid body in equalibrium

$$\Sigma\vec{\tau} = -IabB\cos\theta\,\hat{k} + mg\frac{b}{2}\sin\theta\,\hat{k} = 0$$

$$IabB\cos\theta = mg\frac{b}{2}\sin\theta$$

$$\frac{\sin\theta}{\cos\theta} \div \tan\theta = \frac{2aIB}{mg}$$

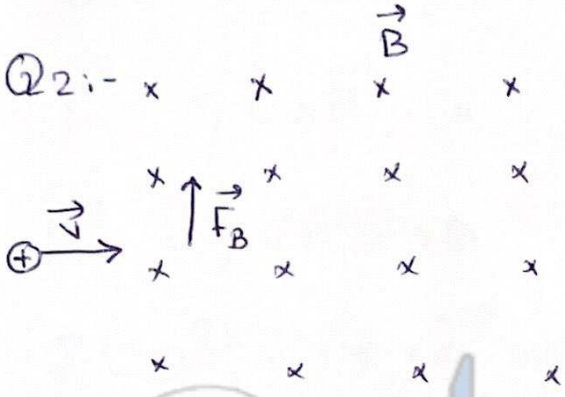
$$\theta = \tan^{-1}\left(\frac{2aIB}{mg}\right)$$

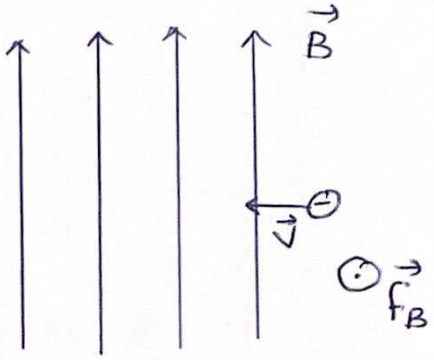
$$= \tan^{-1}\left(\frac{2\times 0.2\times 3.5\times 0.01}{0.05\times 9.8}\right)$$

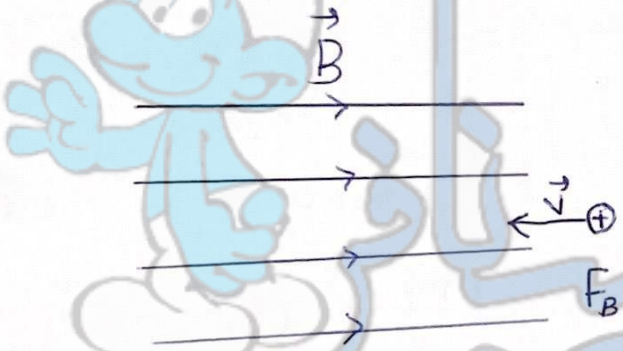
$$\theta = 1.64^\circ$$

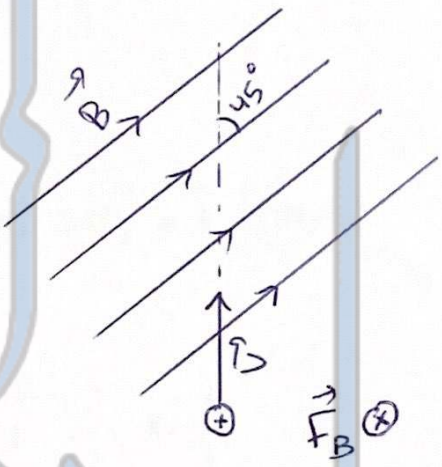
Solved Problems
chapter (29)

Q2:-

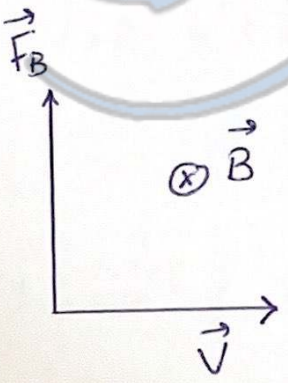


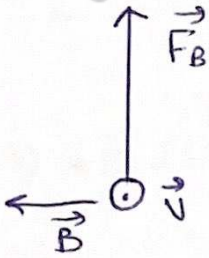


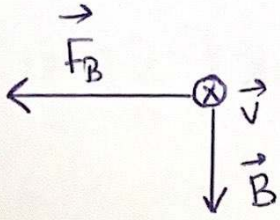




Q3:-







Q6:- $q = +1.6 \times 10^{-19}\ C$ $\quad v = 4 \times 10^{6}\ m/s$

$|\vec{B}| = 1.7\ T$ $\quad F = 8.2 \times 10^{-13}\ N$

$\vec{F} = q\vec{v} \times \vec{B}$

$|\vec{F}| = qvB \sin\theta$

$8.2 \times 10^{-13} = 1.6 \times 10^{-19} \times 4 \times 10^{6} \times 1.7 \sin\theta$

$\sin\theta = 0.74$

$\theta = 48.9^{\circ}$

Q8:- $q = 1.6 \times 10^{-19}\ C$ $\quad \vec{v} = 2\hat{i} - 4\hat{j} + \hat{k}\ (m/s)$

$\vec{B} = \hat{i} + 2\hat{j} - \hat{k}\ (T)$

$$\vec{F} = q\vec{v} \times \vec{B} = 1.6 \times 10^{-19} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 1 \\ 1 & 2 & -1 \end{vmatrix}$$

$= 1.6 \times 10^{-19} [\hat{i}(4-2) - \hat{j}(-2-1) + \hat{k}(4--4)]$

$= 1.6 \times 10^{-19} [2\hat{i} + 3\hat{j} + 8\hat{k}]$

$= [3.2\hat{i} + 4.8\hat{j} + 12.8\hat{k}] \times 10^{-19}\ N$

Q9:- Proton $V = 5.02 \times 10^{6}$ m/s $\theta = 60^\circ$

$B = 0.18 \hat{i}$ T

a- $F = qvB \sin\theta$

$= 1.6 \times 10^{-19} \times 5.02 \times 10^{6} \times 0.18 \times \sin 60^\circ$

$\vec{F} = 1.25 \times 10^{-13} \hat{k}$ N

b- $\vec{a} = \dfrac{\vec{F}}{m} = \dfrac{1.25 \times 10^{-13}}{1.67 \times 10^{-27}}$

$= 0.75 \times 10^{14}$ m/s^2

Q11:- Proton $\vec{V} = 1 \times 10^{7} \hat{k}$ m/s $\vec{a} = 2 \times 10^{13} \hat{i}$ m/s^2

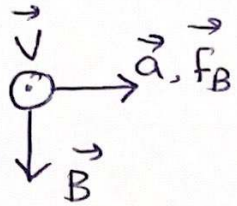
$F = ma = qvB \sin\theta$

$B = \dfrac{ma}{qv \sin\theta}$

$= \dfrac{1.67 \times 10^{-27} \times 2 \times 10^{13}}{1.6 \times 10^{-19} \times 1 \times 10^{7} \times \sin 90}$

$B = 2.09 \times 10^{-12}$ T

$\vec{B} = -2.09 \times 10^{-12} \hat{j}$ T



Q13:- electron $B = 2\ mT$ $V = 1.5 \times 10^{7}\ m/s$

a- $r = \frac{mv}{qB}$

$= \frac{1.67 \times 10^{-27} \times 1.5 \times 10^{7}}{1.6 \times 10^{-19} \times 2 \times 10^{-3}}$

$= 0.78 \times 10^{-4}\ m$

b- $T = \frac{2\pi r}{v}$

$= \frac{2(3.14)(0.78 \times 10^{-4})}{1.5 \times 10^{7}}$

$= 3.27 \times 10^{-11}\ sec$

Q24:- Proton $B = 0.45\ T$ $r = 1.2\ m$

a- $w = \frac{qB}{m} = \frac{1.6 \times 10^{-19} \times 0.45}{1.67 \times 10^{-27}} = 0.43 \times 10^{8}\ rad/s$

b- $w = \frac{v}{r} \Rightarrow v = wr$

$v = \frac{0.43 \times 10^{8}}{2\pi} \times 1.2\ (m/s)$